



3.0 Rollers

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3.0 Rollers

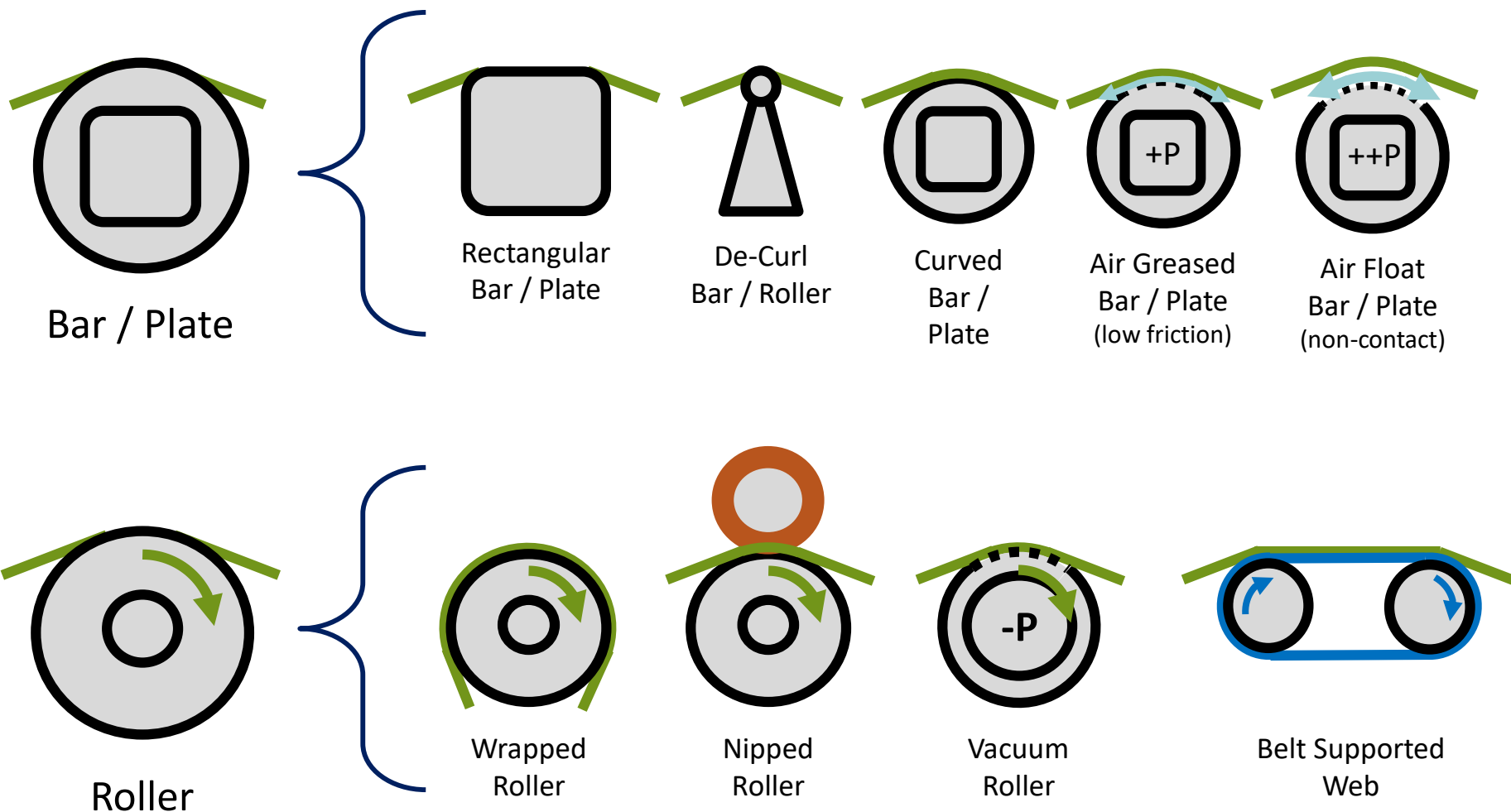
3.1 Roller Design

3.2 Roller Alignment

3.3 Web-Roller Traction

3.4 Air Flotation

Web Support / Transport





3.1 Roller Design

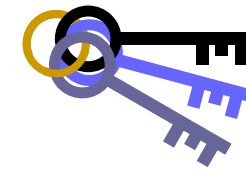
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3.1 Learning Objectives

- ▶ What key factors should be included in a roller specification?
- ▶ What positive functions does a roller serve?
- ▶ What potential problems can a roller create?
- ▶ What is a span?
- ▶ What are the minimum and maximum limits of span length?



Roller Pros and Cons

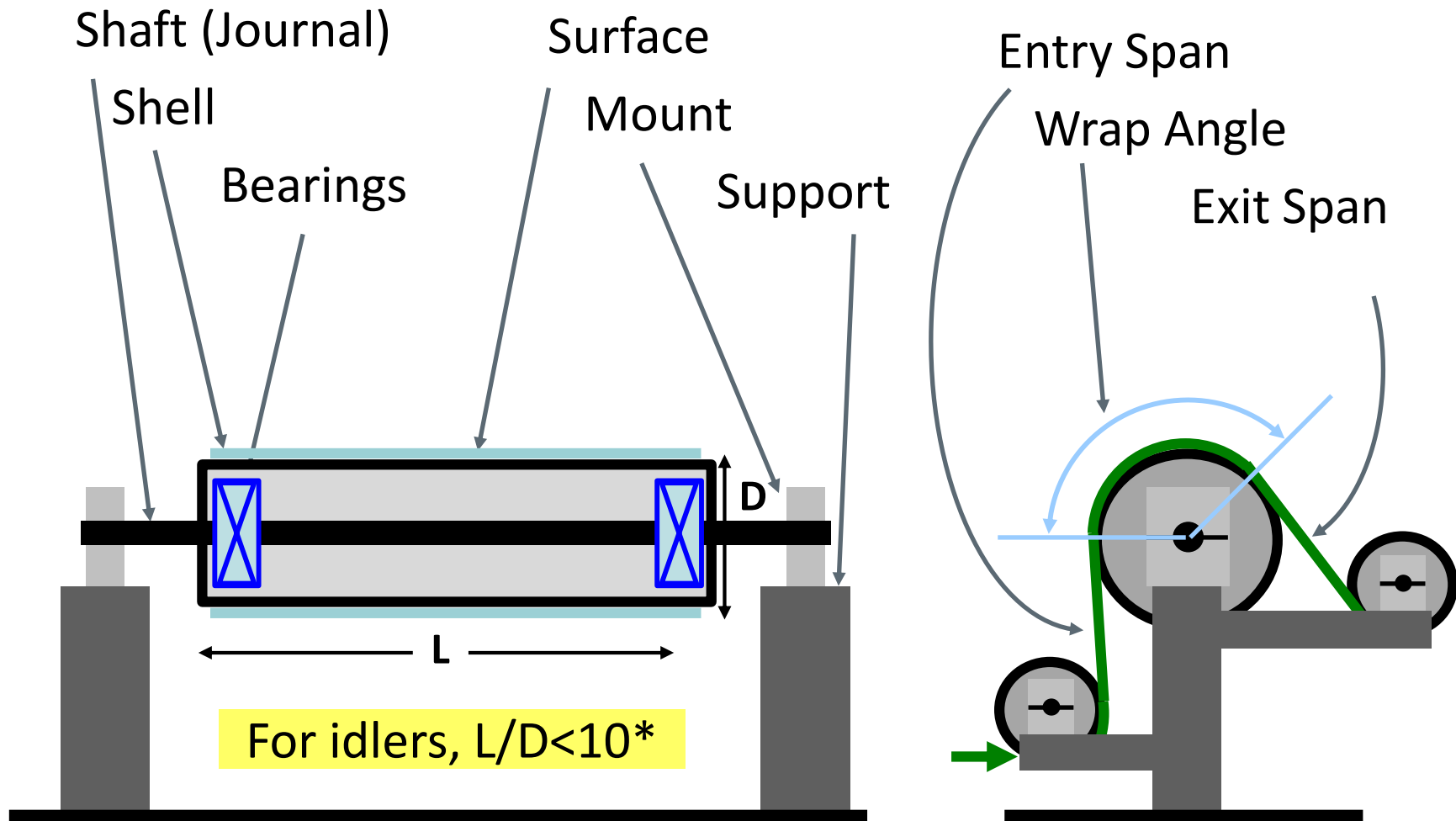
Roller Benefits:

- Redirect the web angle.
- Oppose droop in horizontal spans.
- Promote centerline tracking.
- Provide traction forces to bend the web.
- Create changes in web tension.
- Provide feedback to web tension control.
- Spread a web or slit strands.
- Induce a shape that adds lateral web stiffness.

Roller Detriments:

- Scratching.
- Wrinkling.
- Static charging.
- Excessive drag or inertial torque load.
- Require maintenance for bearings, surface, and alignment.

Idler Roller Design Terminology



*The true design criteria is deflection under load.

General Idler Roller Properties

Idler rollers should be:

- ▶ Low mass to enable easy installation, maintenance, and acceleration to process speed.
- ▶ Cylindrical (of uniform diameter or engineered with specific diameter variations).
- ▶ Free turning.
- ▶ Straight with minimal deflection under gravity or external loads from tension, nipping, or centrifugal forces from self-imbalance.

Roller Data Sheet

Understanding rollers and roller performance starts with data.
Recommended: Create a roller log sheet, including:

Roller Web Path Data Sheet

- Roller ID
- X,Y Position
- X,Y Reference Pt.
- Rotation
(CW or CCW)
- Tension Zone

Roller Design Data Sheet

- Style Name
- Shell Details
- Shaft Details
- Mount Details
- Bearing Details
- Surface or Covering
Details
- Cover replacement

Bearing / Alignment Data Sheet

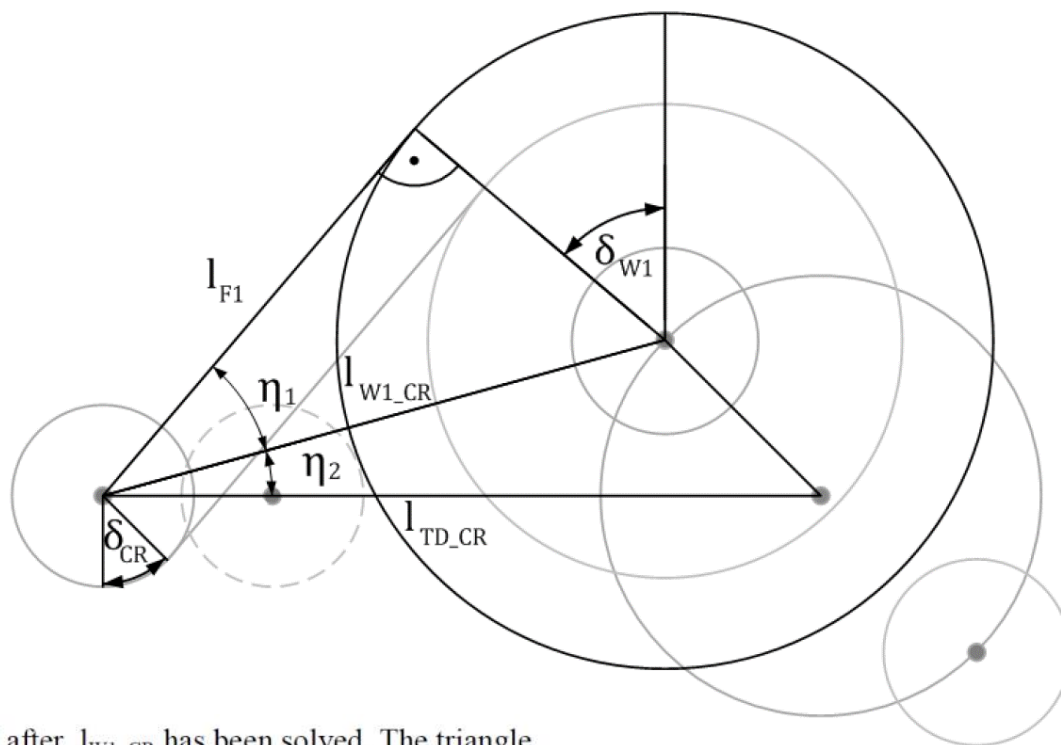
- Roller ID
- Spindown Start
Speed
- Spindown Time
- Alignment Date
- Pre-Alignment Error

Web Path Lengths

FAST WINDER CHANGE WITH REDUCED INFLUENCE ON FILM TENSION

By

Stefan Schuster and Günter Oedl
Brückner Maschinebau
GERMANY



The length l_{F1} can only be calculated after l_{W1_CR} has been solved. The triangle given by the centers of CR, TD and W1 is solved by the formula {5}.

$$l_{W1_CR} = \sqrt{r_{TD}^2 + l_{TD_CR}^2 - 2 * r_{TD} * l_{TD_CR} * \cos(\varphi_{TD})} \quad \{5\}$$

For the length of the free span between the contact roller and the winder 1 can easily be solved by constructing an auxiliary rectangular triangle through center of the contact roller and the center of the winder 1 and a cathetus of $r_{W1} + r_{CR}$. With other words the length l_{F1} is the same as a tangent from the center of contact roller to an auxiliary circle around the center of winder 1 and a radius of $(r_{W1} + r_{CR})$.

Roller Options – Live vs. Dead Shaft, Idler vs. Driven

Dead Shaft Idler Roller



Shaft does not rotate.

Live Shaft Idler Roller



Shaft rotates with shell.

Live Shaft Driven Roller



Roller Options – Live vs. Dead Shaft, Idler vs. Driven

Dead Shaft Idler Roller



Live Shaft Driven Roller



Live Shaft Idler Roller



Torque Driven (Clutched) Roller



Dual Bearing Idler Roller



Tendency Driven Roller



Roller Mount and Support Options



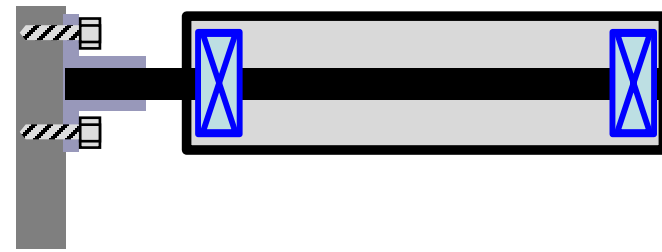
Pillow Block Dual Side Frame



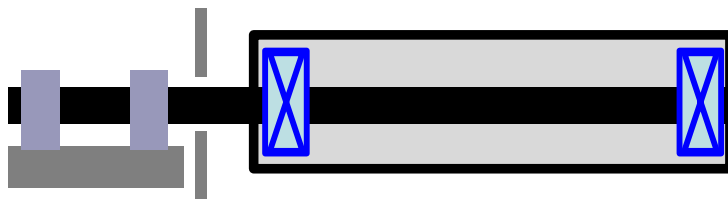
Flange Mount Dual Side Frame



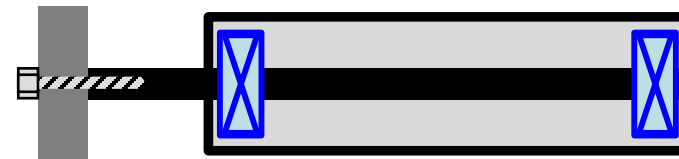
Pillow Block Cantilever Beam



Flange Mount Cantilever Shaft



Dual Support Cantilever Shaft



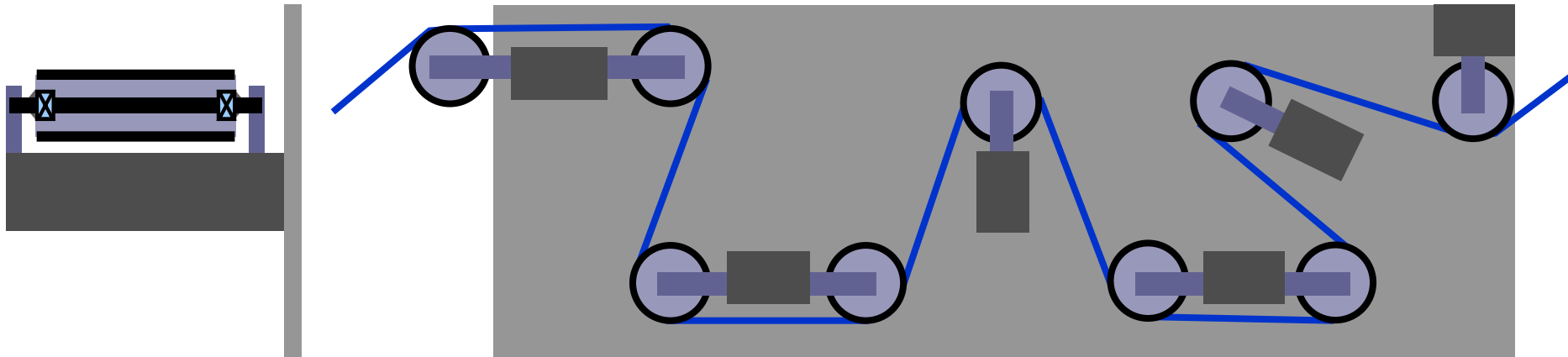
Bolted Cantilever Support

Cantilevered Rollers 2

Greatest Advantage: Ease of Threadup

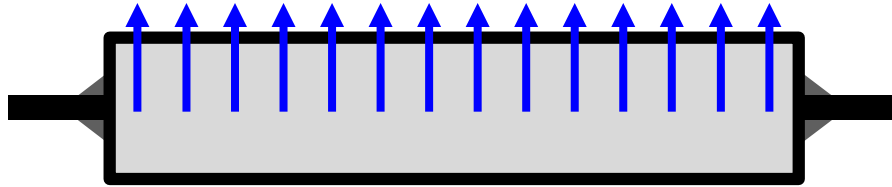
Greatest Disadvantage: Deflection

Correct layout of dual-end supported rollers off a cantilevered structure will keep threadup benefits without deflection problems.



Roller Profiles

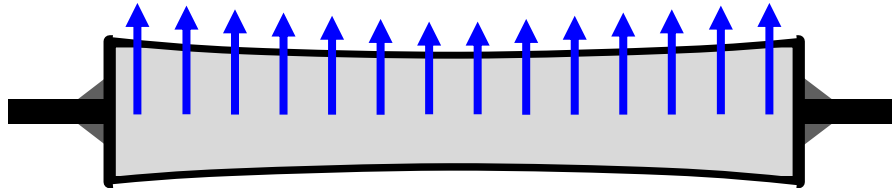
Each roller profile has a unique combination of surface speed and angle profile.



Cylindrical:

Uniform speed profile

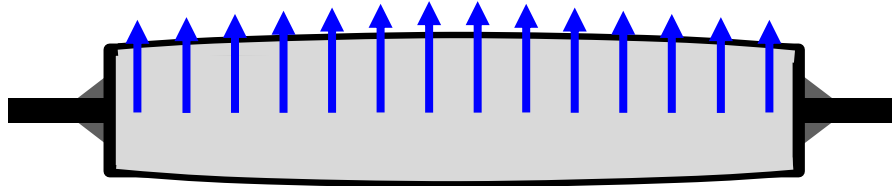
All lanes parallel to machine direction (MD)



Concave (a.k.a. Reverse Crowned):

Faster edges vs. center

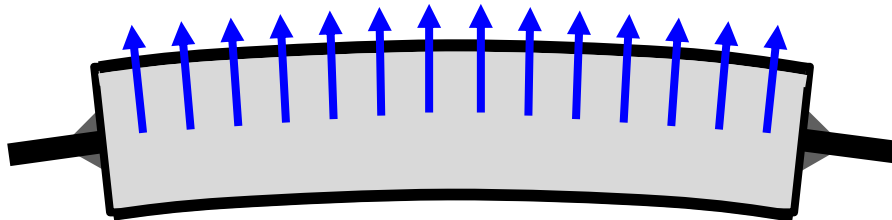
All lanes parallel to MD



Convex (a.k.a. Crowned):

Faster center vs. edges

All lanes parallel to MD



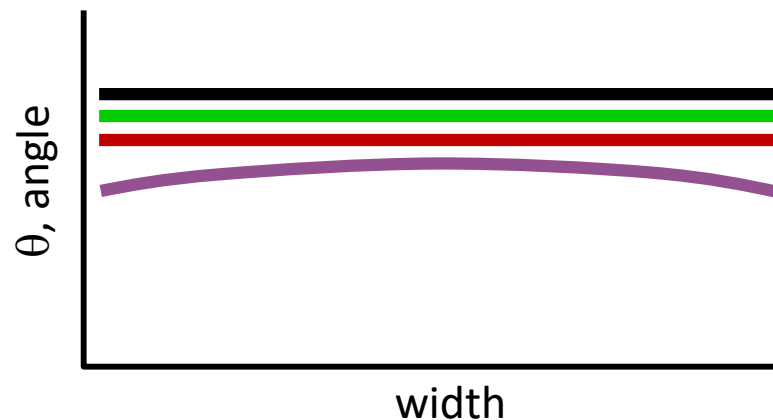
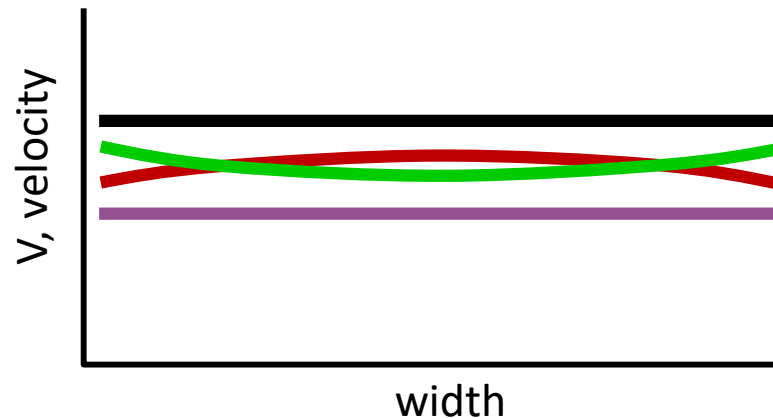
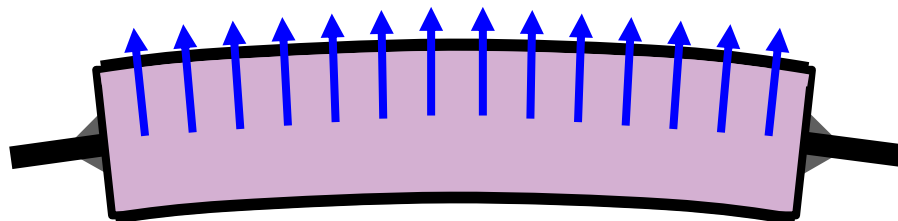
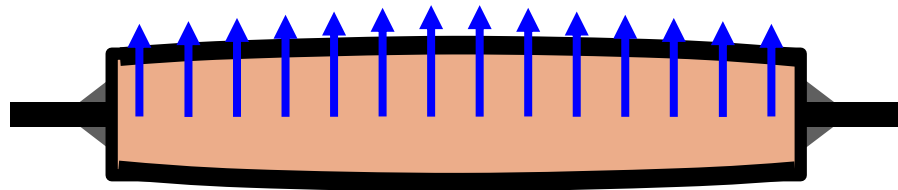
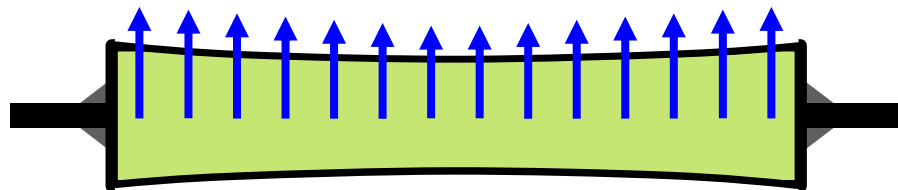
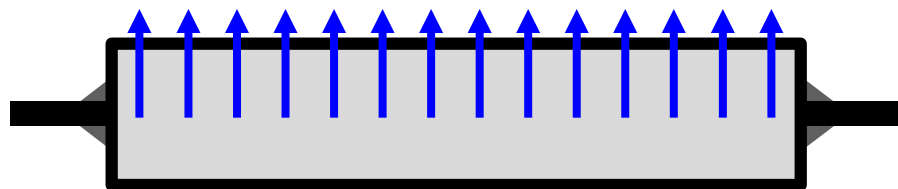
Bowed (a.k.a. Curved Axis, 'Banana'):

Uniform speed profile

Progressive outward angle of lanes

Roller Profiles

Each roller profile has a unique combination of surface speed and angle profile.



Roller Shell and Surface Materials



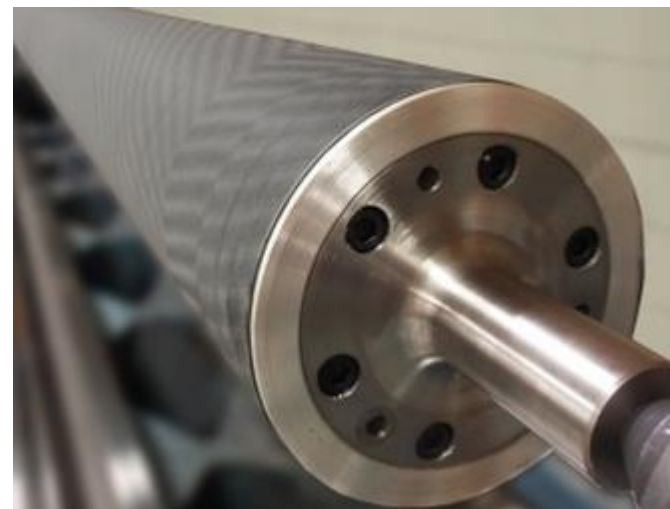
Courtesy of Webex Inc.

Hard Surfaces

- Anodized Aluminum
- Electroless Nickel
- Chrome Plating
- High Release Tungsten Carbide (Plasma)

Shell Materials

- Aluminum
- Steel
- Carbon Fiber Composite



Apply Friction Wrap

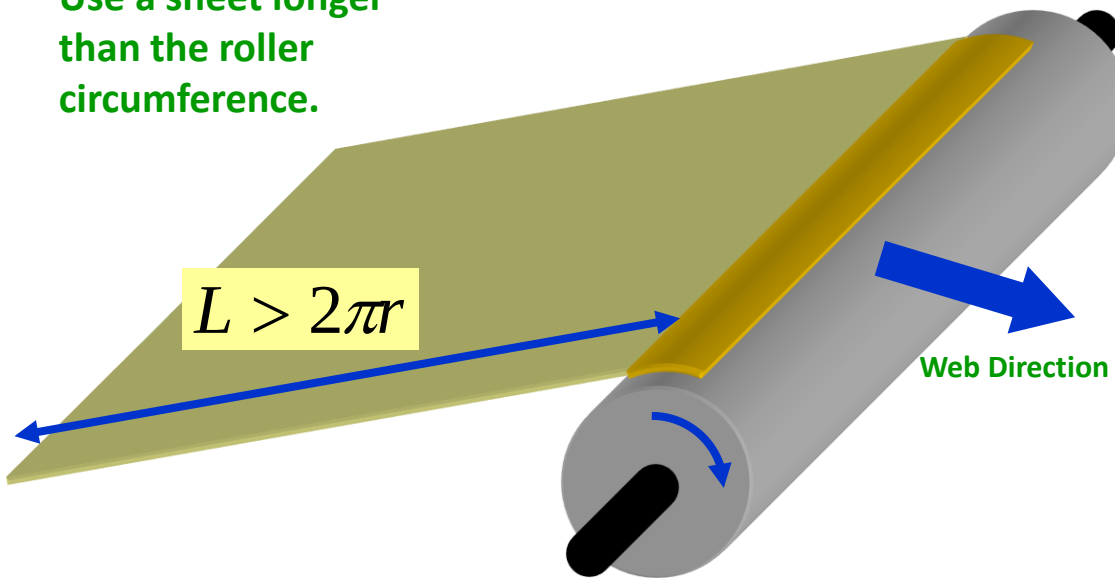
Use a roller wrap sheet to:

1. Increase traction, preventing slip
2. Decrease traction, preventing wrinkles
3. Change surface energy, reducing peel force.

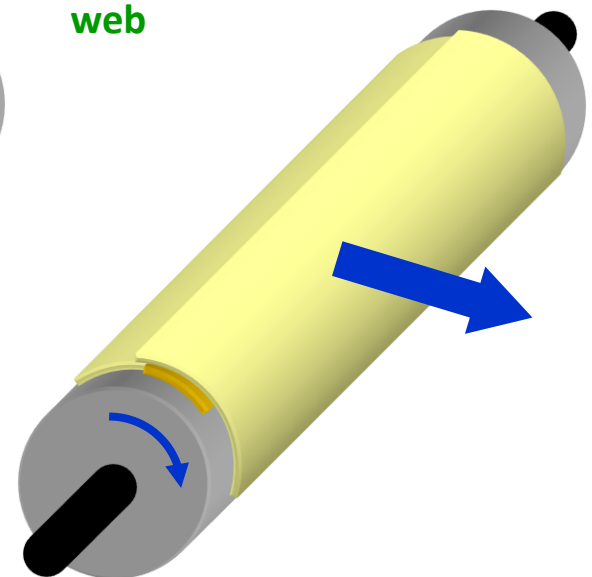
Use a sheet longer than the roller circumference.

Tape the leading edge

$$L > 2\pi r$$



Apply the wrap by rotating the roller in the same direction as the running web



Roller Internal Structure

Thin-Walled Cylinder with Gudgeons

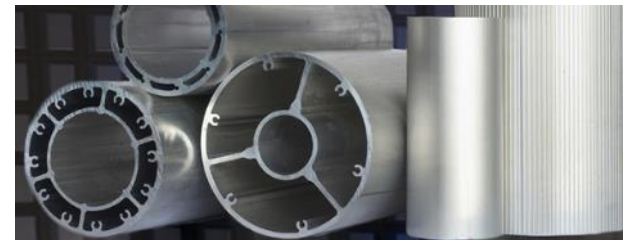


Double-Walled Spoked Cylinders



Courtesy of Double E

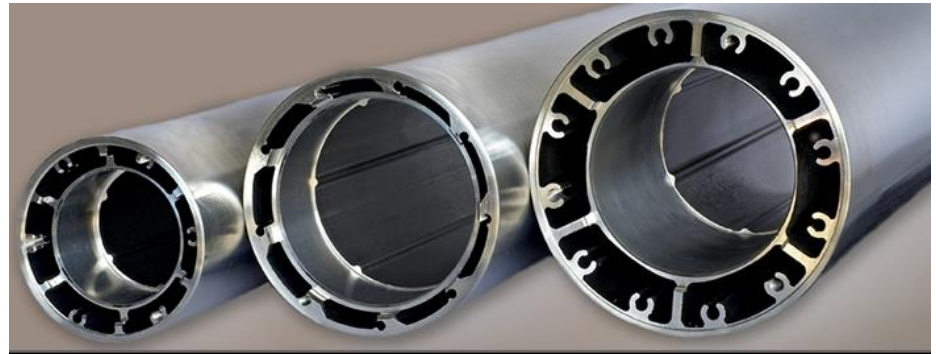
- The double-walled cylinder has increased shape stiffness (like an I-beam).



Courtesy of Componex

Roller Internal Structure

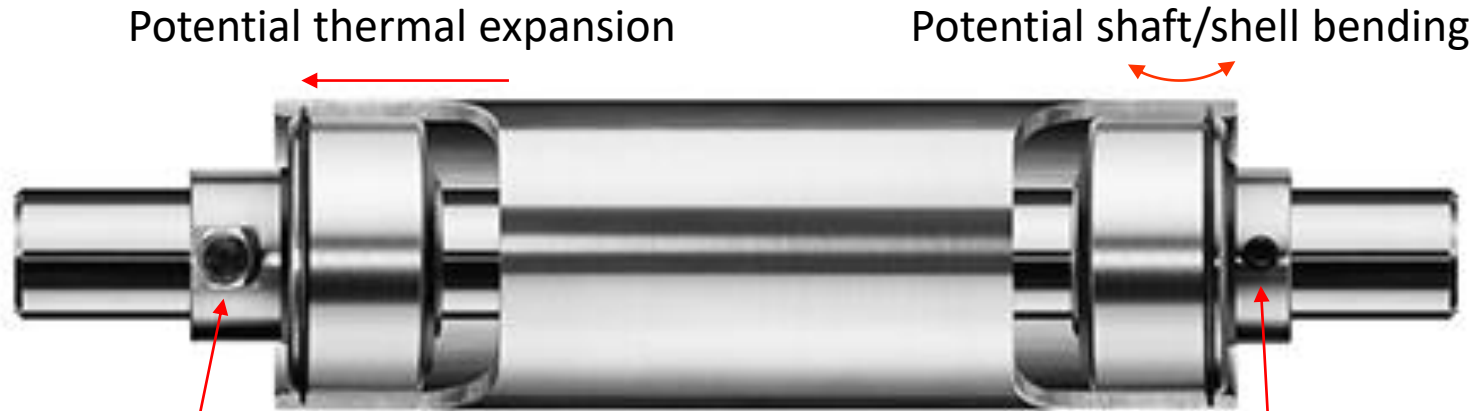
Double-Walled Spoked Cylinders



Courtesy of Componex

- Inner cylinder can be bored to fit bearing press fit, eliminating gudgeons.
- Narrower bearing-to-bearing distance reduces deflection.

Componex Roller Winertia Bearing System



Plunger screw

The plunger screw applies friction to the shaft to keep the inner race of the bearing from spinning when the roller is in operation. When thermal expansion causes the roller to grow, the plunger allows the inner race to slide down the shaft, eliminating side load on the bearing.

Set screw

The set screw on the bearing locks the roller to the shaft.

Courtesy of Componex

Balancing?

- ▶ Roller balance can be significant.
- ▶ Static balancing all roller and core cylinders
- ▶ Dynamic balancing may not be carried out unless otherwise specified by the customer.
 - ▶ G6.3(Balance Grade) for rollers for film and paper industries (less than 400m / min)
 - ▶ G2.5(Balance Grade) for high revolution rollers (400 - 1500m / min)

Roller Balancing

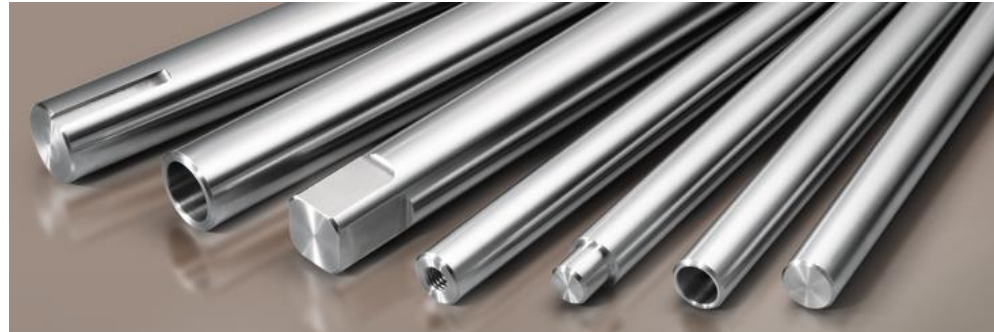
- ▶ For low speed applications (less than 1000 fpm or 300 mpm), good manufacturing practices can meet balancing needs.
- ▶ Static balancing can prevent 'heavy' side of rollers and once-around tension upsets.
- ▶ Dynamic balancing is recommended at high rpm applications to prevent roller damage and tension upsets.



Courtesy of Menges Roller

Idler Roller Shafts

- ▶ Hollow shafts have similar bending stiffness, but greatly reduce the shaft weight.



Reference	Actual Size	Weight
.75"	(.7495/.7485)	.125 lbs/in
1.0"	(.9995/.9985)	.25 lbs/in
1.25"	(1.2495/1.2485)	.35 lbs/in
1.5"	(1.4995/1.4985)	.50 lbs/in
2.0"	(1.9995/1.9985)	.89 lbs/in
25mm	(.9838/.9828)	.22 lbs/in
40mm	(1.5743/1.5733)	.55 lbs/in

Courtesy of Componex

Roller “Rules of Thumb”

To minimize deflection...

- Wrapped Roller
 $L/D < 10$
- Nipped Roller (< 100 PLI)
 $L/D < 5$

To minimize drag issues...

- Turning torque < (radius)(1% of tension)

To check balance...

- A spun roller should stop randomly
(no heavy side).

Roller Specifications

- ☐ Shell (a.k.a. core): material, diameter, face length, balance
- ☐ Journals: (left and right) material, diameter, profile, length
- ☐ Covering or coating: material, thickness hardness, roughness, release properties
- ☐ Bearings: live/dead shaft, type, lube, eccentricity, seal
- ☐ Alignment: specification, straightness, deflection, adjustment, parallelism, runout, mount style
- ☐ Environment: temperature, atmosphere, chemicals

The Great Span Length Debate

- Costs (a roller every $\frac{1}{2}$ web width is expensive)
- Ergonomics and Safety (not too close if manually threaded)
- Gravity (in horizontal spans, consider catenary sag)
- Loose Edges and Misalignment (longer spans more tolerant of misalignment)
- Traction (is there enough wrap angle to go around)
- Air Turns (pre-air turn rollers are a good idea)
- Air Floatation (who needs rollers?)
- Tension Control (usually not a factor)
- Guiding (longer spans take less force to shift)
- Wrinkling (long or extremely short have less wrinkles)
- Spreading (long is usually better)
- Winding (avoid long spans into winding).

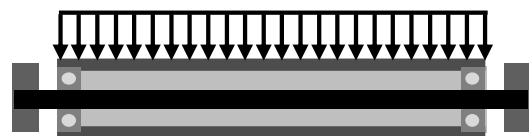
Great Span Debate: Advice

- ▶ Spans are often discussed in L/W ratios (span length to web width), but this is only reasonable when assuming a limited range of widths. (An L/W of 1-2 doesn't make sense when width are narrow.)
- ▶ Consider five broad width ranges:
 - Strands: less than 2-in (<50mm)
 - Narrow: 6 to 20 inches (150-500mm)
 - 'Standard': 40 to 60 inches (1.0-1.5m)
 - Wide: 80 to 160-in (2-4m)
 - Extra Wide: greater than 160-in (>4m)
- ▶ TJW advice on limiting long spans:

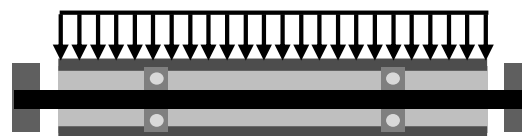
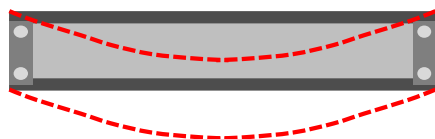
W=1-6 in. (25-150mm),	Length < 80-in. (<1.5m)
W=24-80 in. (0.6-2m),	Length < 100-in. (<2.5m)
W=100-200 in (2.5-4m),	Length < 160-in (<4m)

(Double these lengths for pure vertical spans where gravity does not pull the web out of plane.)

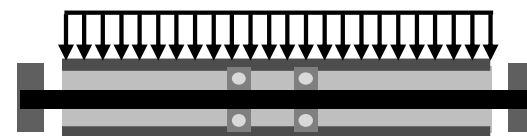
Shell Deflection and Bearing Position



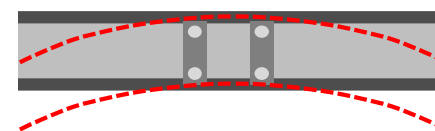
60-in.



34-in.

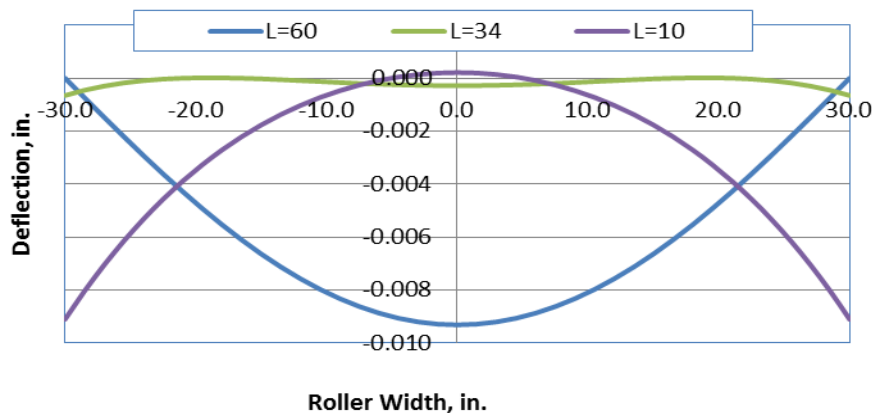


10-in.



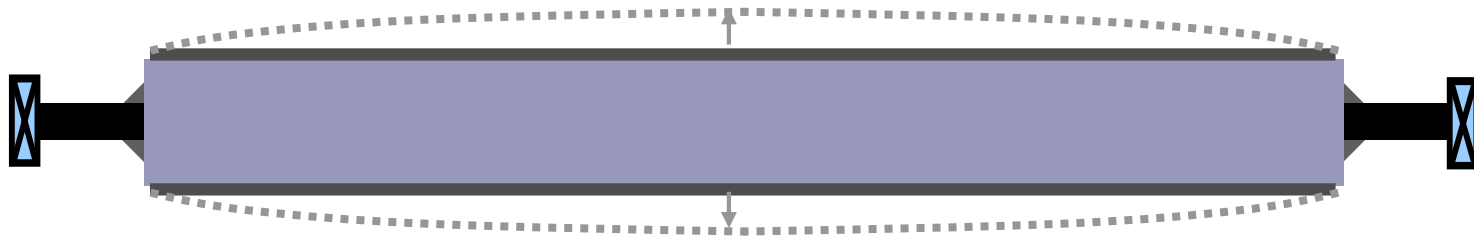
Red dashed lines show direction and magnitude of deflection from straightness.

Roller Deflection vs. Support Location



Critical Velocity

A high rotational speeds a roller will begin whipping when the roller's mass and spring constant hit their resonance frequency.

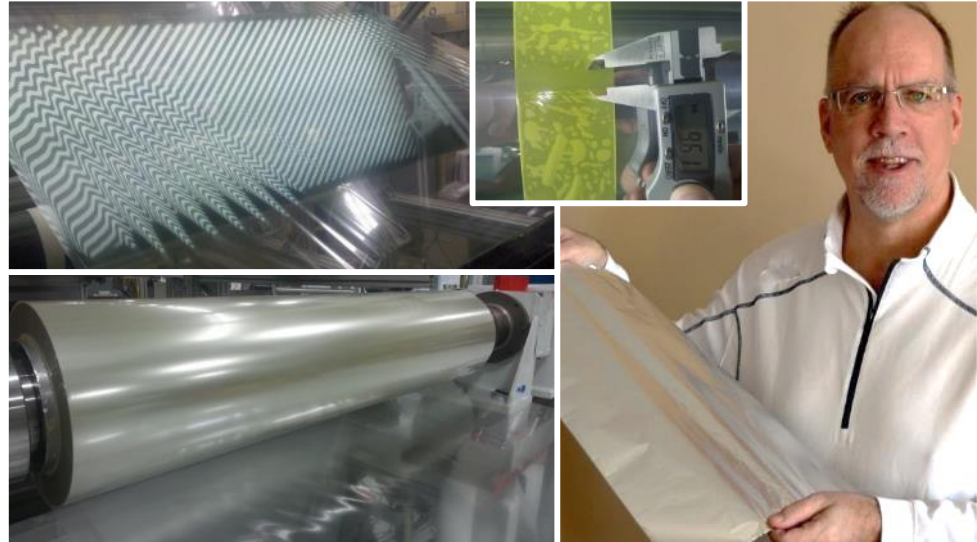


$$V_{crit} = 15.71 fD$$

Function of
diameter-cubed

$$f = \frac{\pi}{2} \sqrt{\frac{gEI}{wl^4}}$$

Function of
diameter-squared



3.3 Roller Alignment

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3.2 Learning Objectives

- ▶ What are level and square (tram)?
- ▶ Why is uniform diameter important to roller alignment?
- ▶ What methods are used to measure roller alignment?
- ▶ What is the right specification for roller alignment?
- ▶ What level of misalignment will create problems?
- ▶ How often should equipment be re-aligned?
- ▶ What makes equipment friendly to setting and holding roller alignment?

Roller Alignment

Level = Perpendicular to gravity

Square = Perpendicular to centerline

What is required?

Angle errors less than 0.2-5.0 mils/in
or mm/m (0.2-5.0 mradians)

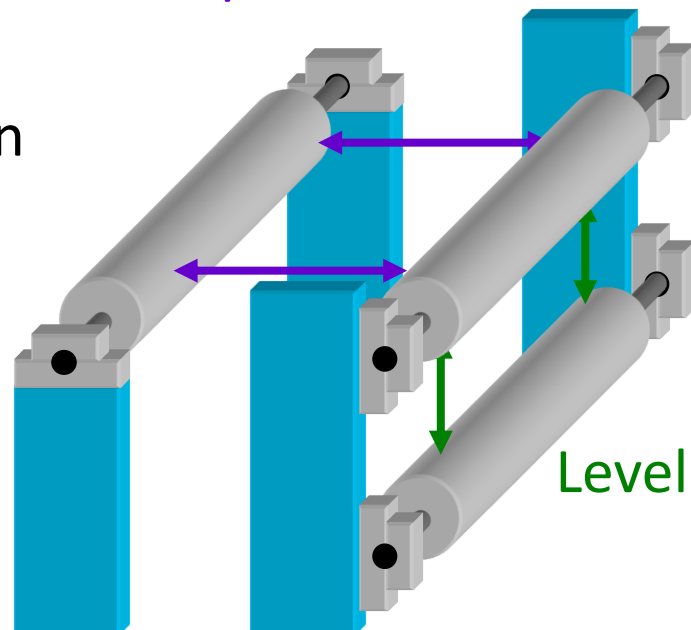
What is measurable?

0.05-0.10 mradians

Common Specification?

<0.2 milli-radians

Square to Centerline



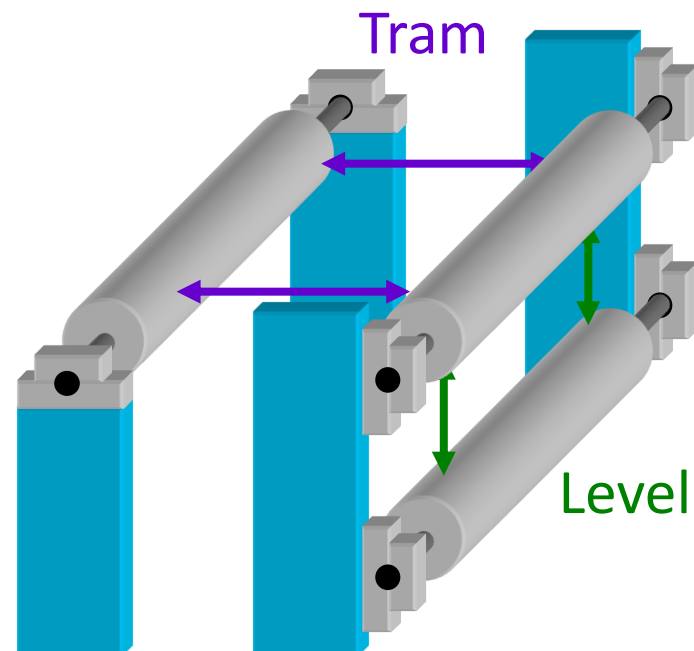
Roller Alignment

Common Specification? < 0.2 m-radians

This specification should be no surprise to most converting equipment suppliers and not incur unreasonable added expense.

Extreme webs can relax this specification.

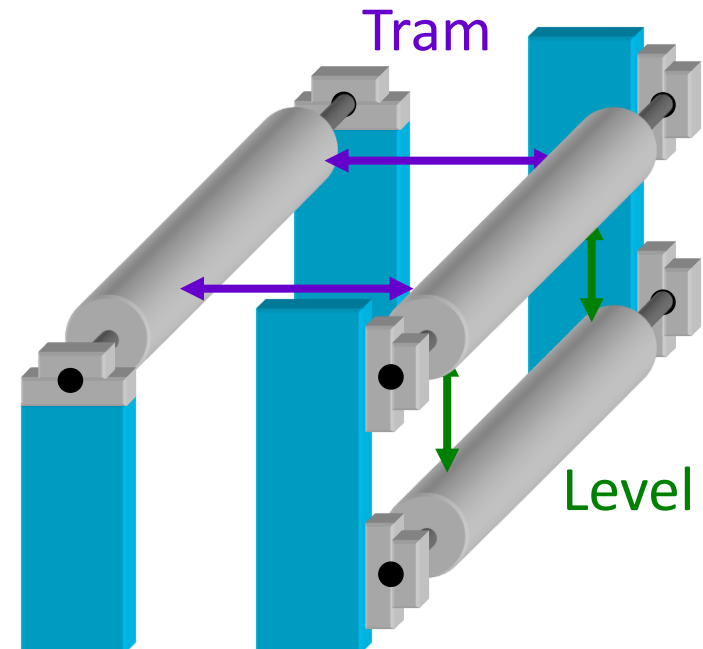
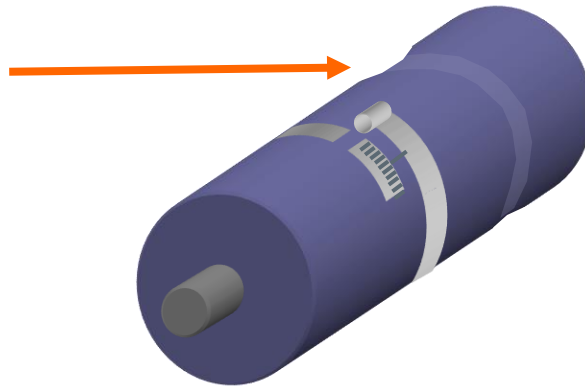
- Products that are especially stiff (don't wrinkle or bend easily) are not easily damaged or shifted with misaligned rollers.
- Products with low modulus (over 1% stretch under typical tensions) can relax this specification since small equipment errors do not create large tension changes.



Roller Alignment

Even small diameter variations can create a big error in roller alignment measurements.

**Diameter
Variations**



Level = Perpendicular to gravity

Square = Perpendicular to centerline

What is a Pi Tape?

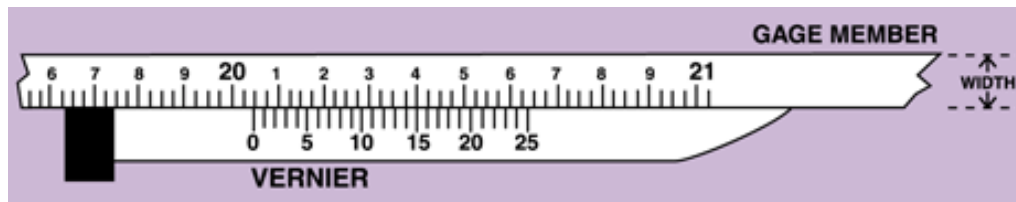
A Pi Tape™ is a precision steel tape used to measure internal or external diameters.



www.pitape.com

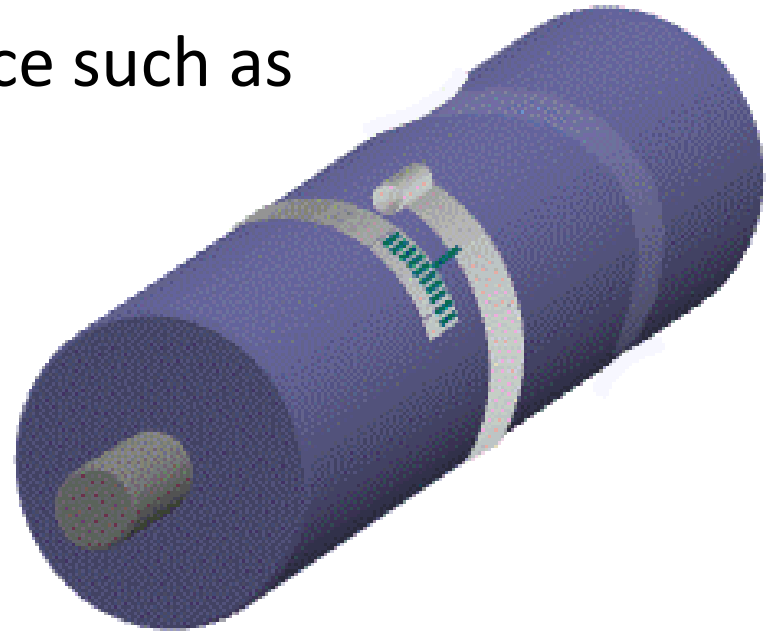


The length units are spread out by a factor or 3.1415, so a Pi Tape reads diameter directly. A Vernier scale allows measurement to ± 0.001 inch (or ± 25 microns).



What is a Pi Tape?

Unlike a typical tape measure, a Pi Tape has no camber or cross-tape curvature allowing it to lay flat when wrapping around a curved surface such as a roller.



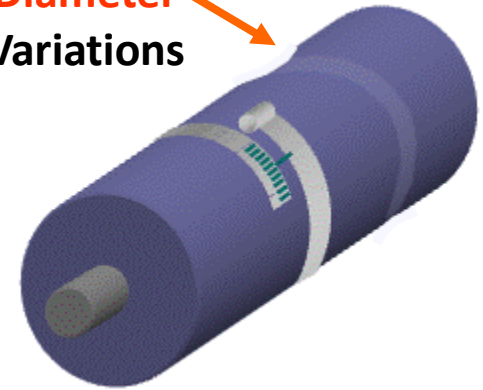
Why Measure Roller Diameter?

Reason #1:

Cross-roller diameter variations cause errors in roller level and alignment measurements.

Even small diameter variations can create a big error in roller alignment measurements.

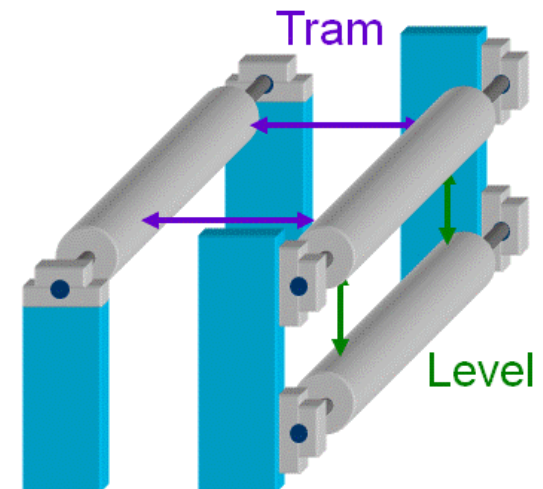
Diameter
Variations



Goal:

New metal rollers should be machined to better than ± 2 mils.

Diameter variations should cause less than a 20% error in measurement.



Goal: What is Good Enough?

Reason #2:

Cross-roller diameter variations will create crossweb tension variations. A large diameter section of a roller will shift load of tension to that web position, possibly damaging the web.

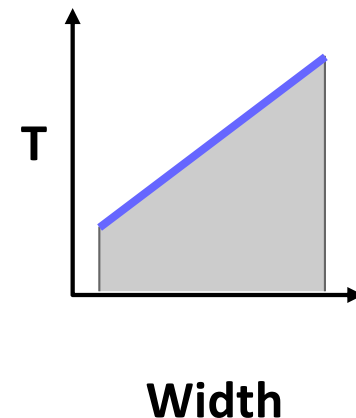
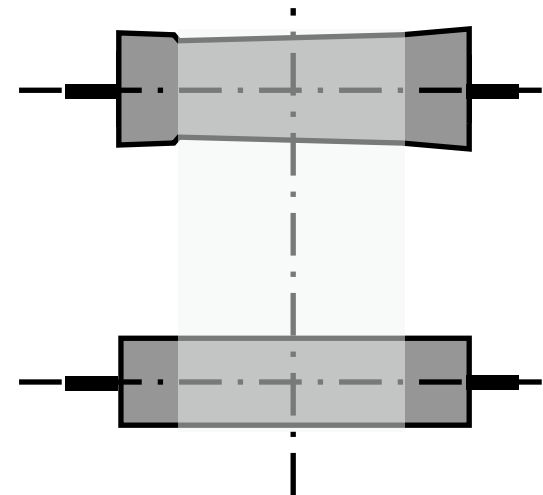
Goal:

To avoid damaging your web products, diameter variations should be less than yield strain (plus a safety factor)

Examples: If yield strain is 1%

60 mils on 6-inch rollers

80 mils on 8-inch rollers



Why Measure Roller Diameter?

Reason #3:

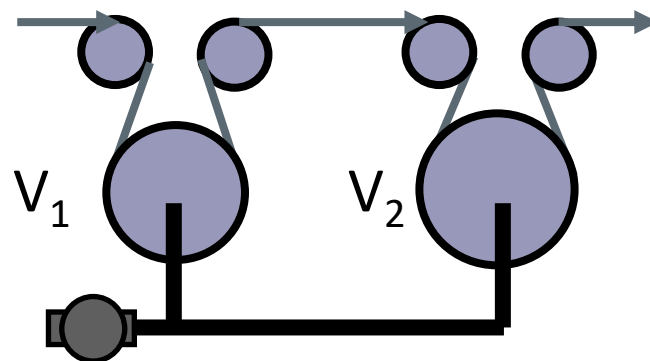
In a section, such as the cooling section, where a series of rollers are driven as a set, small roller-to-roller diameter variations can lead to tension increases, or worse, total loss of tension.

The stretch in the web between open-loop speed controlled roller will increase or decrease proportional to the percent speed change. If a web has low stretch (0.1-0.2%), then a small decrease in roller-to-roller diameter can cause slack web.

Goal:

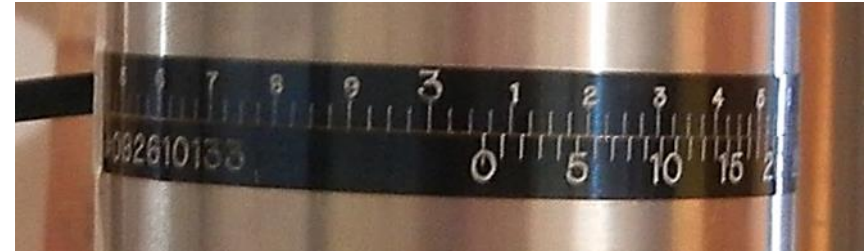
If tension creates a low web stretch (0.2%), then diameter decreases need to be less than 0.1%.

Example: 6 mils on 6-inch rollers
12 mils on 12-in rollers



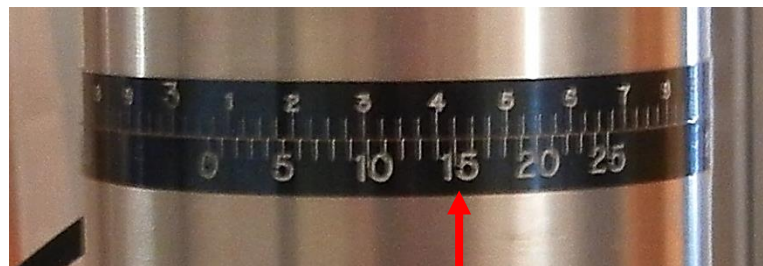
How to... Pi Tape

1. Make certain the tape is free of damage and the tape and object to be measured are both clean..
2. Wrap the tape around the object to be measured. The Vernier scale should be just below the gauge scale. Tighten the tape around the object with 5 pounds tension for O.D. tapes.
3. Each line on the gage member represents 0.025", while each line on the Vernier represents 0.001".
4. Locate the "zero" on the vernier scale and note the highest value achieved on the gage scale above it (the highest value to the left of the zero). In this example, that value is 3.050".

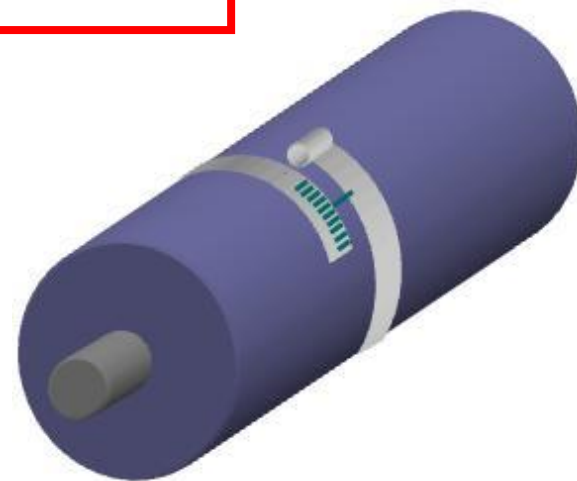


Pi Tape Diameter Measurements

5. Next, observe the vernier scale's value at the point where it lines up exactly with a marked division line on the gauge scale. In this example, that value is 16 (.016").



6. Finally, to obtain the diameter of the object, simply add the two values together:
 $3.050'' + .016'' = 3.066''$

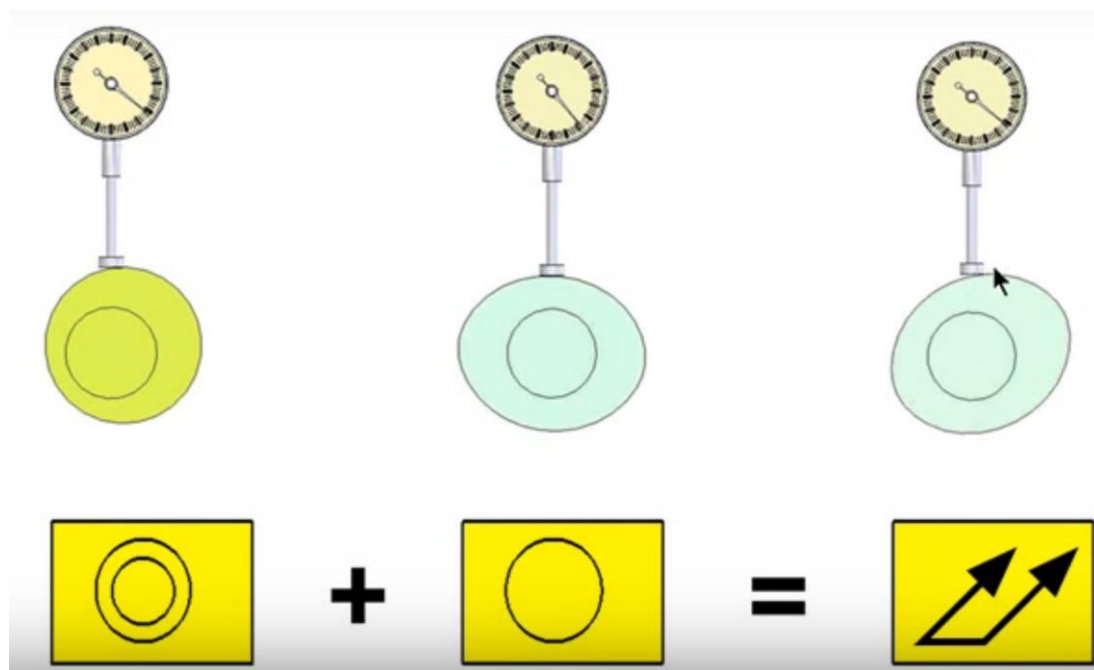


Tips for Pi Tapes

- Make sure the tape and roller are clean.
- Make sure the tape has not been stepped on or kinked, which may destroy the accuracy.
- Use a tape measure to mark off where you will plan to measure diameter across the roller's width.
- For an idler roller, mark an MD line on it to aid in aligning the Pi Tape® to a parallel position.
- For larger diameter rollers, hold the Pi Tape® in the proper parallel position with masking tape.
- When not in use, wipe clean and apply a light rust preventive oil and store in the tape container.

Measure Runout (Eccentricity + Concentricity)

Use a micrometer to measure roller surface position vs. revolution.



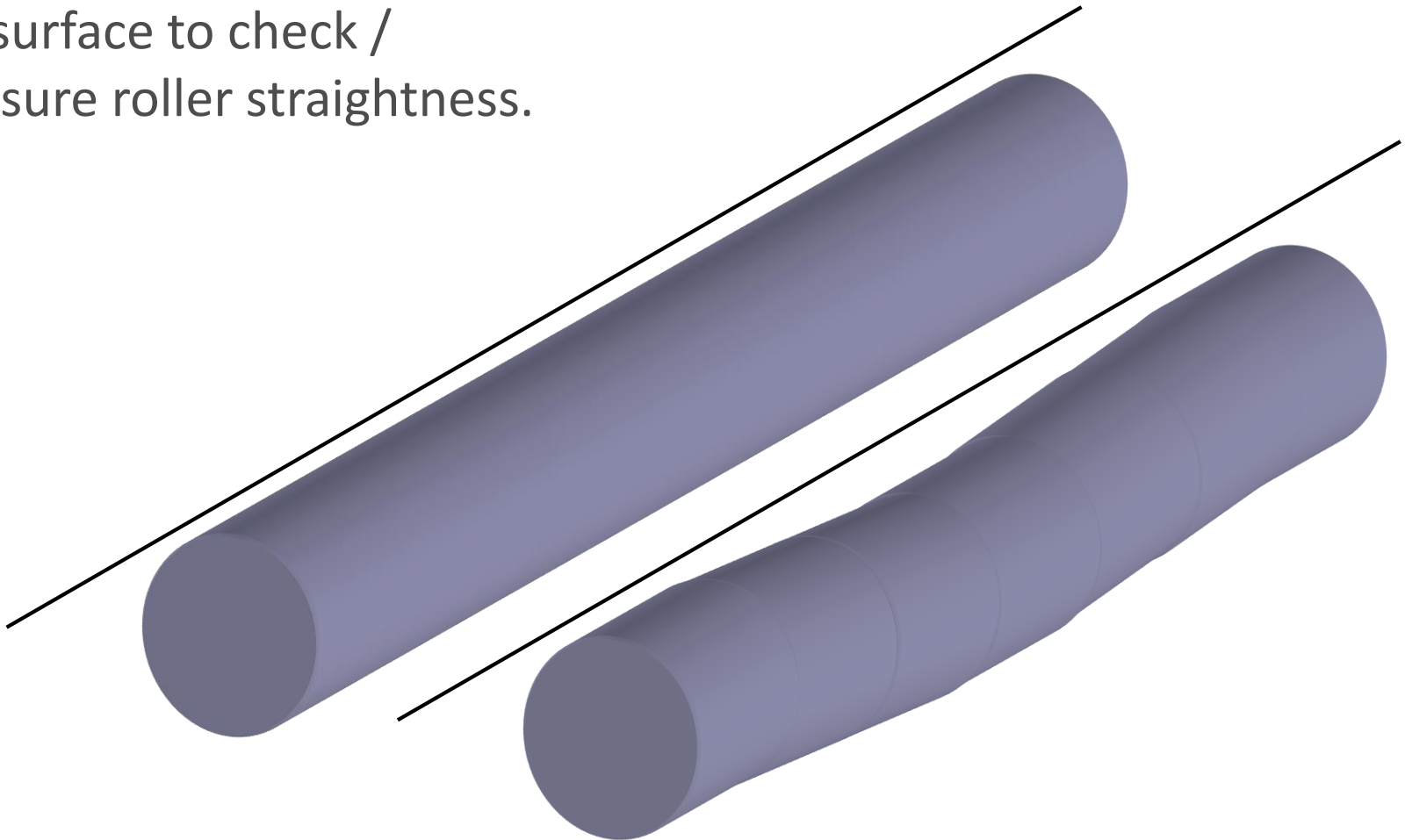
Runout of a rotating roller can be caused by:

- Off-center bearing,
- Non-circular shell,
- Bent shell.

Image courtesy of Tec-Ease Inc.

Roller Straightness

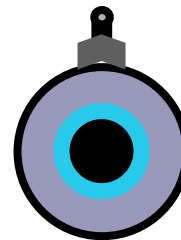
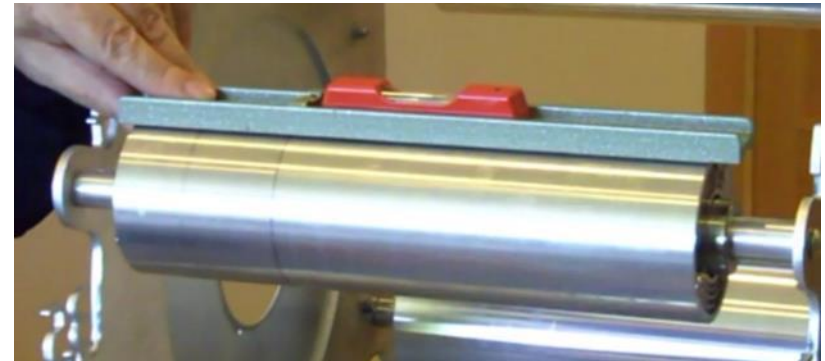
Use a straight edge, laser, or flat surface to check / measure roller straightness.



Measuring Roller Level

Roller alignment always begins measuring level.

Precision spirit levels are accurate to ± 0.5 mils/ft or 0.05mm/m



Standard level from EDA (France)
Model 114-0.05mm/m shown.

~\$250

at-laumonier.fr/eda/EN/prod_stand.php

Precision vs. Carpenter's Level



V-Groove?

Yes

No

Surfaces?

Flat or Round

Flat Only

Calibrated?

Yes

No

Secondary Level?

Yes

Maybe

What is a Tramming Stick?

Tramming Stick

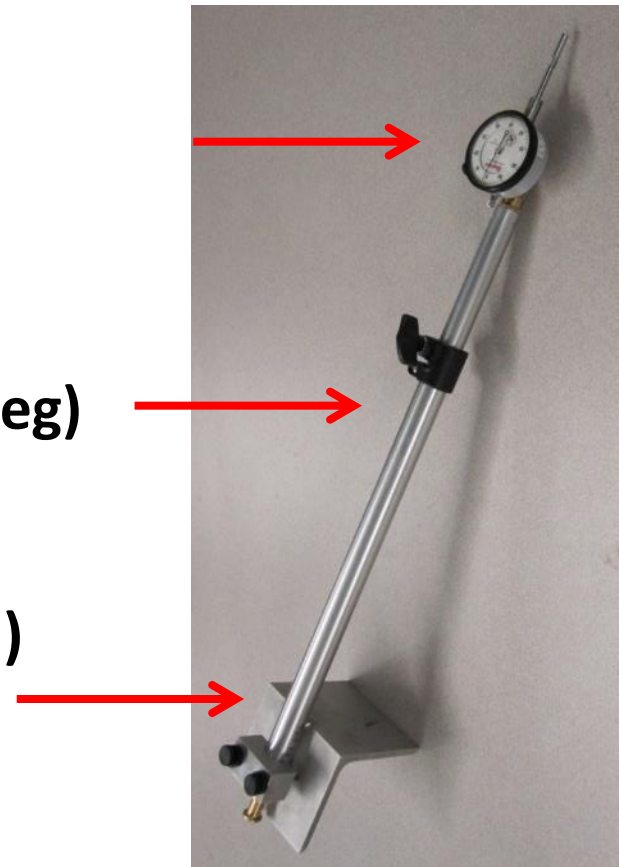


A tramming stick is custom device designed to measure the distance between rollers.

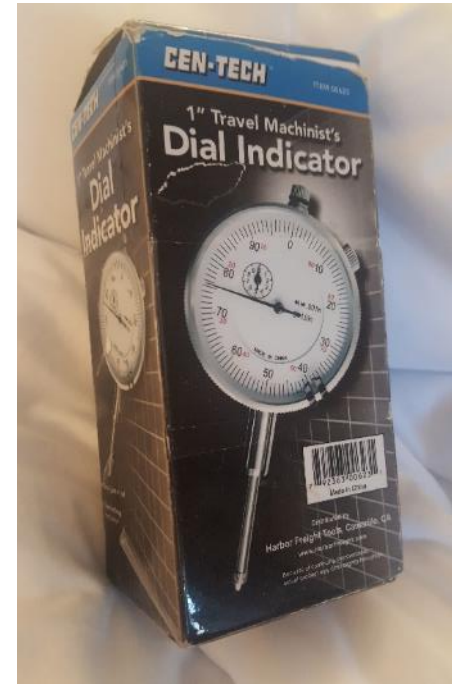
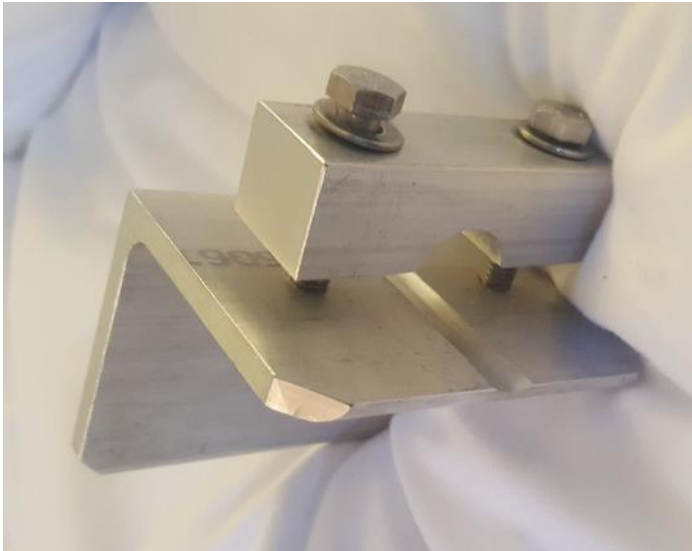
What is a Tramming Stick?

A tramming stick has three components:

1. **Dial indicating micrometer**
(usually with a range of 0.5-1.0 inches reading to 0.001-inches.)
2. **Telescoping rod** (such as a tripod leg)
3. **Angle iron or crow's foot** (to mount squarely on the reference roller.)



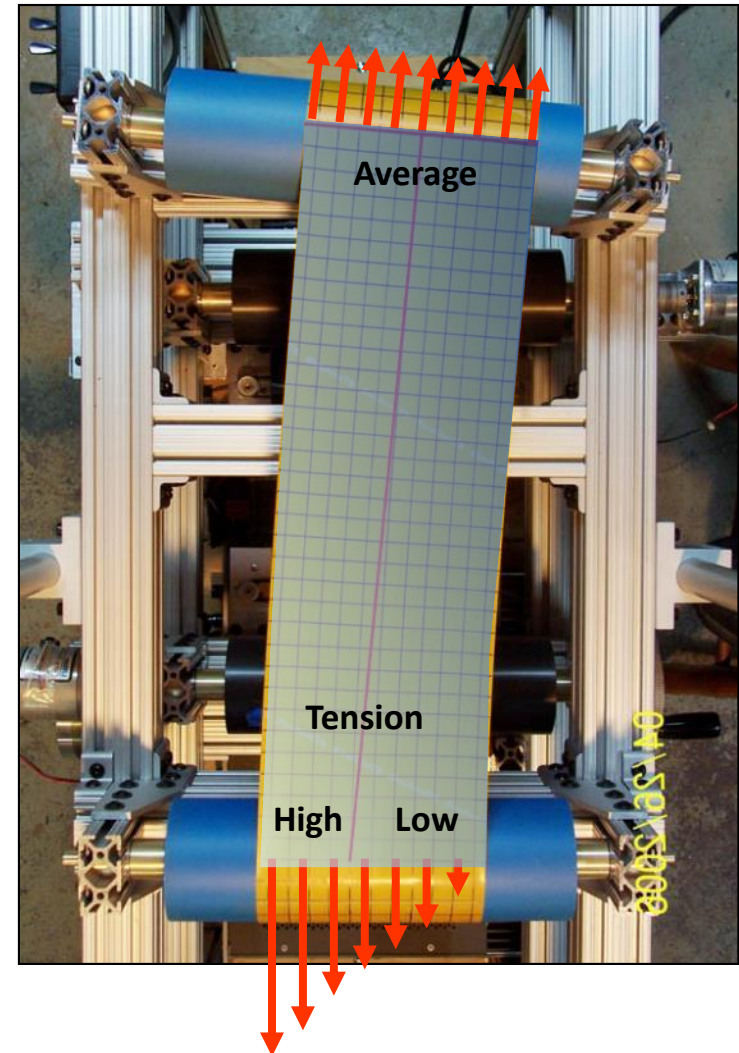
Tramming Stick Details



Why Measure Roller Alignment?

Problems from misaligned rollers?

1. CD tension variations
2. Web shifting
3. Buckling the web in the span
(= Troughing)
4. Buckling during the roller contact
(= Wrinkling)
5. Permanent buckles in the web
(= Creases)



How to... Use a Tramming Stick

1. Ensure the rollers are clean, cylindrical, and level.
2. Place the angle iron square near an edge of the first roller.
3. Extend the telescoping rod to a length where the distance to the second roller is within the dial micrometer's range.
4. While ensuring the tramming stick is square to the first roller, sweep it across the face of the second roller and note the shortest distance indicated by the micrometer.
5. Repeats steps 2-4, on the other side of the roller set.
6. Adjust the rollers until the two sides are equal or within specifications.

Stick and Pi Tape Alignment

Measure and compare distance between two rollers at A-A and B-B.

Example:

$$L_A = X + 0.655''$$

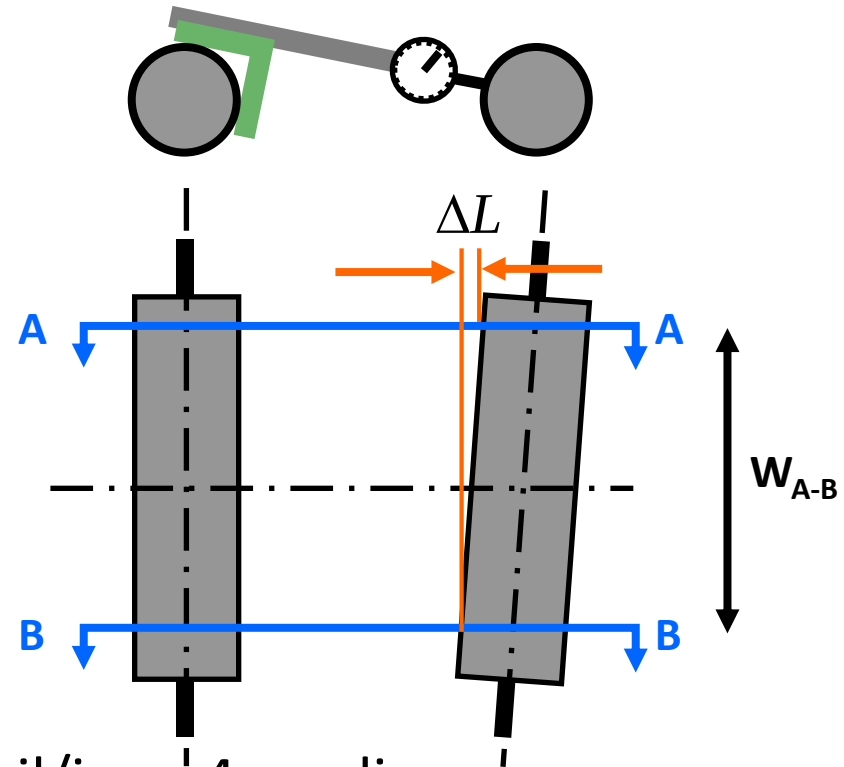
$$L_B = X + 0.615''$$

$$\Delta L = 0.040''$$

$$W_{A-B} = 10''$$

Misalignment = 40 mils / 10" or 4 mil/in or 4mradians

Alignment Error = 2 mils/in (about 20 mils/ft)
(about 10x worse than industry spec.)



How to... Pi Tape Alignment

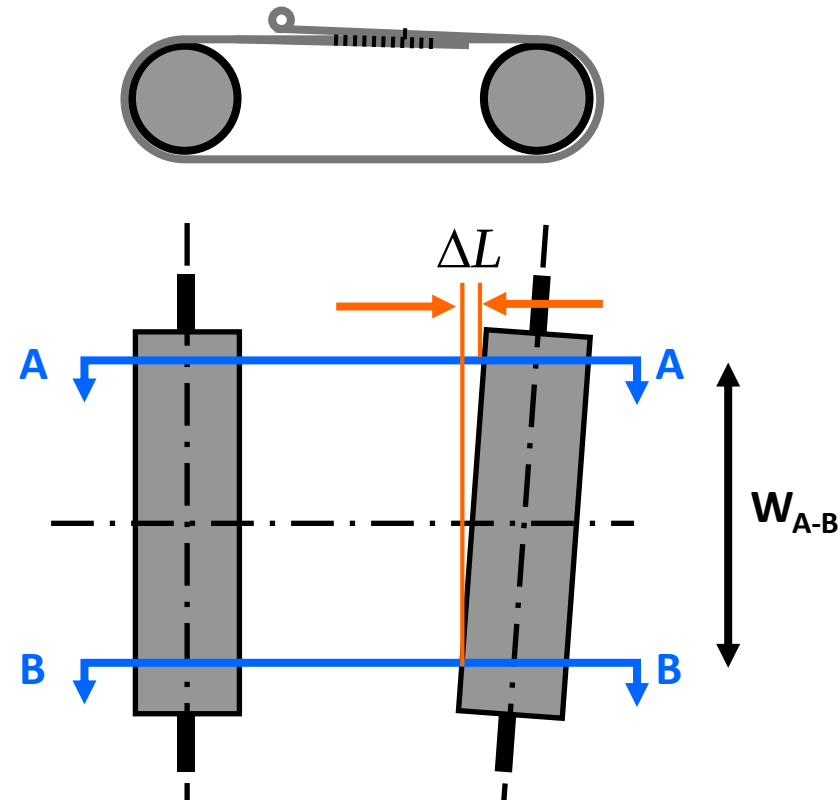
1. Ensure the rollers are clean, cylindrical, and level.
2. Select a Pi Tape long enough to wrap around two rollers and the span between them.
3. Wrap the Pi Tape around one end of the two roller set.
4. Ensure the Pi Tape is square on both rollers.
5. Use the Pi Tape to read the total length of wrapping both rollers and the span (twice) between them.
6. Repeats steps 2-5, on the other side of the roller set.
7. Convert the Pi Tape readings to relative length by dividing by 6.28.
8. Adjust the rollers until the two sides are equal or within specifications.

Stick and Pi Tape Alignment

Pi Tape readings need to be converted to account for the 3.14-to-1 Pi Tape scale factor and that this method measure the alignment error twice.

$$L_{TRAM} = \frac{L_{PI_TAPE}}{2\pi} = \frac{L_{PI_TAPE}}{6.28}$$

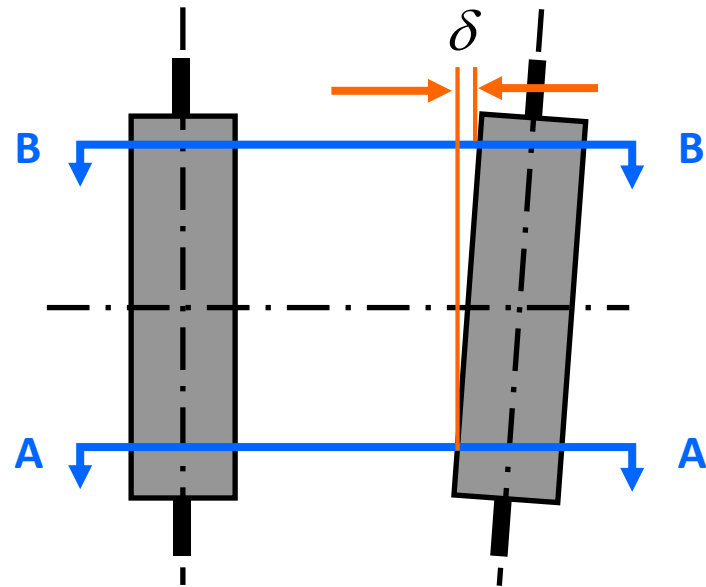
If a Pi Tape measures an alignment difference fo 62.8 mils, the roller misalignment is 10 mils.



Stick and Pi Tape Alignment

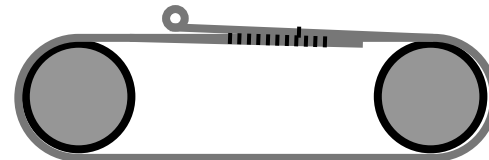
Measure and compare distance between two rollers at A-A and B-B.

Alignment relies on good diameter, straightness, eccentricity, and level.



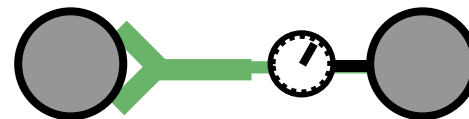
Option 1:

Measure with Pi Tape™



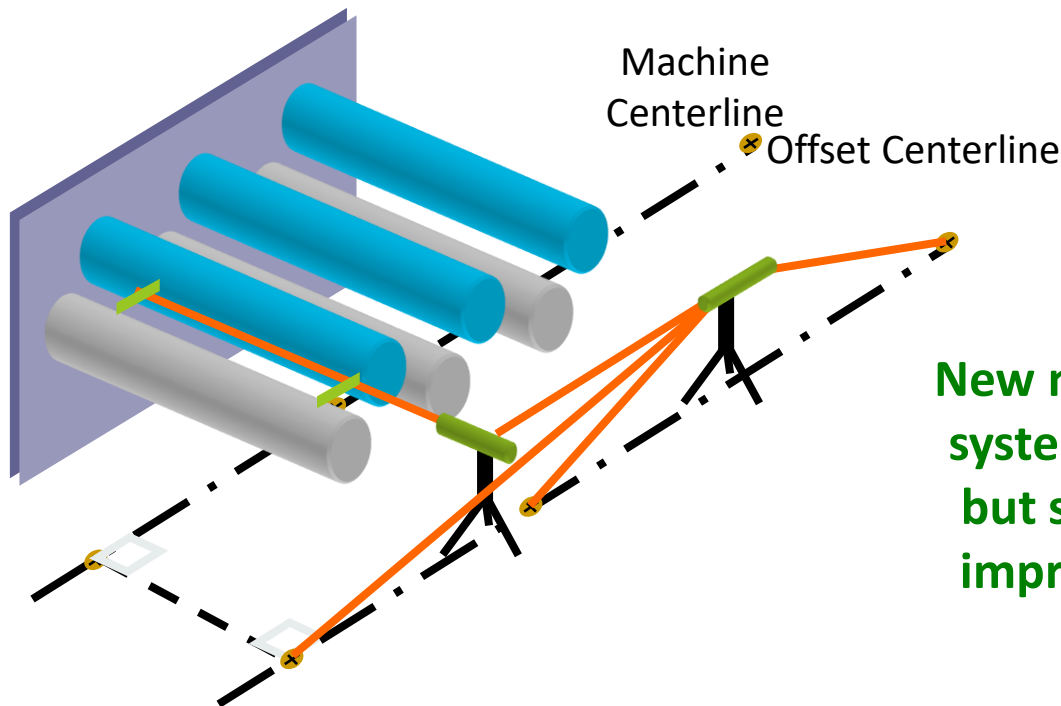
Option 2:

Measure with tramming stick



Offset Optical Alignment

Offset alignment measures perpendicular roller position from an offset centerline.



Optical transits (a.k.a. a Theodolite) have been used for many years.



New multi-axis laser systems cost more, but save time and improve accuracy.



Image courtesy of Hamar Laser Instruments

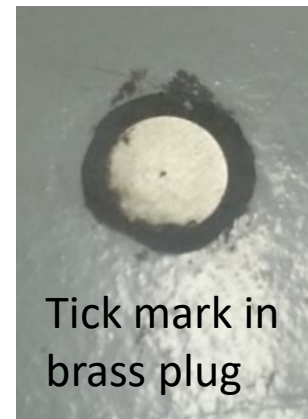
Optical Alignment Details

Offset centerline defined by 'tick' marks on brass plugs in concrete floor.

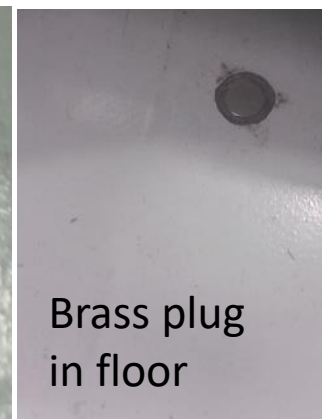
First transit 'bucked in' to offset centerline.

Second transit set perpendicular to first transit.

Second transit head translates to measure differential of roller ends.



Tick mark in
brass plug

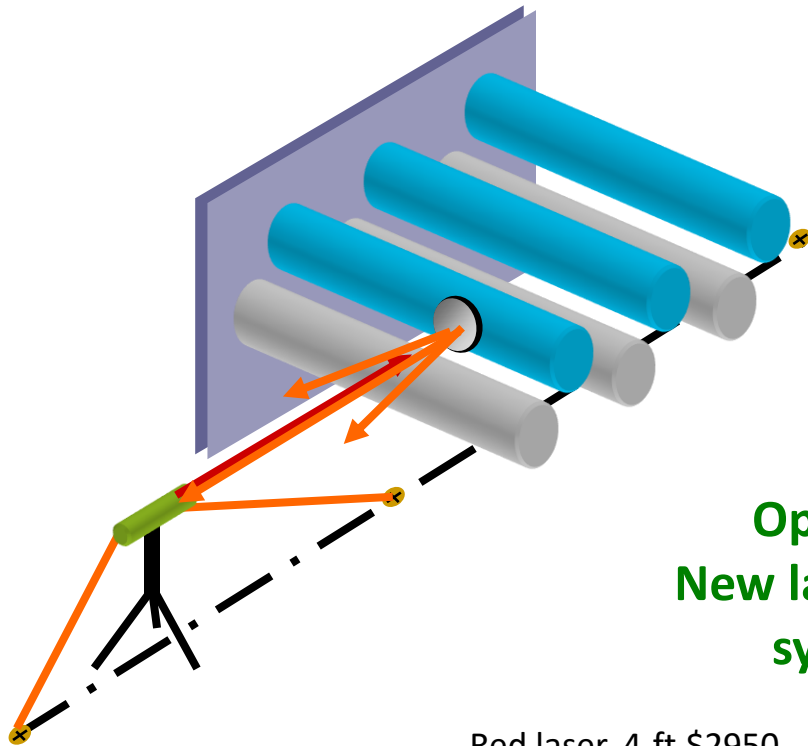


Brass plug
in floor



Centerline Optical Alignment

Centerline alignment check parallelism between machined centerline and perpendicular line projected from the roller face.



**Option 2:
New laser based
system.**

Red laser, 4-ft \$2950

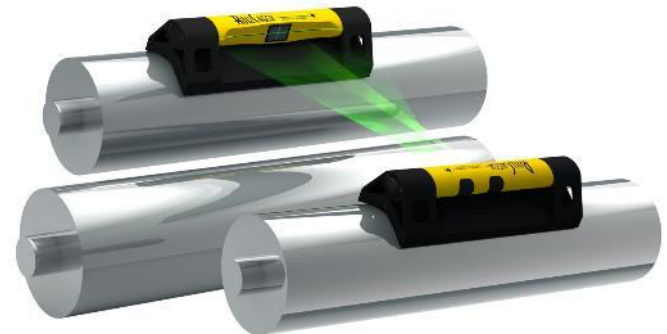
Green laser, 10-ft, \$8450

**Option 1:
A theodolite
and mirror.**



Laser Reflector /
Level Receiver

Laser Emitter (2)
/ Square Reference



www.seiffertindustrial.com
[Roll Check YouTube](#)

Laser-Gyroscopic Alignment

Three precision laser gyroscopes sense shifts in x-y-z angles vs. reference roller.

Accuracy to $4\mu\text{m}/\text{m}$.

Pruftechnik Paralign™

www.pruftechnik.com

New service available in US,
estimated \$5-10K / day.

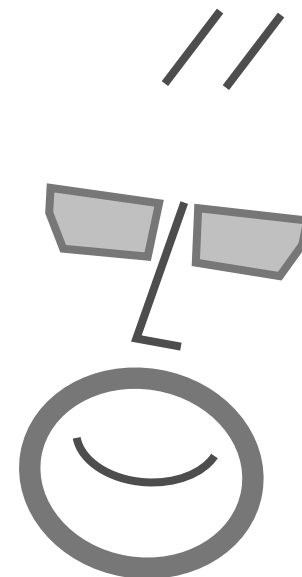


Image courtesy of Pruftechnik Service, Inc.

Tensioning Benefits Qs

How often should equipment be re-aligned?

- ~~A) Every six months.~~
- B) When the web asks for it.
- C) Never.



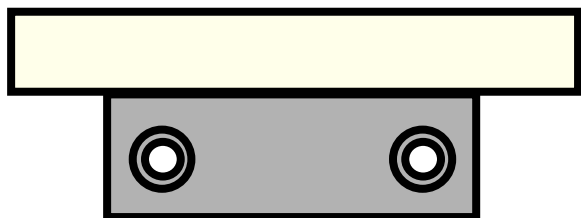
Roller Mount Options

Best options are split cap, either

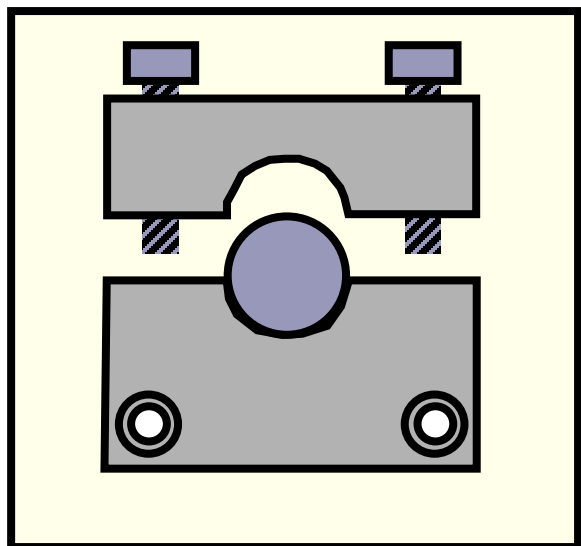
Flange Mount

or

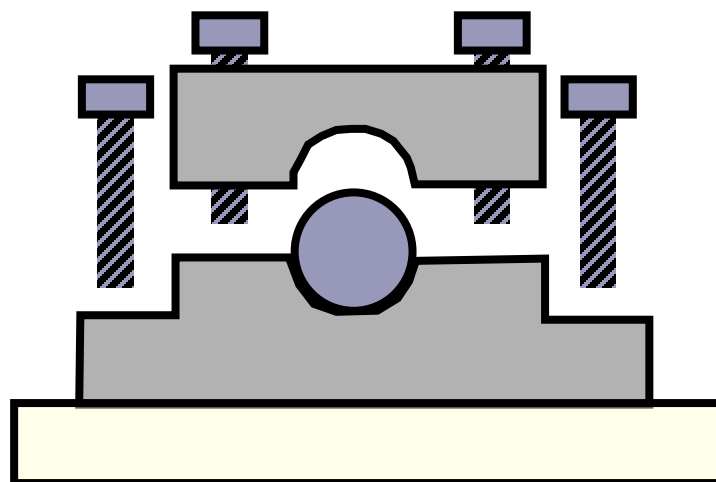
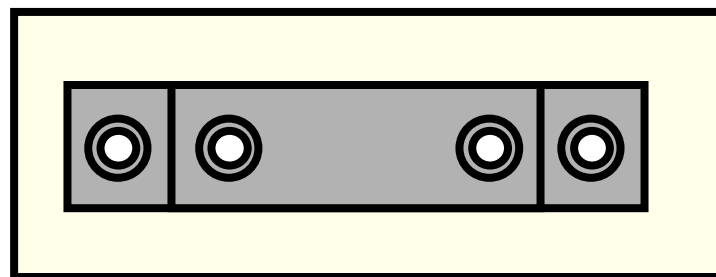
Pillow Block Style.



Top
View

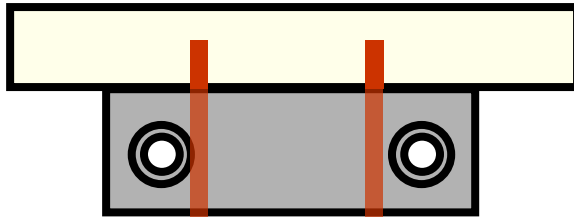


Front
View

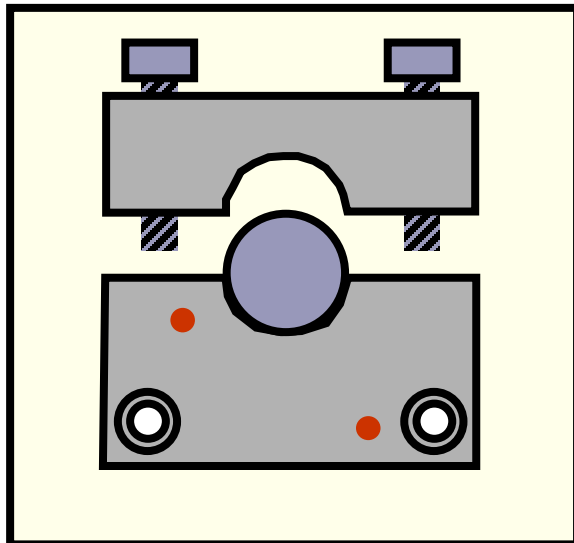
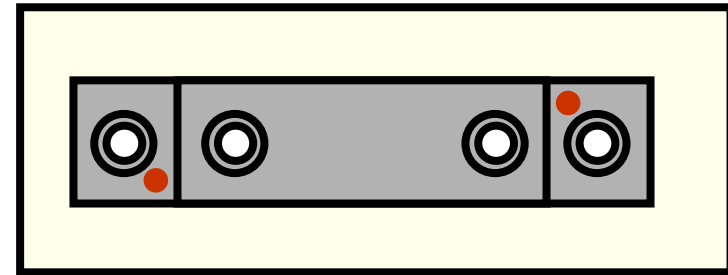


Pinning

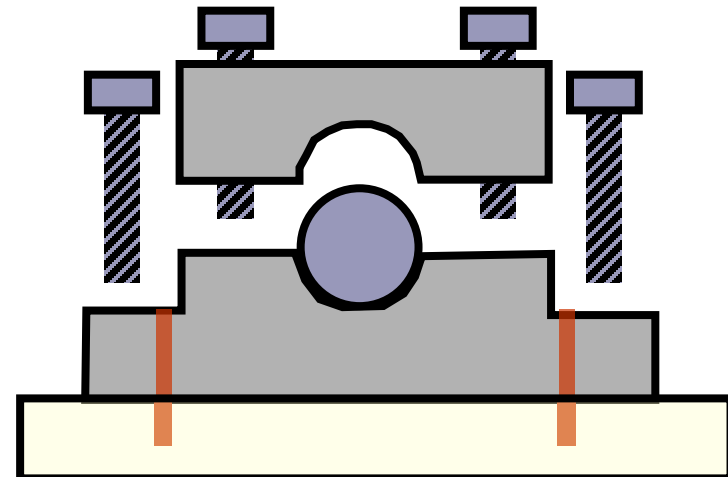
Bolts prevent motion in one direction. To maintain alignment, drill and pin the machine side of these roller mounts.



Top
View



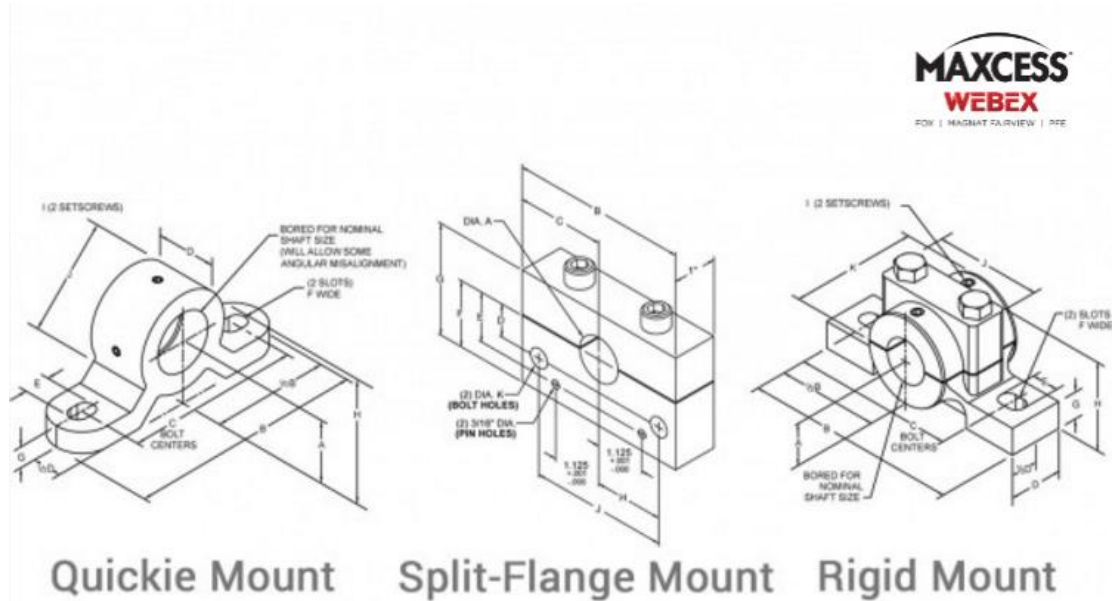
Front
View



Split Cap Roller Mounts



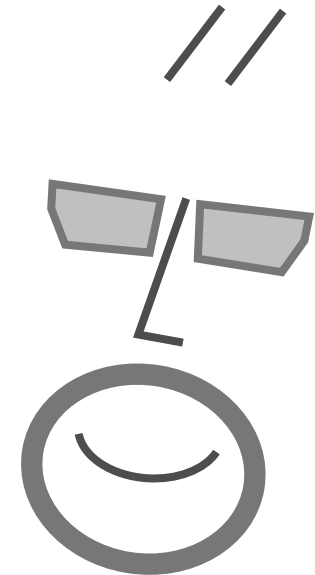
Courtesy of Timken



Courtesy of Maxcess Webex

Tensioning Benefits Qs

How do moving components maintain alignment?



Pivoting 😊 vs Linear Sliding ☹️

Pivot rollers should be preferred to linearly sliding rollers for:

- Dancers
- Accumulators (though large accumulators are nearly all linear)
- Nipping Roller
- Winder Turrets

Why?

- Friction effects on the process are reduce greatly by the pivot's mechanical advantage.
- Pivot usually maintains better alignment.
- Linear systems should follow the Saint Venant's principle of sled or bearing length vs width (Sled length should be at least 60% of width).

Saint-Venant's Principle



"... the difference between the effects of two different but statically equivalent loads becomes very small at sufficiently large distances from the load."

- Saint-Venant: *Linear Bearings*:
 - $L/D > 1$, 1.6:1 very good, 3:1 super ideal
 - Every year some students try to make $L/D < 1$ and their machines jam!
 - Wide drawers guided at the outside edges can jam
 - Wide drawers guided by a central runner do not!

Courtesy of MIT FUNDaMENTALS website on mechanical design.

<http://web.mit.edu/2.75/fundamentals/FUNdaMENTALS.html>

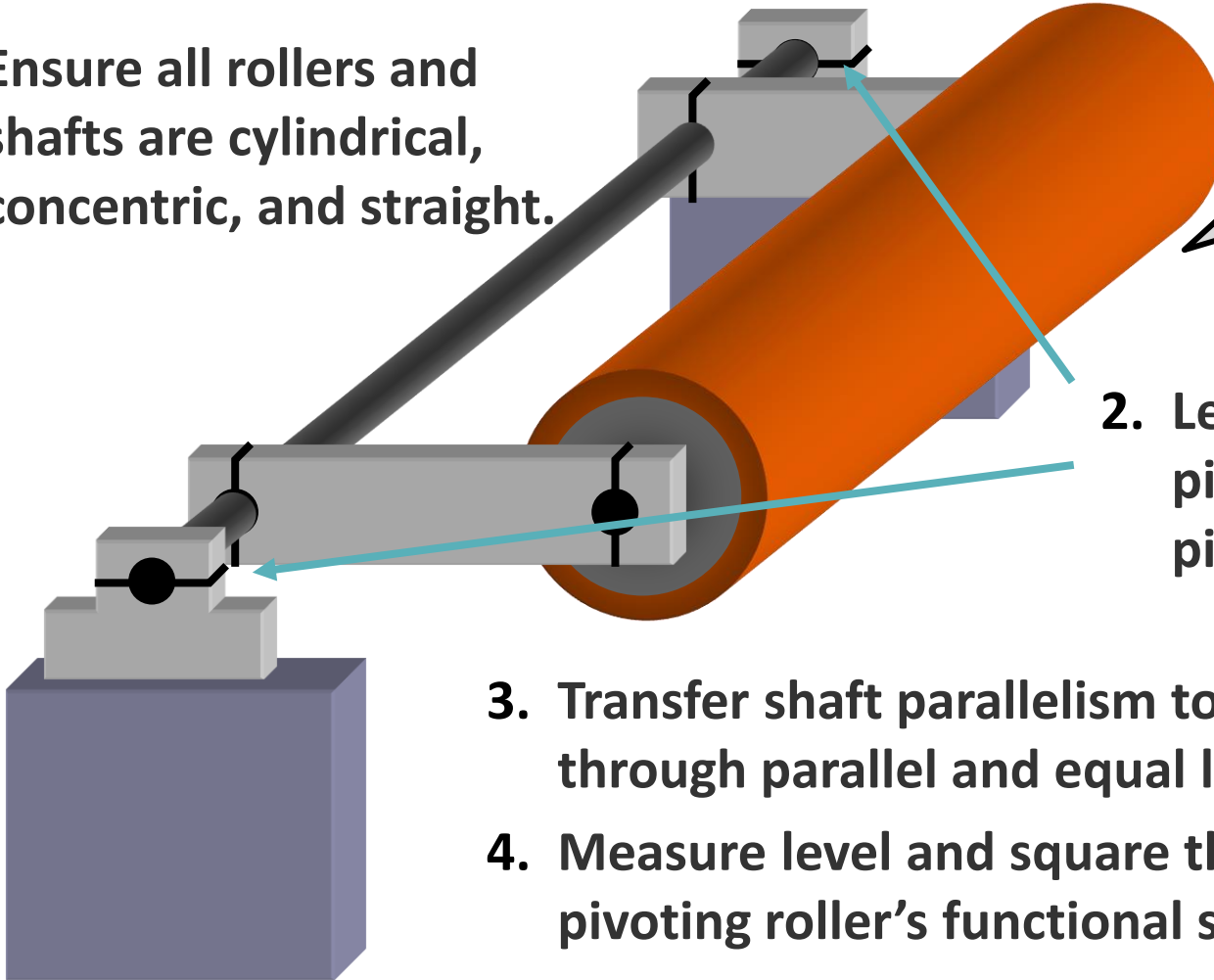
Ensure Pivoting Alignment

1. Ensure all rollers and shafts are cylindrical, concentric, and straight.

Most important for dancers and turret winders.

2. Level and square pivoting roller's pivot shaft.

3. Transfer shaft parallelism to nipping roller, through parallel and equal length arms.
4. Measure level and square throughout the pivoting roller's functional stroke.





Web-Roller Traction

When Does a Roller Slip?

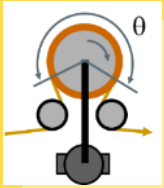
Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

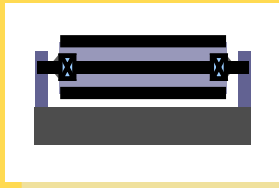
tjwalker@tjwa.com www.webhandling.com 651.686.5400

WH Fundamentals

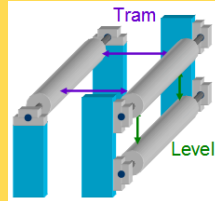
3: Rollers & Traction



Traction



Roller Basics



Alignment

+ Rollers and Traction

- ✓ Understand when webs slip on rollers
- ✓ Design rollers for driving, idling, and nipping applications
- ✓ Why is roller parallelism important?

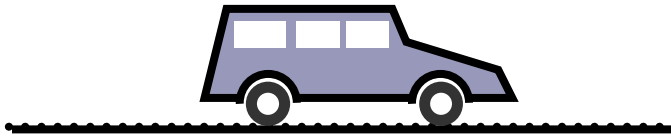
3.3 Learning Objectives

- ▶ When will web and roller speeds be different?
- ▶ How is web-roller slip measured?
- ▶ What causes scratching?
- ▶ What is traction?
- ▶ How much traction does a driven roller have/need?
- ▶ How much traction does an idler roller have/need?
- ▶ What is microslip and where does it occur?
- ▶ How does air lubrication affect traction?
- ▶ What is bearing drag and how it is measured?

Good Traction = Control

- ▶ Avoid slip (and scratching).
- ▶ Avoid nipping solely for traction.
- ▶ Understand entrained air and how roller surfaces control lubrication.
- ▶ Idler rollers do not need excessive traction.
- ▶ Understand the traction required from bearing drag, roller inertia, and other factors.
- ▶ Understand roller design options and which factors are important to the web (drag, lubrication, alignment, deflection).

Traction Analogy



Tire / Road

- Tire material
- Road material
- Load, load distribution
- Speed
- Acceleration & Inertia
- Lubricant (viscosity, depth)
- Temperature (adhesion)
- Geometry (road curve)
- Tire geometry
- Tread
- Debris

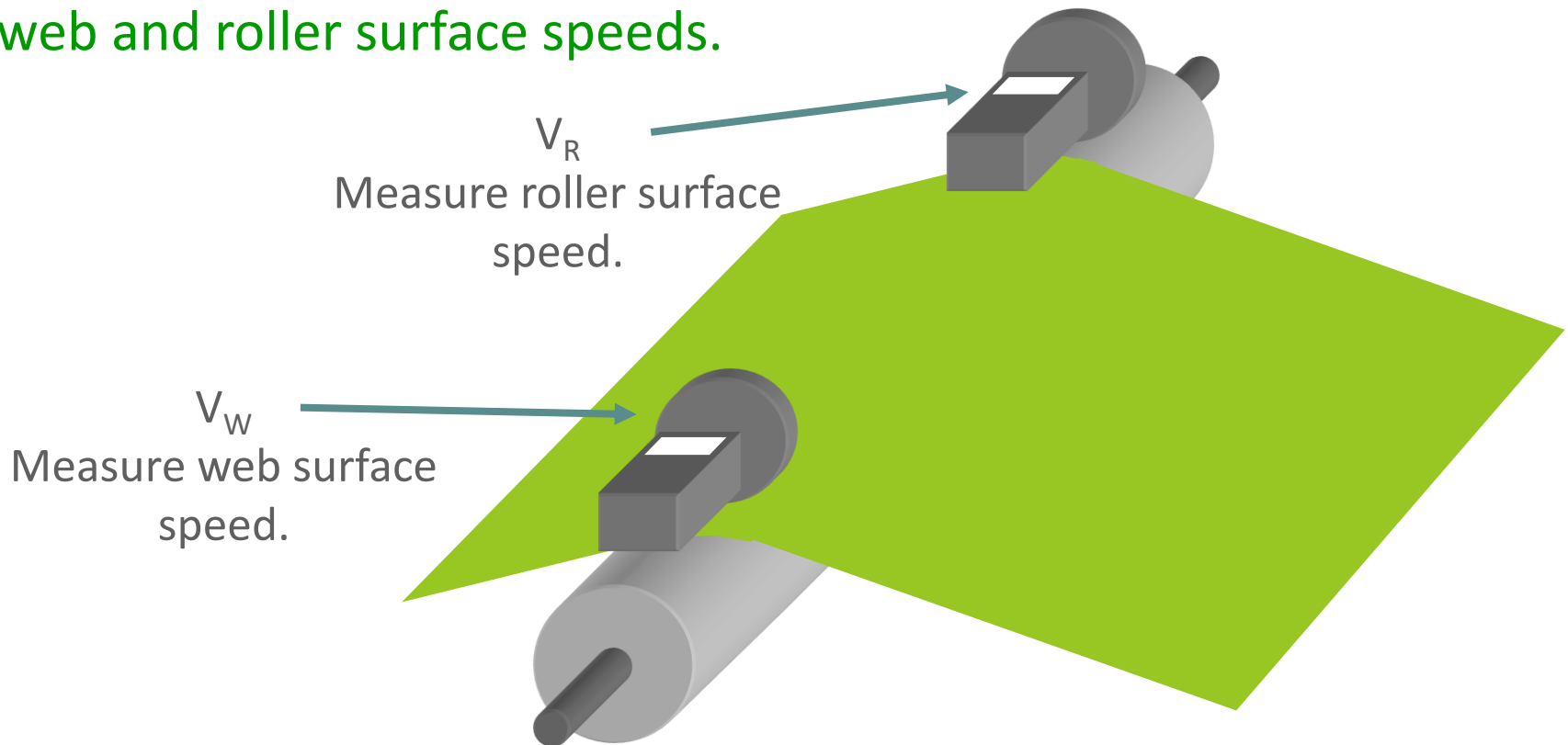


Roller / Web

- Roller material
- Web material
- Tension, T-distribution
- Speed
- Acceleration & Inertia
- Lubricant (viscosity, height)
- Adhesion
- Geometry (wrap angle)
- Roller geometry
- Surface roughness
- Debris

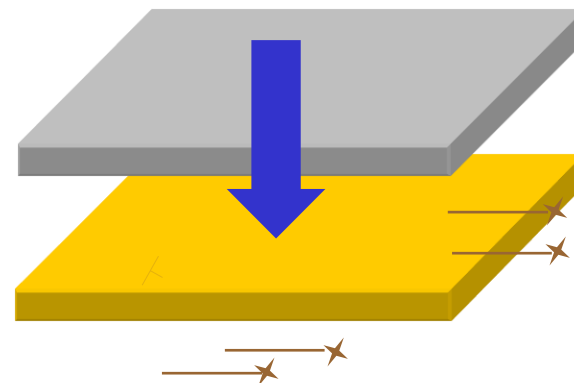
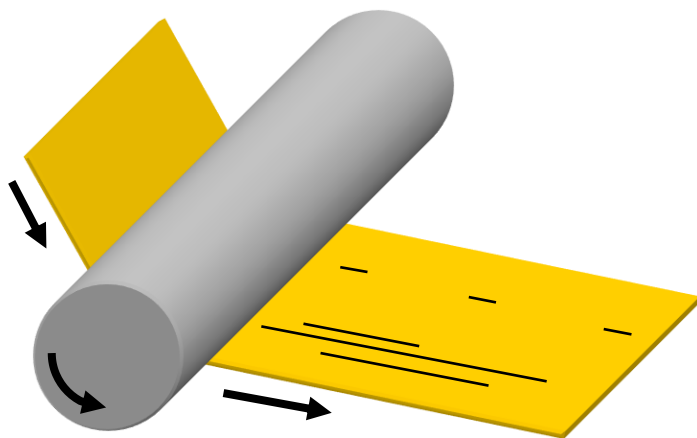
Measure MD Slip

On any roller, check for difference in web and roller surface speeds.



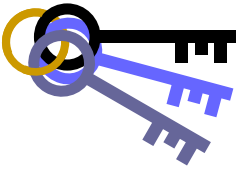
What Causes Scratching?

Will this roller scratch the web?



Prerequisites for Scratching:

1. Contact
2. Normal Force
3. Slippage
4. Stress Concentration
5. Relative Hardness



KEY CONCEPT



Safety Factor = What you have / What you need

Traction Safety Factor (TSF)

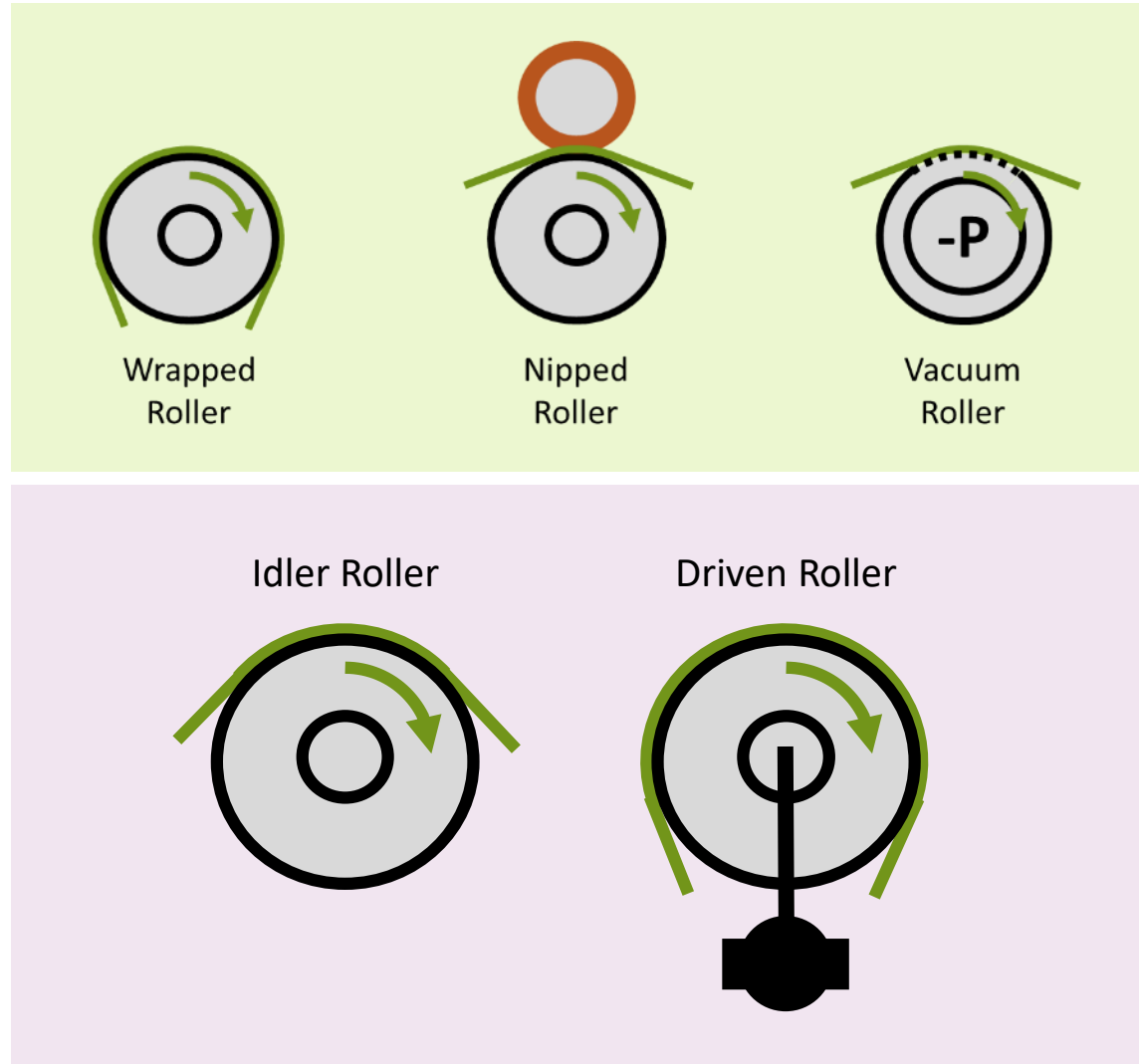
The TSF is the ratio of the available traction torque or force divided by the torque or force required for tension changes between tension zones or total force or torque required by an idler roller.

Uses:

- Predict when drive rollers will slip and lose tension control.
- Predict when idler rollers will slip and create scratches, including accelerating and decelerating conditions.

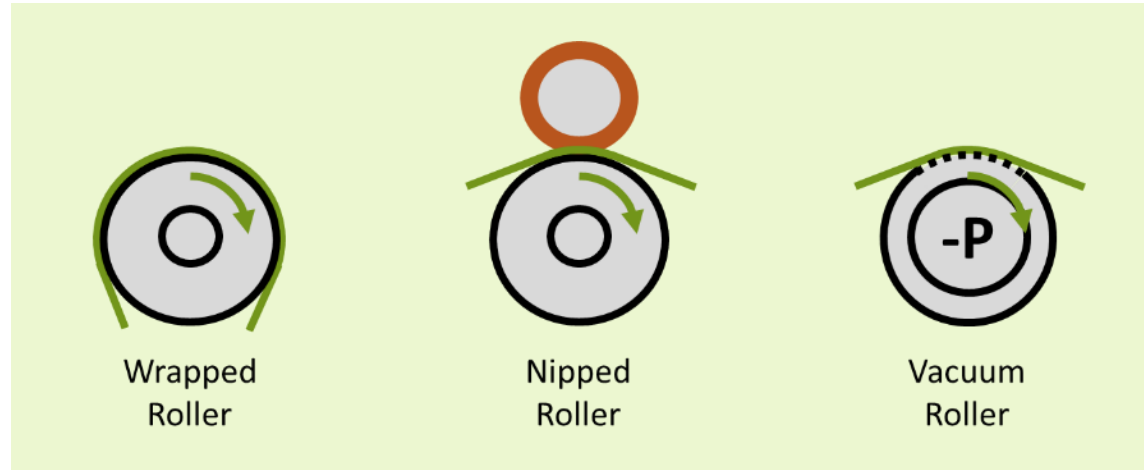
Traction Requirements

$$\text{TSF} = \frac{\text{Available Traction}}{\text{Required Traction}}$$



Traction Requirements

$$\text{TSF} = \frac{\text{Available Traction}^*}{\text{Required Traction}}$$

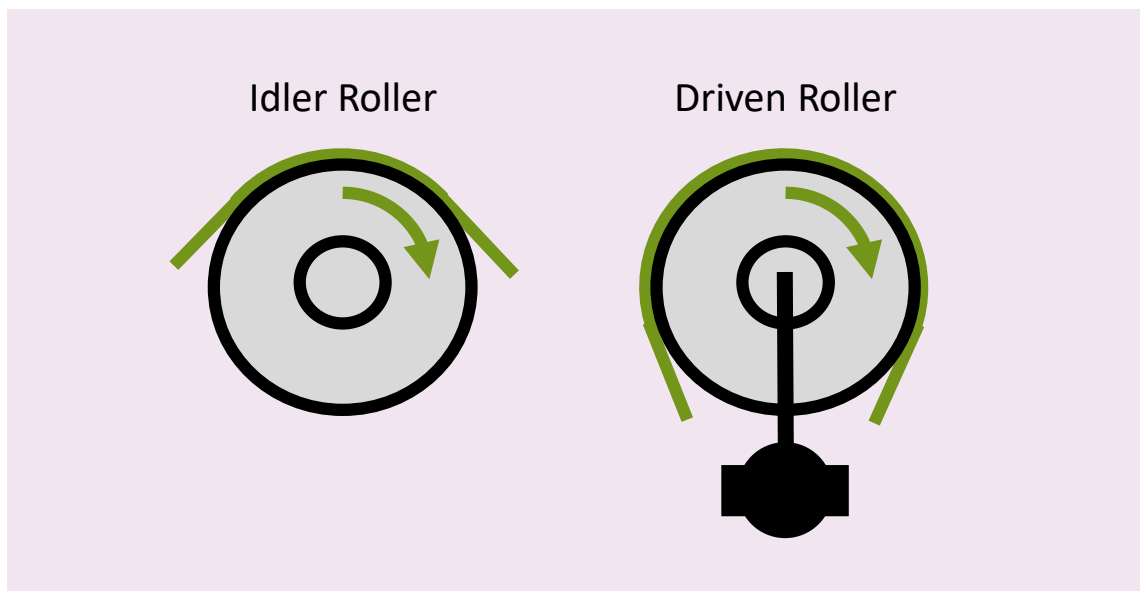


* Traction available will vary with: roller surfaces, webs, speed, tension, radius, etc.

Traction Requirements

$$TSF = \frac{\text{Available Traction}}{\text{Required Traction}^*}$$

- * Traction required will vary with: steady-state, acceleration, deceleration, tension set points, draw ratios, etc.



Driven Roller Traction Safety Factor

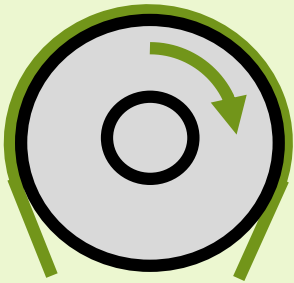
$$TSF > 1$$

- No slip or scratch
- Good tension/speed control

$$TSF < 1$$

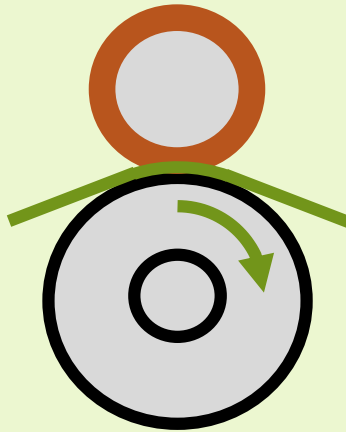
- Slipping, scratching,
- Poor tension/speed control

Web Support / Transport / Tensioning Options



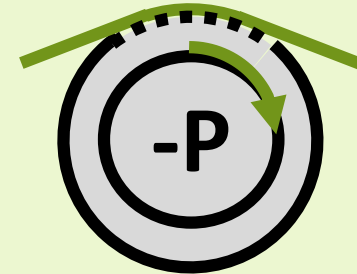
Wrapped
Roller

$$\frac{T_{HI}}{T_{LO}} = e^{\mu\theta}$$



Nipped
Roller

$$\Delta T = \mu N$$



Vacuum
Roller

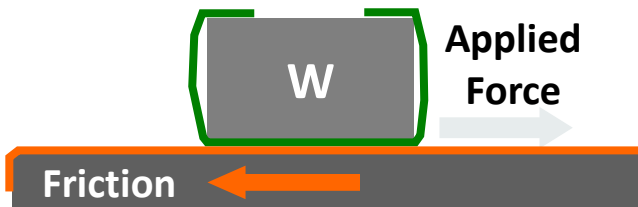
$$\Delta T = f(\mu, P, r, w, \theta, T)$$

Friction Coefficient

The friction coefficient, μ , is the ratio of frictional force relative to normal load. Two common friction coefficient measurement methods are:

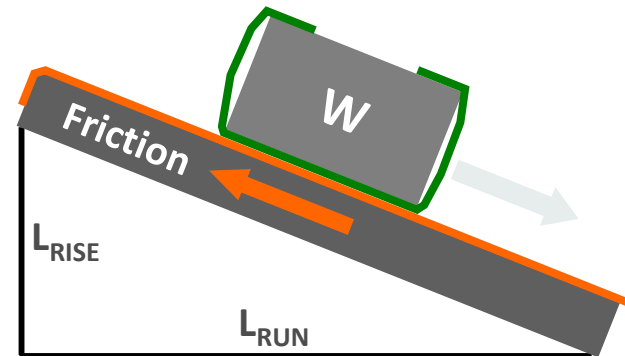
Sliding Block Test

$$\mu_{A-B} = \frac{F}{W}$$

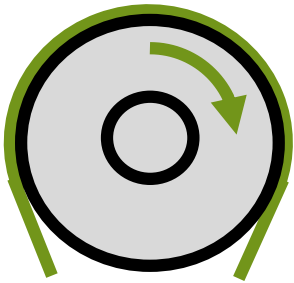


Incline Plane Slide Test

$$\mu_{A-B} = \frac{L_{RISE}}{L_{RUN}}$$



Web Support / Transport / Tensioning Options



Wrapped
Roller

$$\frac{T_{HI}}{T_{LO}} = e^{\mu\theta}$$

Nearly all idler roller and many driven rollers are simple, wrapped rollers. The slip condition on wrapped roller is a function of tension ratio.

Available Traction (without slip)

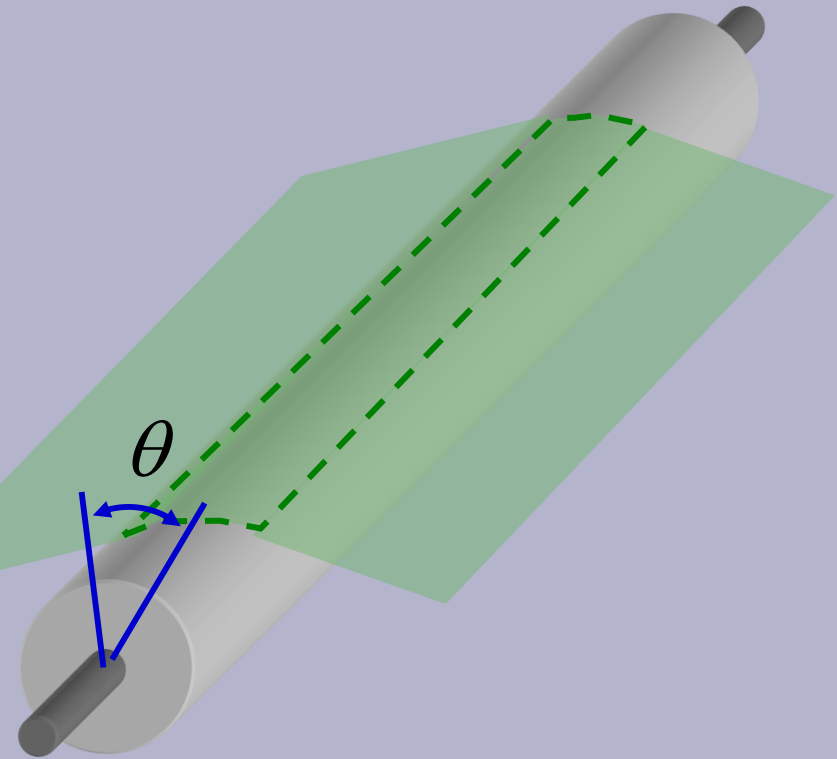
$$F_{COT} = \mu_T F_T \theta$$

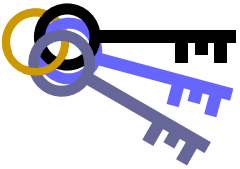
F_T = Force of Tension, N or lbs

μ_T = Traction coefficient, --

θ = Wrap angle, radians

Where is radius?
Don't big rollers
have more traction?





KEY CONCEPT



Web-Roller Pressure

*Pressure between a web and roller will be proportional to tension per width divided by radius. **$P=T/r$***

Use: Calculating the pressure of unnipped laminating

Example: If the top web has 2 PLI on a 4-inch radius roller, the pressure under it will be 0.5 psi.

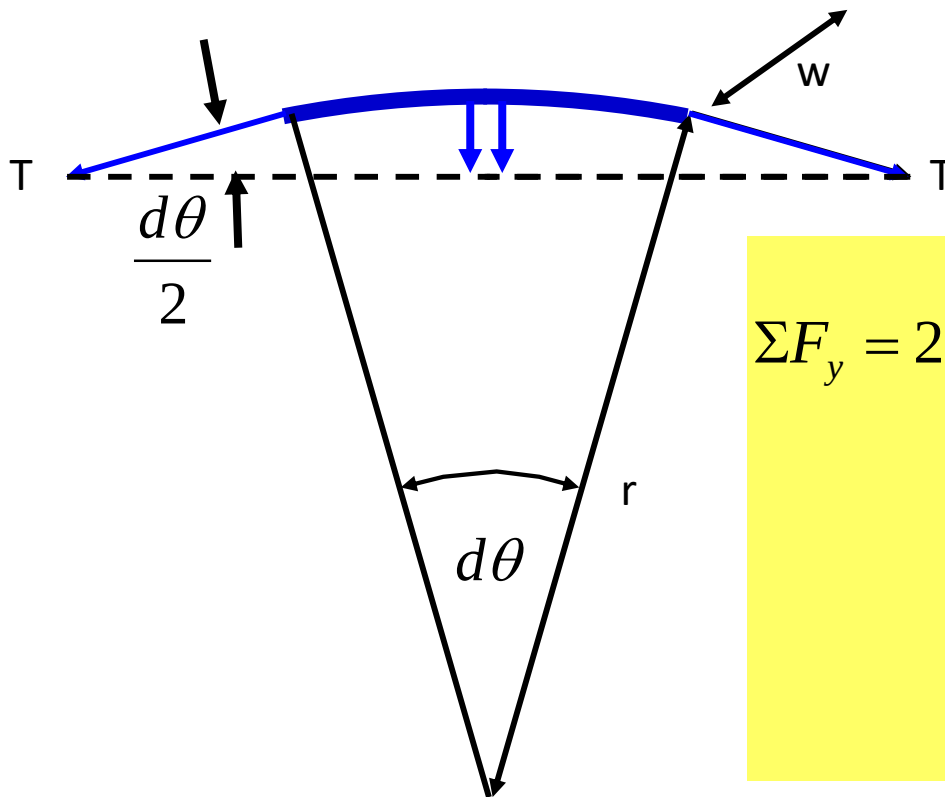
Use: Calculate the pressure needed to lift a web on an air turn.

Example: For a web tension of 1 PLI on a 10-inch radius air turn, the air velocity must create 0.1 psi of pressure.

Web Pressure

This equation seems overly simple.
How is it derived?

$$P = \frac{T}{r}$$



$$\Sigma F_y = 2Tw \sin\left(\frac{d\theta}{2}\right) = 2Tw\left(\frac{d\theta}{2}\right) = Twd\theta$$

$$A = rwd\theta$$

$$P = \frac{\Sigma F_y}{A} = \frac{Twd\theta}{rwd\theta} = \frac{T}{r}$$

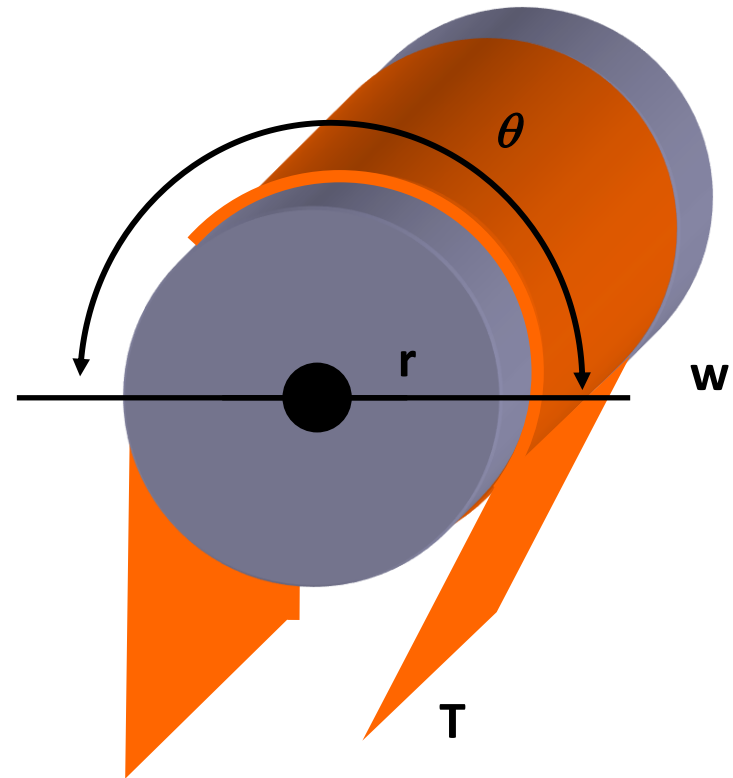
Friction Independent of Area

$$P = \frac{T}{r}$$

$$A = rw\theta$$

$$N = PA = \left(\frac{T}{r} \right) (rw\theta)$$

$$F = \mu N = \mu T w \theta$$

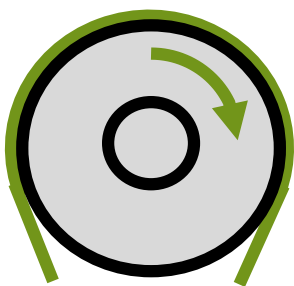


Roller Traction and Radius

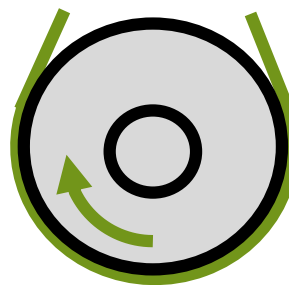
- ▶ Radius Independent
 - ▶ With 'dry' friction, traction is independent of radius
- ▶ Larger Radius = More Traction?
 - ▶ When there is web-roller adhesion, larger contact areas creates more traction.
 - ▶ For vacuum rollers, larger radius creates greater effective area of vacuum and greater traction.
- ▶ Smaller Radius = More Traction?
 - ▶ Smaller rollers have more web-roller pressure and postpone air lubrication traction losses.

Roller Traction and Wrap Orientation

- ▶ For many webs, gravity or web weight is ignored.
- ▶ As web basis weight (mass per area) increases, wrapping 'over' a roller will create web-roller traction independent of tension.
 - ▶ This can be especially important in surface winding.
- ▶ Wrapping 'under' a roller may lose traction from web weight.



'Over' Wrap Roller



'Under' Wrap Roller

Belt Equation

Also known as:

- Capstan Equation
- Euler-Eytelwein Formula

$$\frac{F_{HI}}{F_{LO}} = e^{\mu\theta}$$



Leonhard Euler



F_{HI} = High Tension, lbs or N

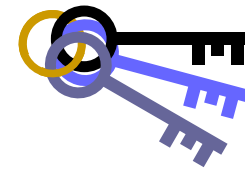
F_{LO} = Low Tension, lbs or N

μ = Web-roller coefficient of friction

θ = Wrap angle, radians



Johann Eytelwein



The Belt Equation

For simple rollers, the belt equation is a useful equation, describing the friction that develops between a web and roller.

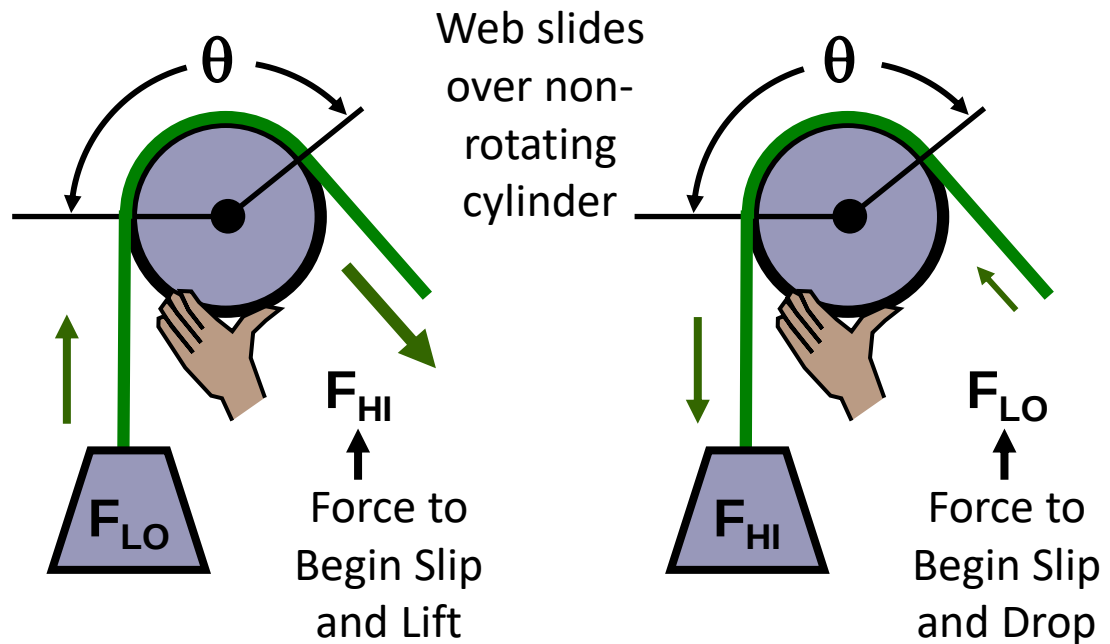
$$\frac{F_{HI}}{F_{LO}} = e^{\mu\theta}$$

F_{HI} = High Tension, lbs

F_{LO} = Low Tension, lbs

μ = Web-roller coefficient of friction

θ = Wrap angle, radians



Wrapped Rollers: Available Traction

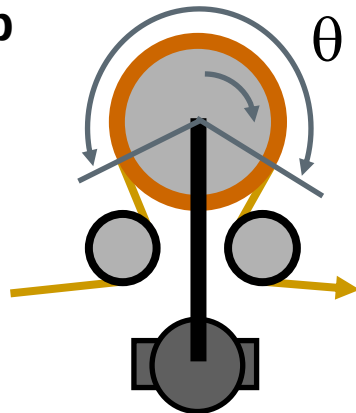
Potential friction of wrapped rollers increases exponentially with friction coefficient and wrap angle.

wrapped drive rollers typically have:

- 1) Rubber covering for high friction coefficient
- 2) Omega- or S-wraps for large wrap angles

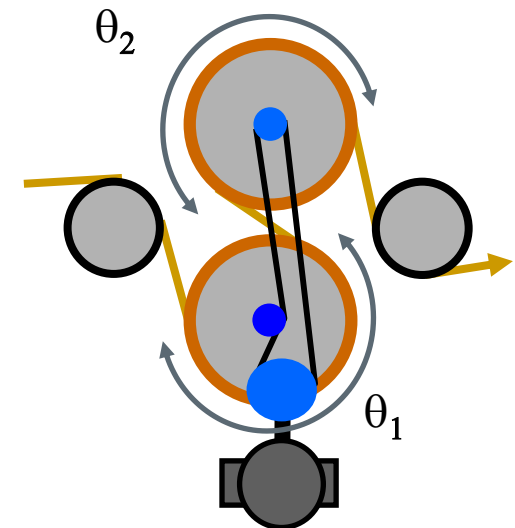
Omega Ω -Wrap

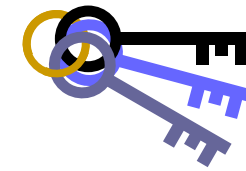
$$\frac{F_{HI}}{F_{LO}} = e^{\mu\theta}$$



S-Wrap

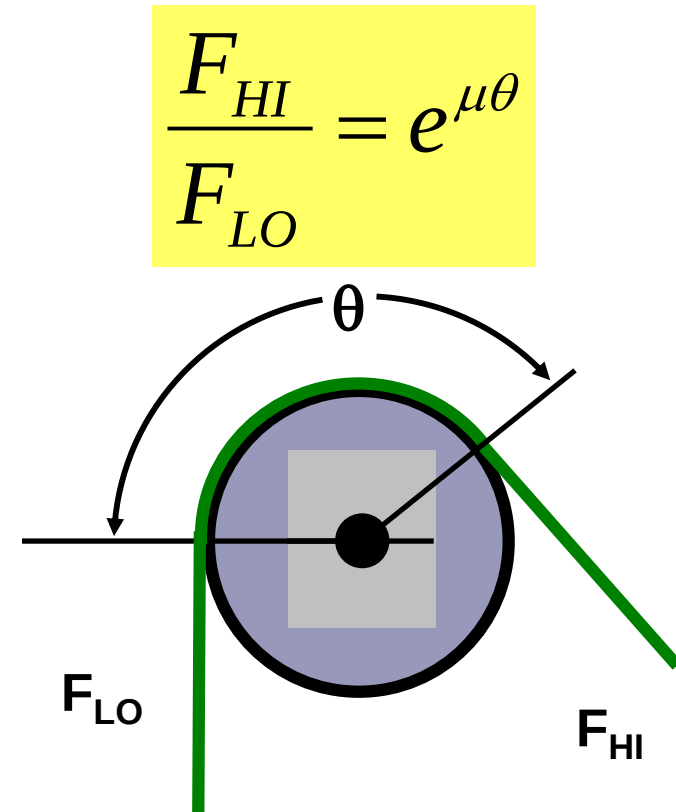
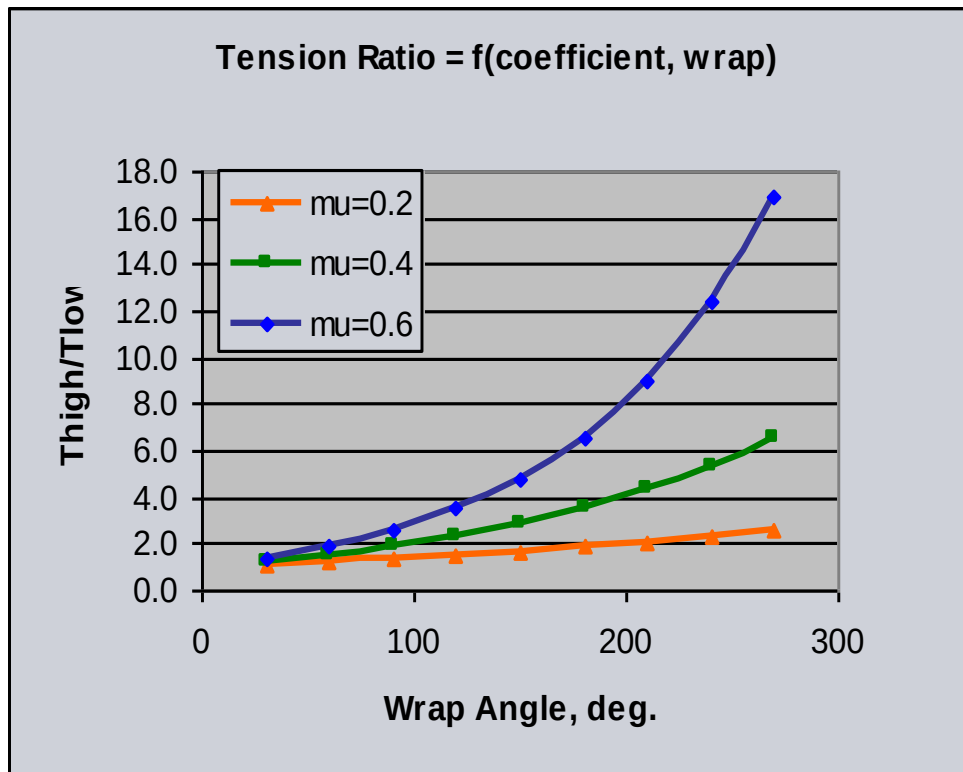
$$\frac{F_{HI}}{F_{LO}} = e^{\mu(\theta_1 + \theta_2)}$$





The Belt Equation

Since the belt equation is exponential, more wrap angle greatly increases the friction.



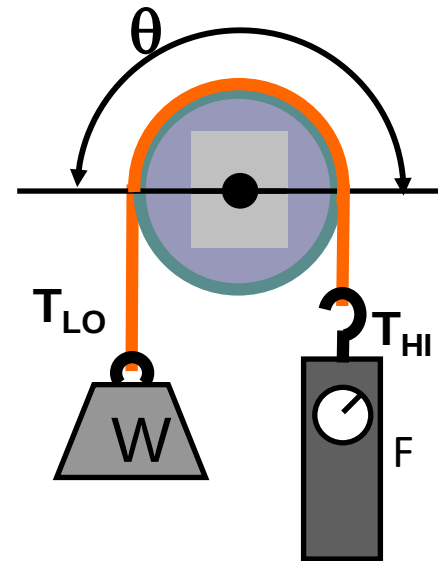
Measure Friction (Web-Roller)

Use the belt equation to determine web to roller friction.

1. Take a strip of web and tape loops on each end.
Attach a mass to one end and weight it with the force gauge.
2. Wrap the target roller with a known wrap angle.
3. If necessary, hold the roller so it doesn't rotate.
4. Pull with the force gauge until the web slides on the roller, note the force, F (Tension ratio is F/W).
5. For a second data point, push with the force gauge, dropping the tension until the web slips the other direction. Note the force, F (Tension ratio is W/F).
6. Enter the tension ratio and wrap angle (in radian) in the belt equation and calculate friction coefficient.

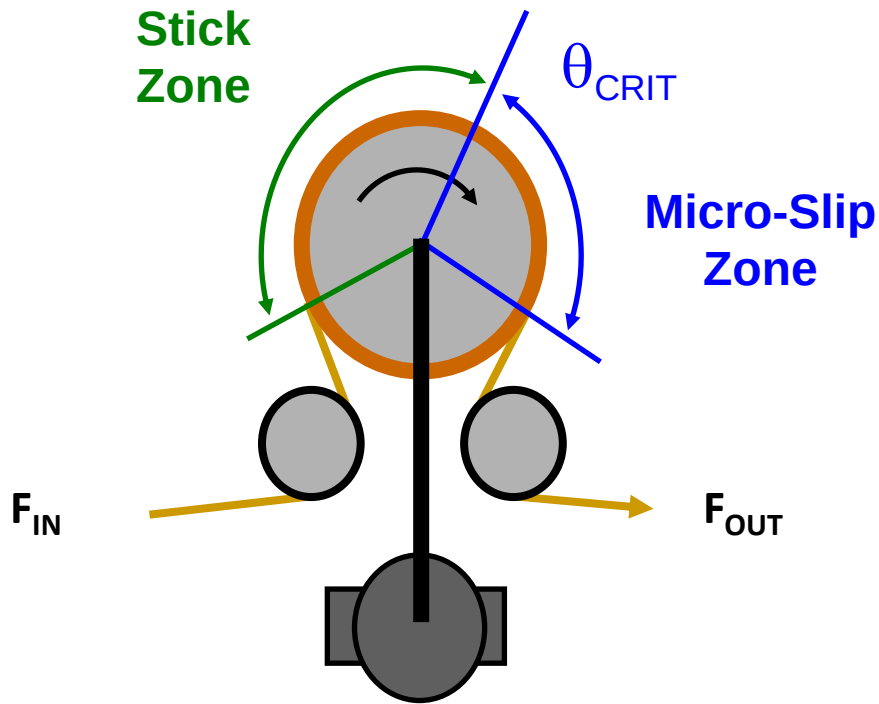
$$\mu = \frac{\ln(F_{HI} / F_{LO})}{\theta}$$

$$\frac{F_{HI}}{F_{LO}} = e^{\mu\theta}$$



Critical Wrap Angle

For a given tension ratio and friction coefficient, there is a critical wrap angle required to prevent full slip.



$$\theta_{CRIT} = \frac{\ln(F_{HI} / F_{LO})}{\mu}$$

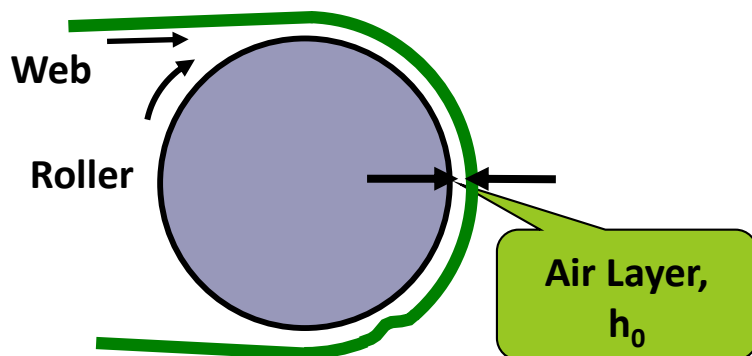
F_{HI} = High Tension, lbs

F_{LO} = Low Tension, lbs

μ = Web-roller coefficient of friction

θ = Wrap angle, radians

Air Lubrication



$$h_0 = 0.643r \left(\frac{6\eta(V_W + V_R)}{T} \right)^{2/3}$$

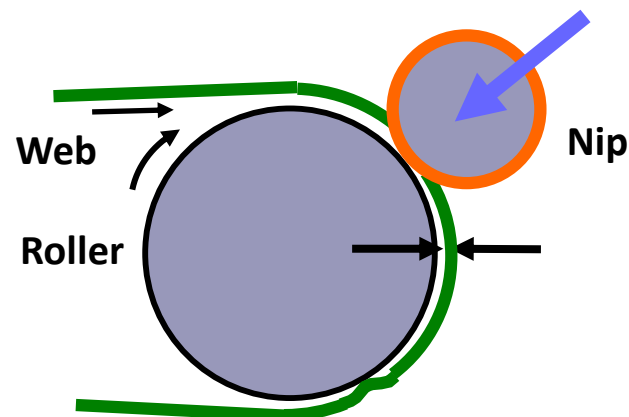
h_0 = Steady-state float height, in.

r = Roller radius, in.

V_W = Web speed, in/s

V_R = Roller speed, in/s

T = Tension, pli



$$h_0 = 4\eta V \frac{r_{eq}}{N}$$

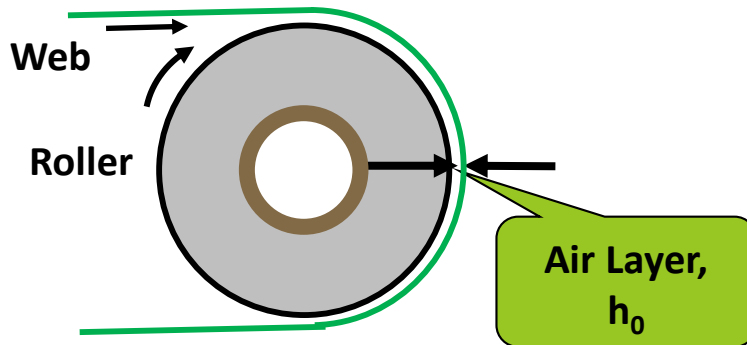
r_{eq} = Equivalent roller radius, in.

η = Air Viscosity,

V = Process speed, in/s

N = Nip load, pli

Entrained Air of Wrapped Roller



Wrapped Roller?

Air increases with:

- ▶ Larger radius
- ▶ Higher speed
- ▶ Lower tension

Entrained air increases proportional to radius.

2X radius = 2X entrained air.

$$h_0 = 0.643r \left(\frac{6\eta(V_W + V_R)}{T} \right)^{2/3}$$

h_0 = Steady-state float height, in.

r = Roller radius, in.

V_W = Web speed, in/s

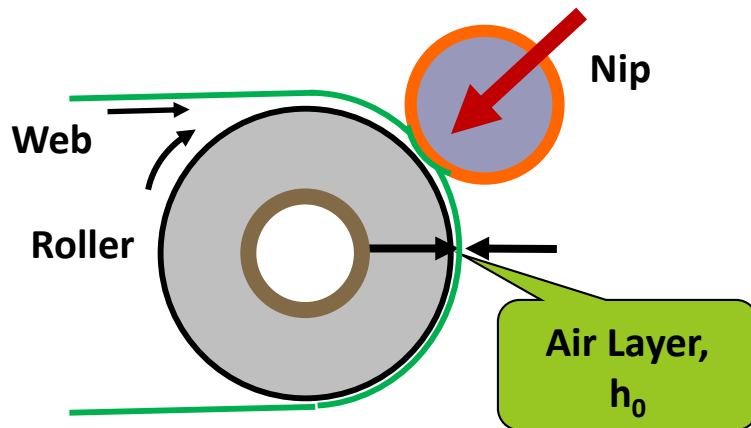
V_R = Roller speed, in/s

T = Tension, pli

Entrained air will be a function of speed-to-tension ratio (V/T):

50 fpm/0.25 pli = 400 fpm/2 pli

Entrained Air in Nipped Rollers



$$h_o = 4\eta V \frac{r_{eq}}{N}$$

- r_{eq} = Equivalent roller radius, in.
 η = Air Viscosity,
 V = Process speed, in/s
 N = Nip load, pli

Nipped Rollers?

Air increases with:

- ▶ Larger 'effective' radius
- ▶ Higher speed
- ▶ Lower nip loads

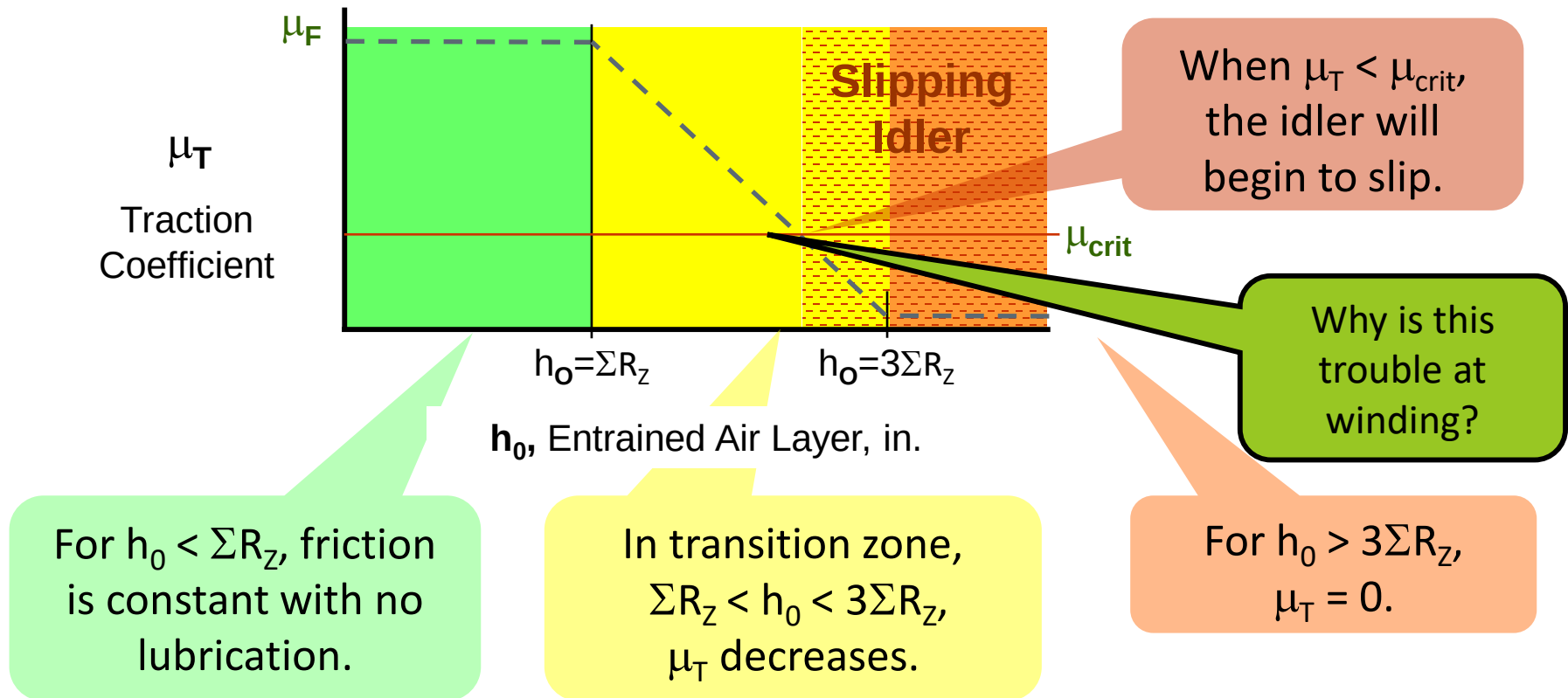
Air increase with radius, but the nip roller radius may have a stronger effect than winding roll radius.

Entrained air will be a function of speed-to-nip ratio (V/N):

100 fpm/1 pli = 500 fpm/5 pli

Traction vs. Air Lubrication

When lubricating air is greater than combined roughness, traction is lost.

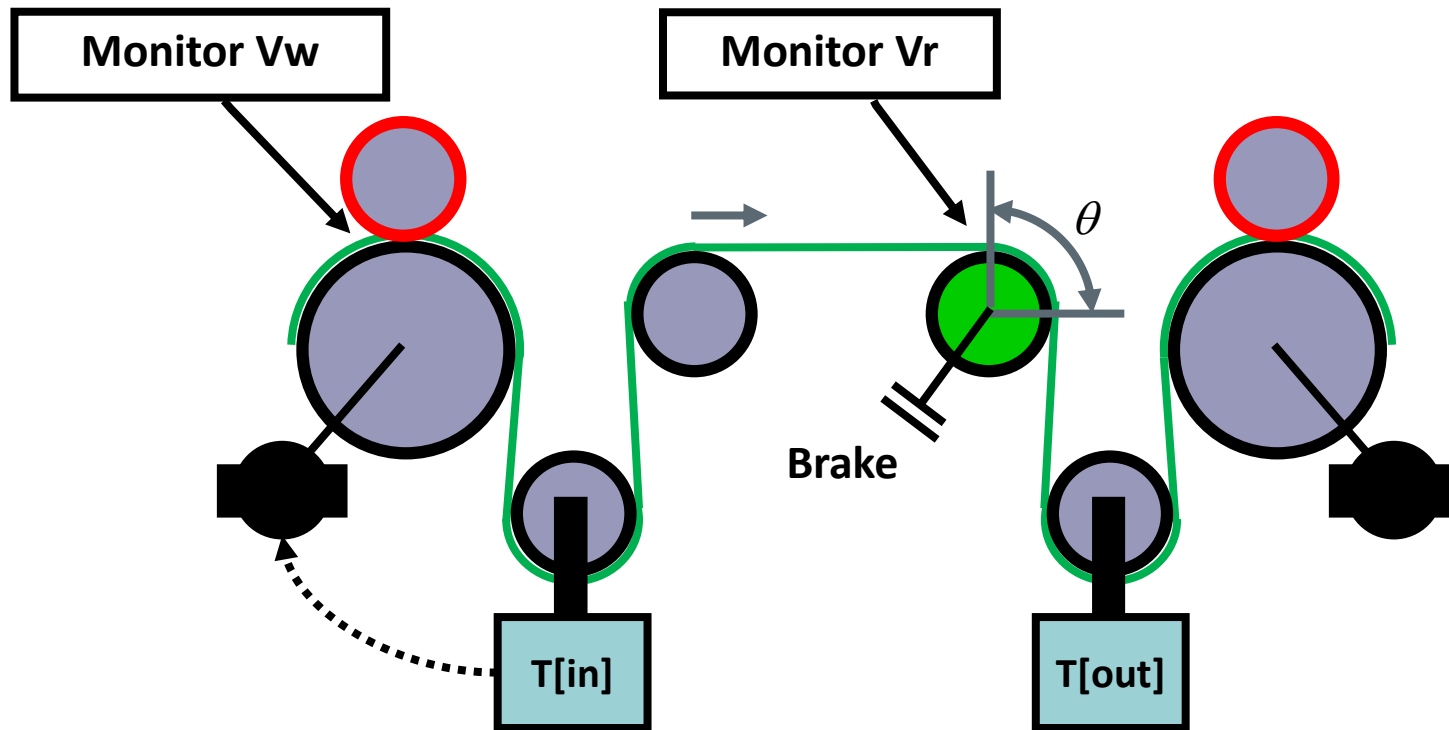


ΣR_z = sum of web and roller R_z roughness, in.

μ_F = friction coefficient, no lubrication

μ_{crit} = traction coefficient require to turn roller

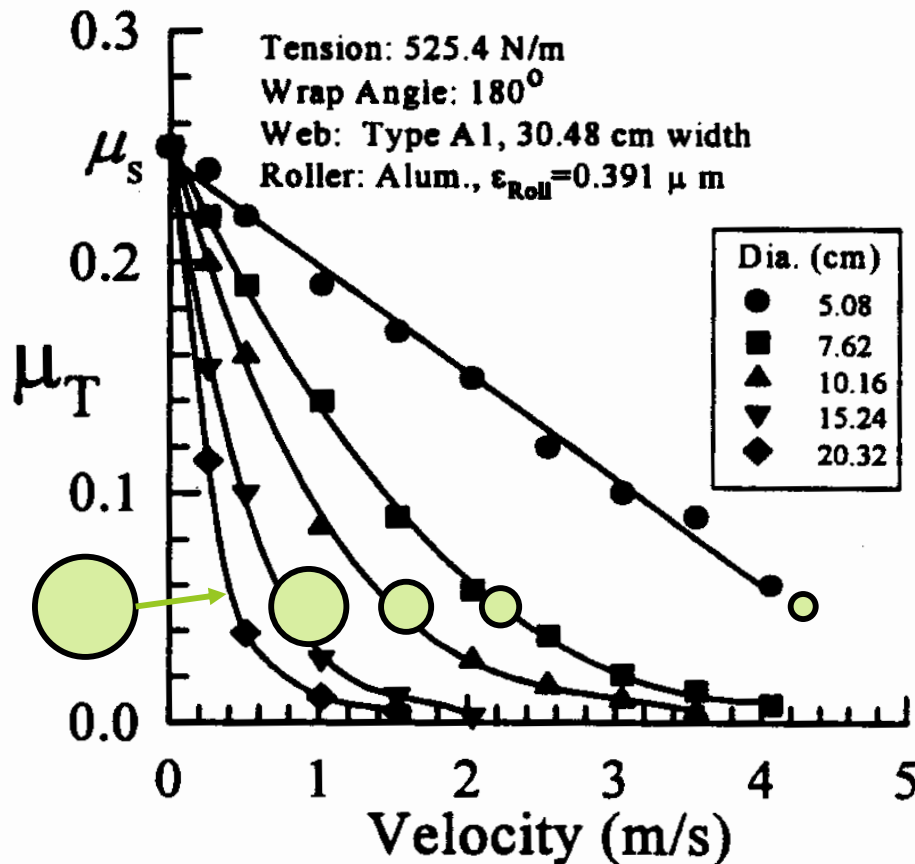
On-Line Traction Measurement



- Tension In is controlled to set point.
- Brake is applied until target roller slips 1-2%.
- Tension Out is noted at slip condition.
- Traction Coefficient is calculated from tensions and wrap.

$$\mu = \frac{\ln(T_{OUT}/T_{IN})}{\theta}$$

Roller Air Lubrication and Diameter



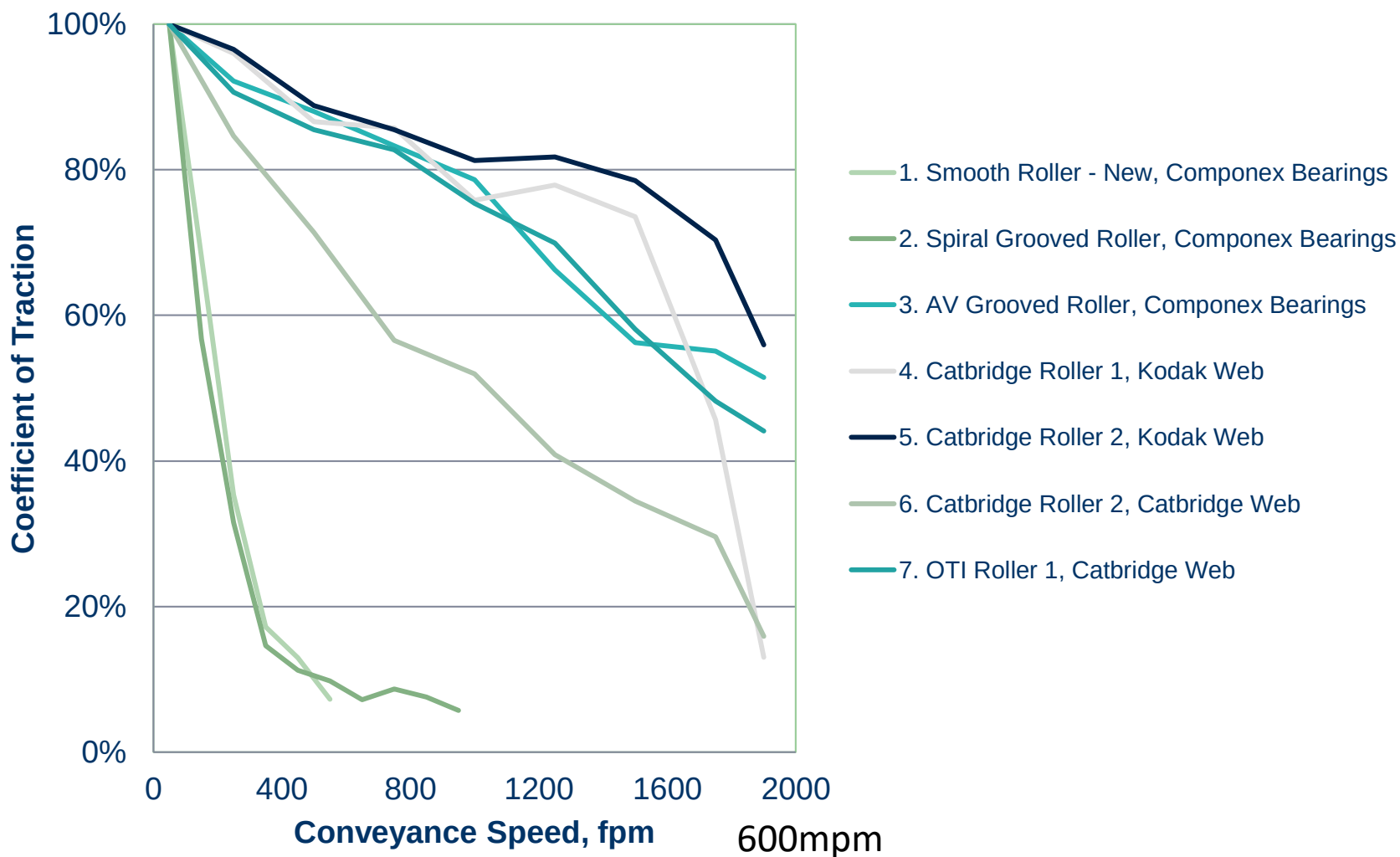
Larger diameter rollers lubricate sooner than small diameter rollers.

Tension = 3PLI

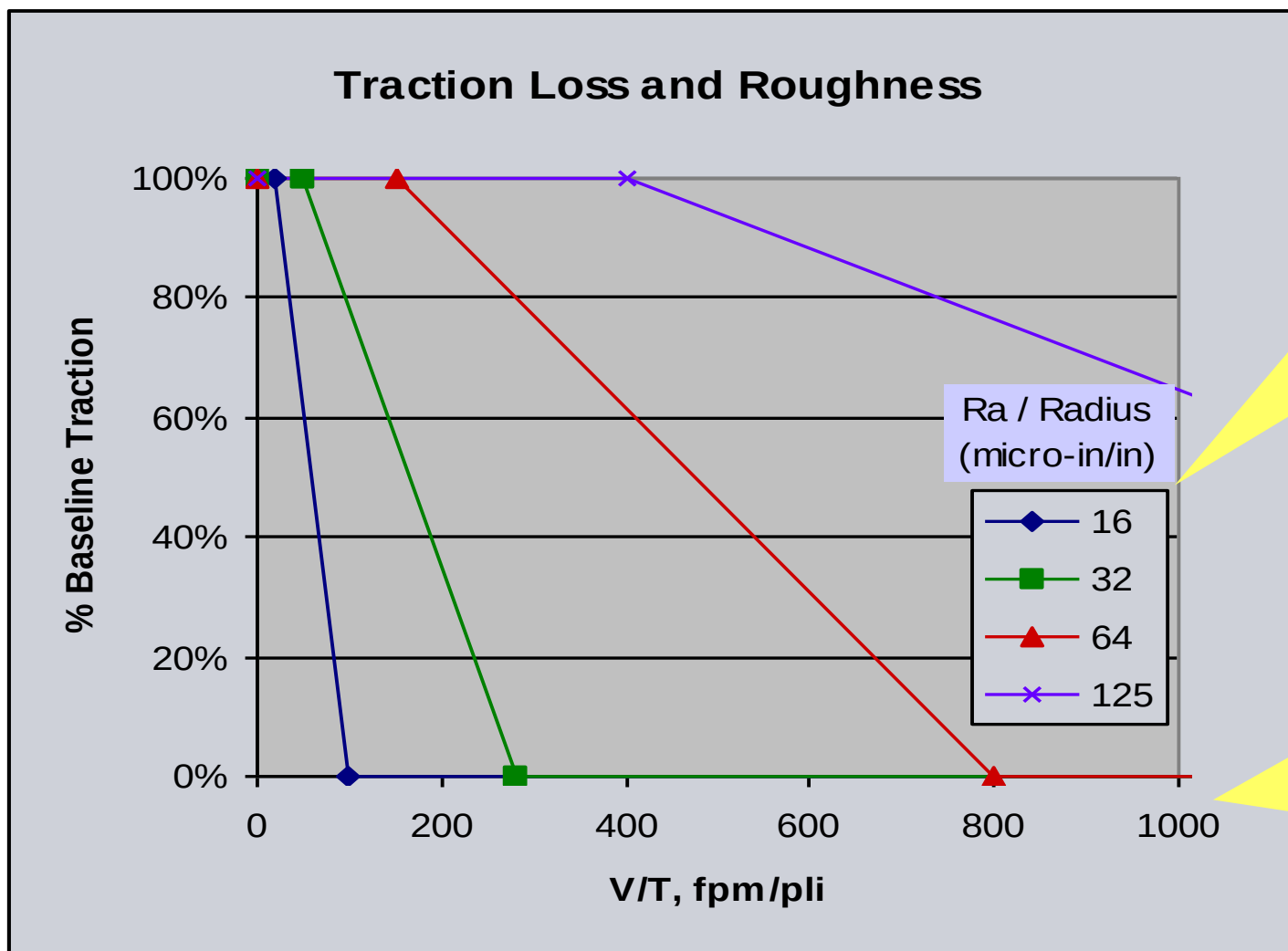
Diameters are: 2, 3, 4, 6, and 8-in

K S Ducotey and J K Good, "The Importance of Traction in Web Handling", *J Tribol* **117** 679-684 (1995).

Coefficient of Traction: Various Rollers



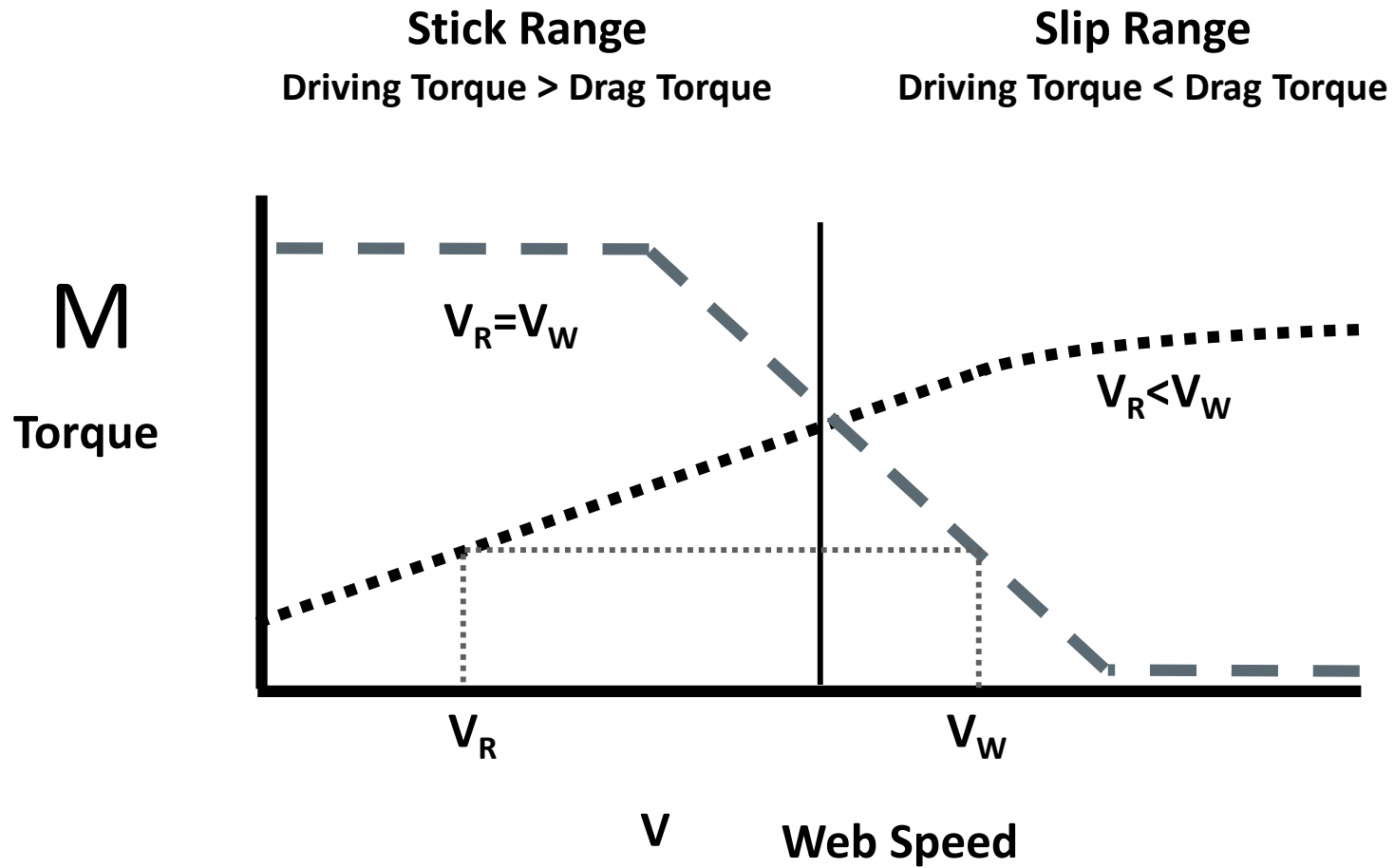
Roughness and Lubrication



Roughness to prevent lubrication is increase with roller radius.

Lubrication increases with higher speed/tension ratio.

$$V_{\text{web}} > V_{\text{roller}} > 0?$$



Example: Belt-Roller Slip



$R = 20 \text{ in. (0.5m)}$

$T = 100 \text{ lb/in (20000N/m)}$

$V = 200 \text{ mph (320km/hr) = 17000 ft/min = 3500 in/s}$

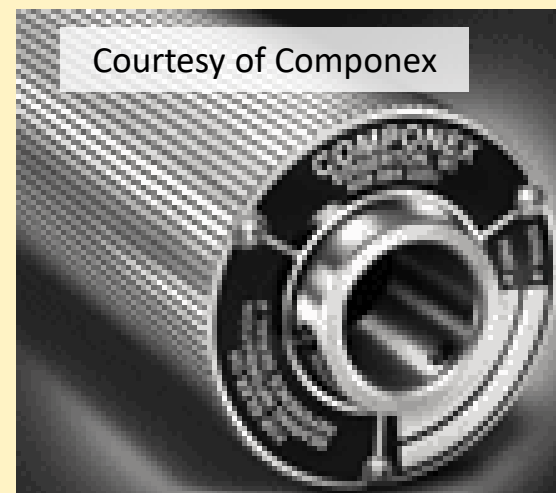
Air layer > 1-mil (25 microns)



Roller Surfaces

- ▶ Ultra Smooth (Chrome)
- ▶ Machined Roughness (e.g. 32 Ra)
- ▶ Sandblasted Surface
- ▶ Macro-Grooved
- ▶ Micro-Grooved
(depth and spacing 0.010-0.50")
- ▶ Plasma Sputtered
- ▶ Roller Wraps (tape, shrink sleeves)
- ▶ Rubber Covered

Roller Surfaces



Macro-Grooving

Micro-Grooving

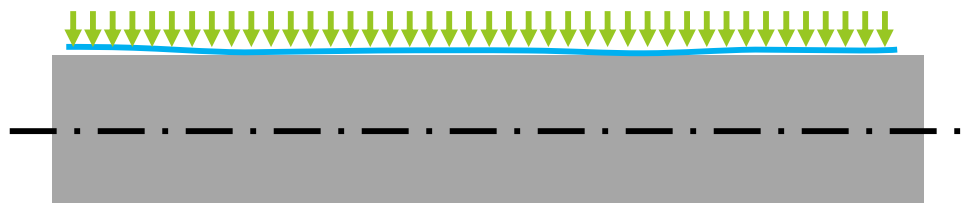


Pressure on Grooved Rollers

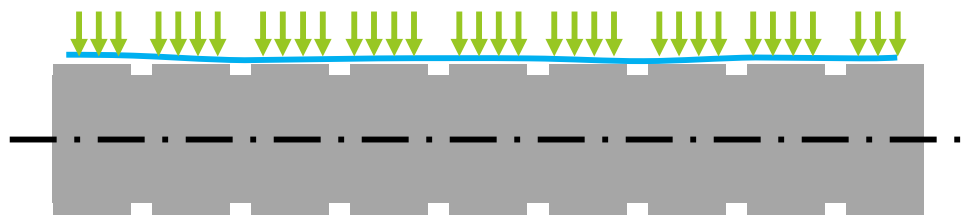
$$P = \left(\frac{w_G + w_L}{w_L} \right) \left(\frac{T}{r} \right)$$

Grooves reduce contact area, increasing contact pressure at the real contact area and drives air into grooves or valleys of surface roughness.

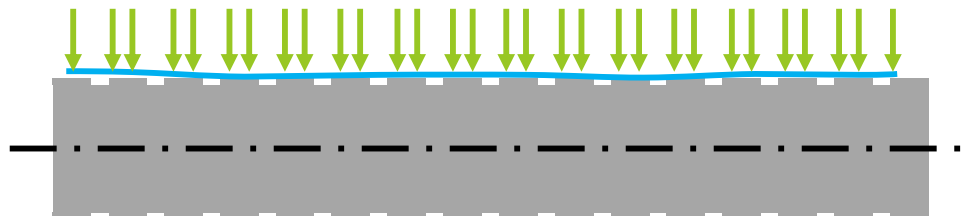
$$P = \left(\frac{0 + 1}{1} \right) \left(\frac{1PLI}{2in.} \right) = 0.5 \text{ psi}$$



$$P = \left(\frac{1 + 4}{4} \right) \left(\frac{1PLI}{2in.} \right) = 0.625 \text{ psi}$$



$$P = \left(\frac{1 + 2}{2} \right) \left(\frac{1PLI}{2in.} \right) = 0.75 \text{ psi}$$



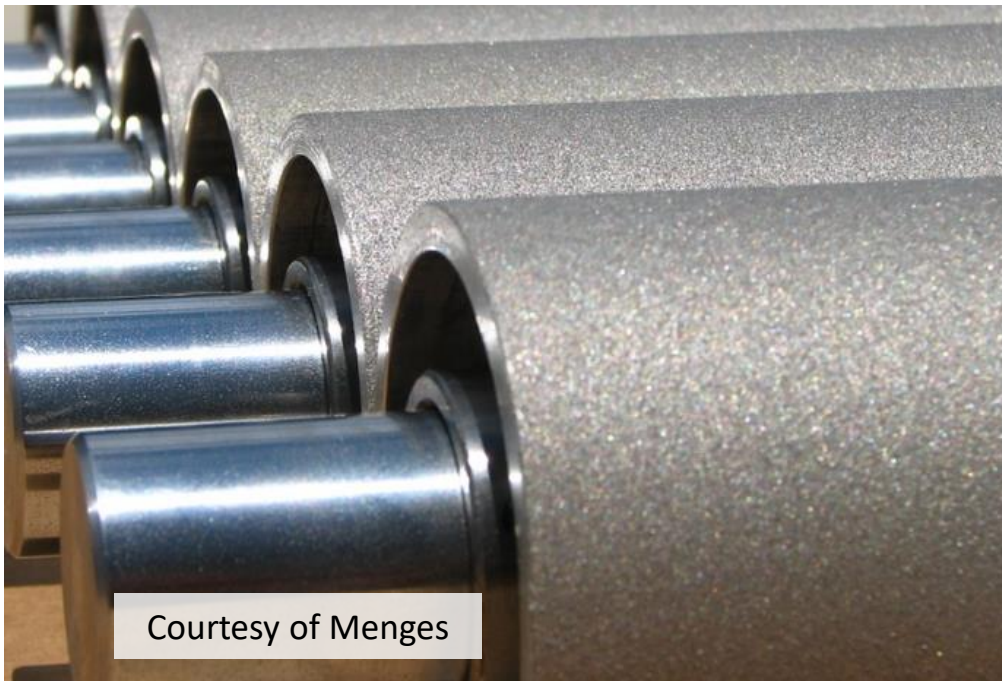
'Plasma' Roller Surface

HVOF spraying Tungsten Carbide / Cobalt Chromium Coating

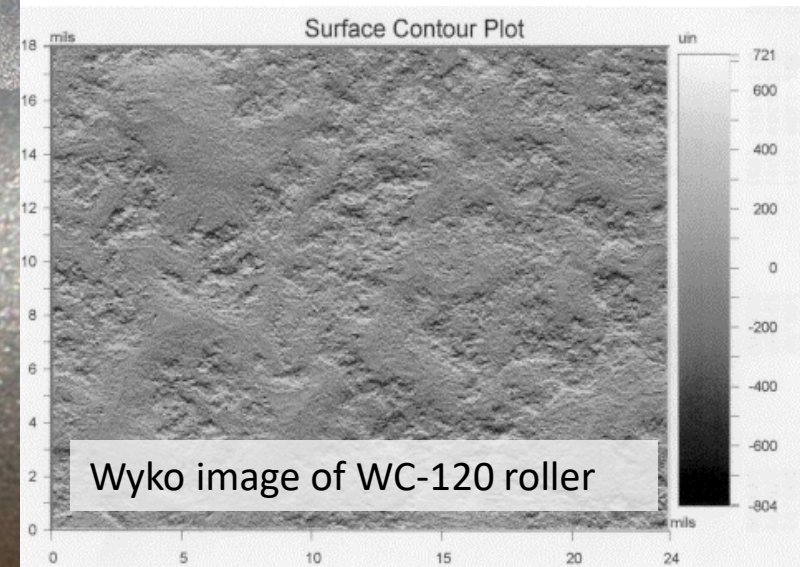
Plasma coatings can include Teflon or Silicone for release or increased COF.



Courtesy of Gordon England



Courtesy of Menges



Wyko image of WC-120 roller

Information courtesy of Optimization Technologies, Inc.

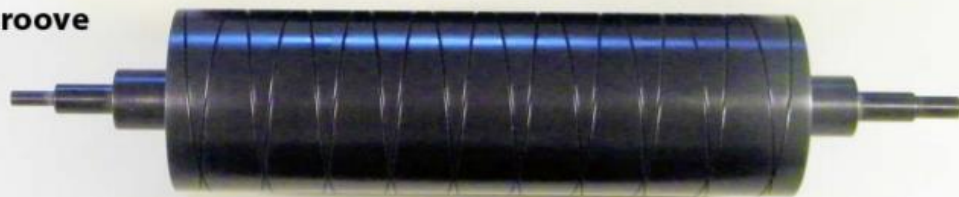
Case Study: Groove Options

Standard



Standard
Machined 32 μ inch [0.8 μ m] Ra surface

Spiral groove

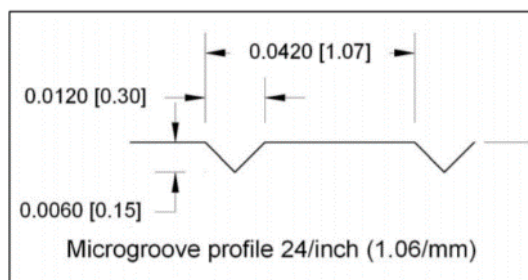
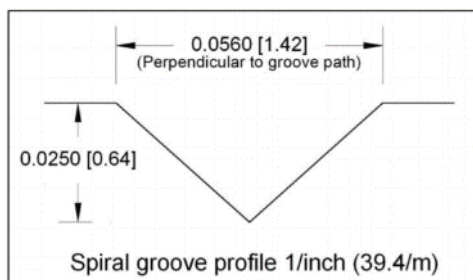


Spiral groove
Two grooves cut end to end, opposing directions, machined 32 Ra surface

Microgroove



Microgroove:
One groove on each half, opposite hand, machined 32 Ra



Getting and Losing Traction

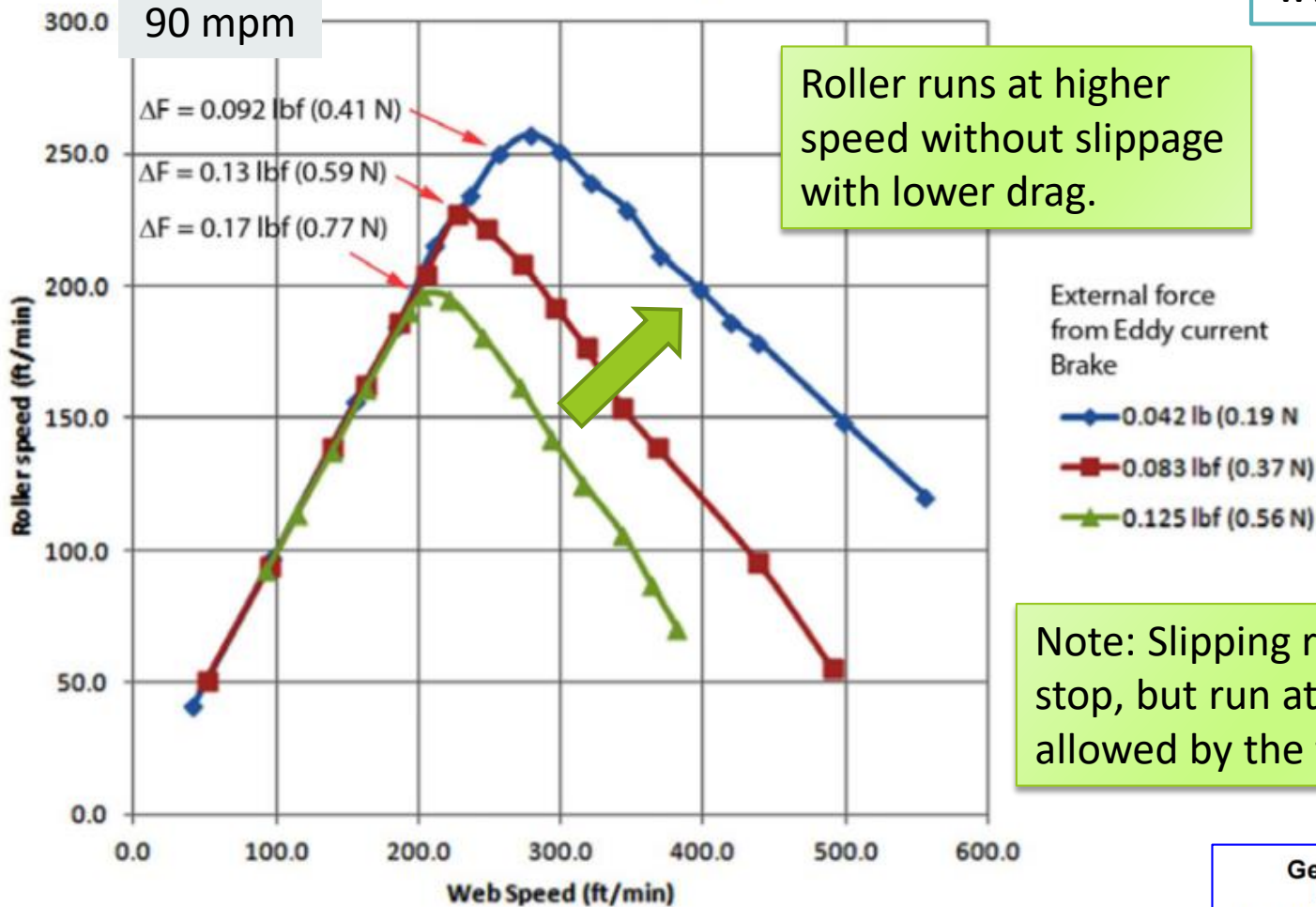
Jerry Brown, Essex Systems
AIMCAL Web Coating and Handling Conference
October 2012

www.normalentry.com

Critical Speed = Web-Roller Slippage

Jerry Brown
www.normalentry.com

Web speed versus roller speed - standard roller



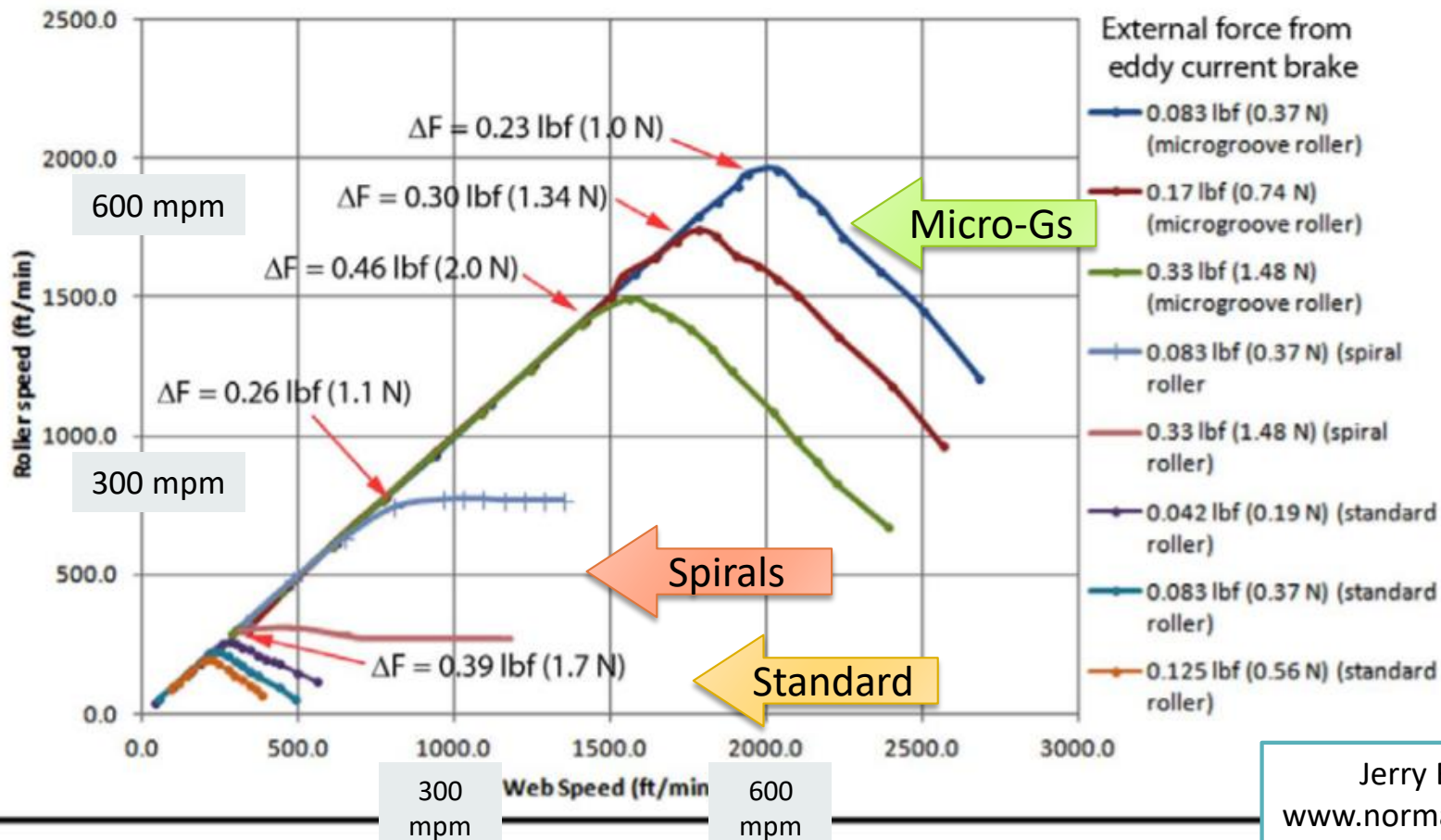
Note: Slipping rollers don't stop, but run at the speed allowed by the traction limit.

Getting and Losing Traction

Jerry Brown, Essex Systems
AIMCAL Web Coating and Handling Conference
October 2012

Case Study: Smooth, Macro-, & Micro-Grooves

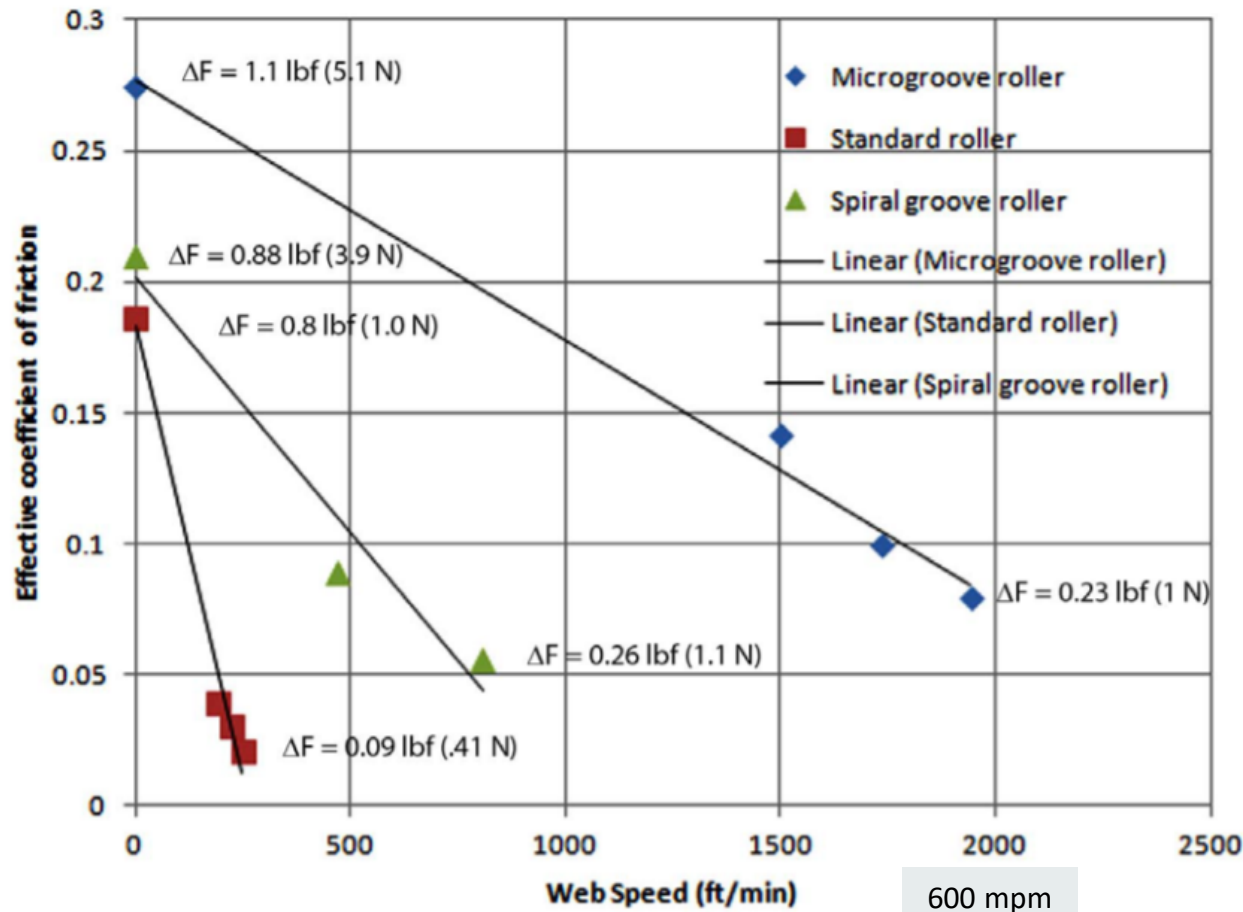
Comparison of web speed vs roller speed for standard, spiral and microgroove rollers



Jerry Brown
www.normalentry.com

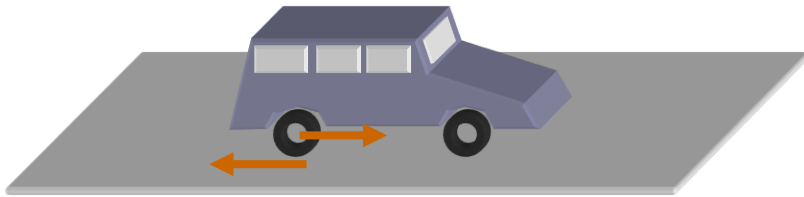
Case Study: Effective COF (Coefficient of Traction)

Effective coefficients of friction for standard, microgroove and spiral rollers

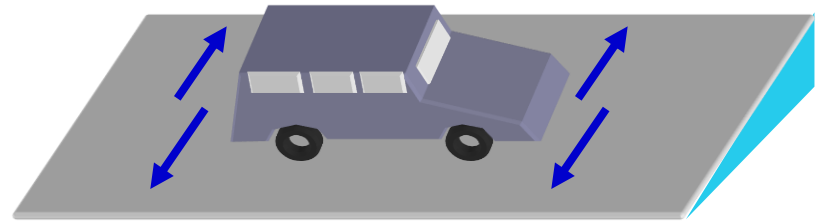


The Friction Circle

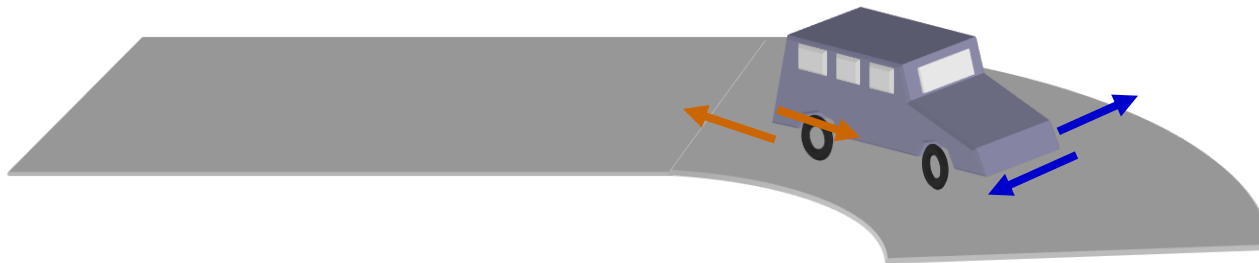
In driving, you use tire-road friction in two directions:



1. To accelerate or brake



2. To hold the car laterally



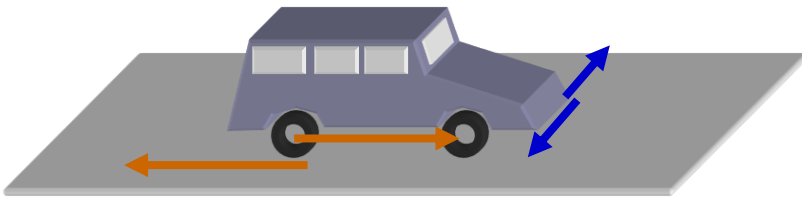
In a turn, you may use friction in both directions at once.

The Friction Circle

The total friction available, F , is equal to μ times N .

μ = coefficient of friction,

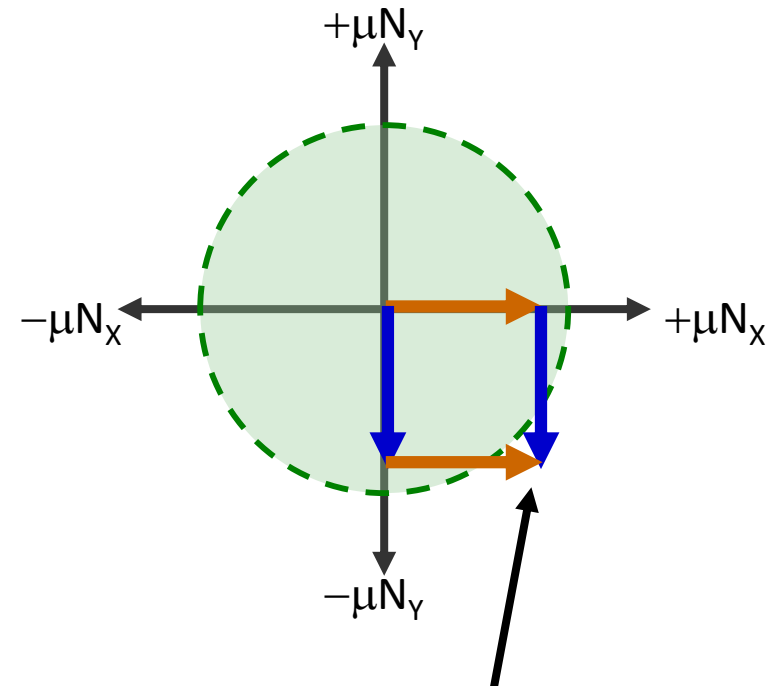
N = normal load (car weight)



Slip occurs if the applied load exceeds the friction available in either direction,

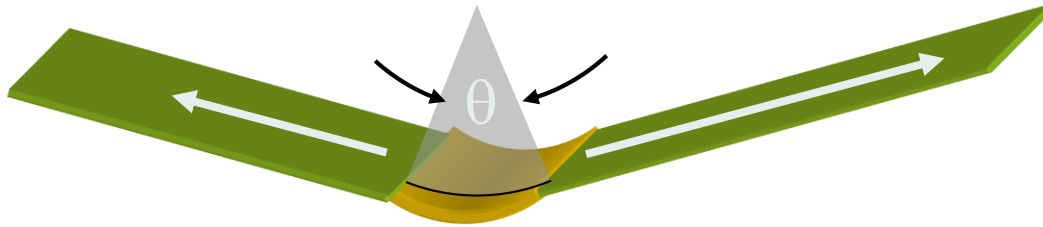
—OR—

If the combined vector sum exceeds the limit.



Neither one of these friction demands alone would create slip, but combined they exceed the friction circle limit.

AW Rollers Limited by Friction



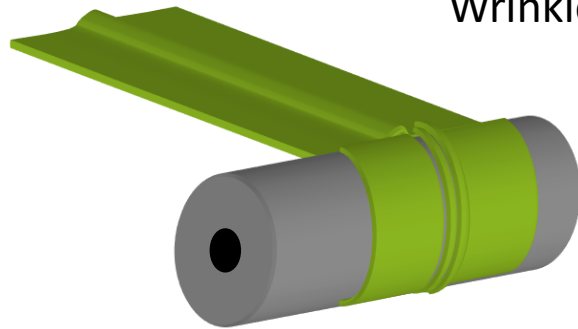
Machine Direction Friction

Transverse Direction Friction

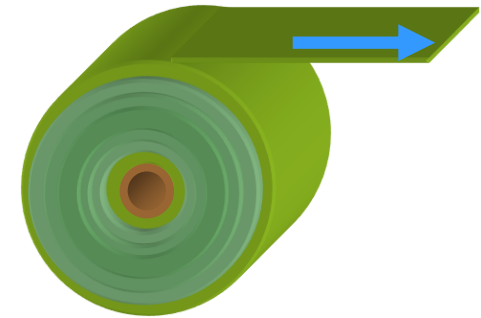


Friction Circle and Web Handling

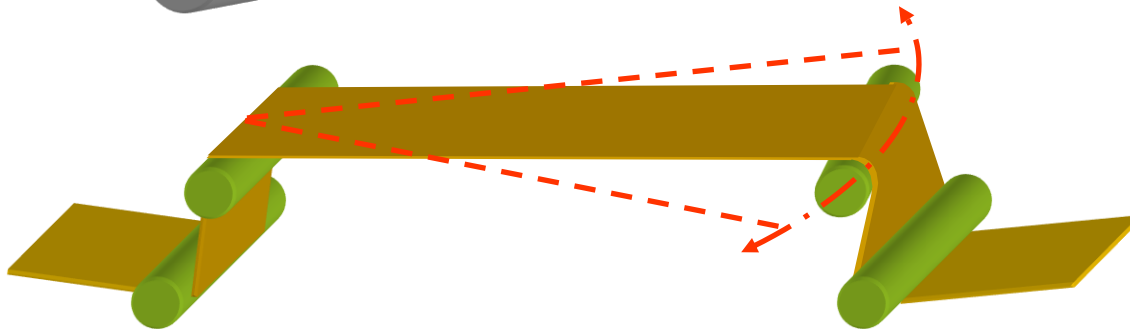
Where does the Friction Circle explain unusual web handling behavior?



Wrinkles disappear just before a roller slips.



Tangential slip in a winding roll sets off lateral motion.



Steering rollers slip before identical idlers.

Each case involves simultaneous use of friction in two directions.

Dual Traction Roller

Dual traction roller

EP 0709328 A2

ABSTRACT

A dual traction roller surface provides a constant, intermediate traction coefficient by combining two different surfaces, one low and one high traction. The dual traction surface eliminates web slippage and scratches without increasing wrinkle sensitivity.

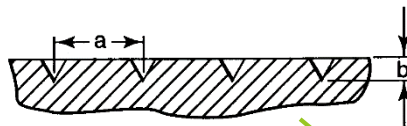


Fig. 5



Fig. 2A



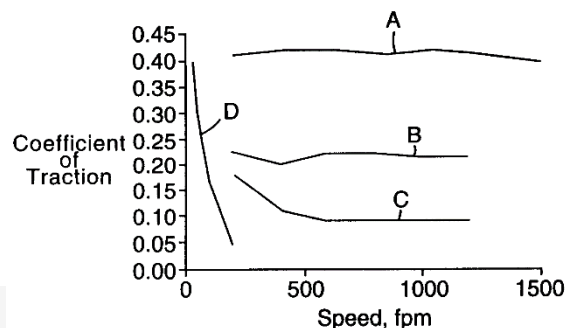
Fig. 2B



Fig. 3



Fig. 4



Publication number EP0709328 A2
 Publication type Application
 Application number EP19950116894
 Publication date May 1, 1996
 Filing date Oct 26, 1995
 Priority date  Oct 27, 1994

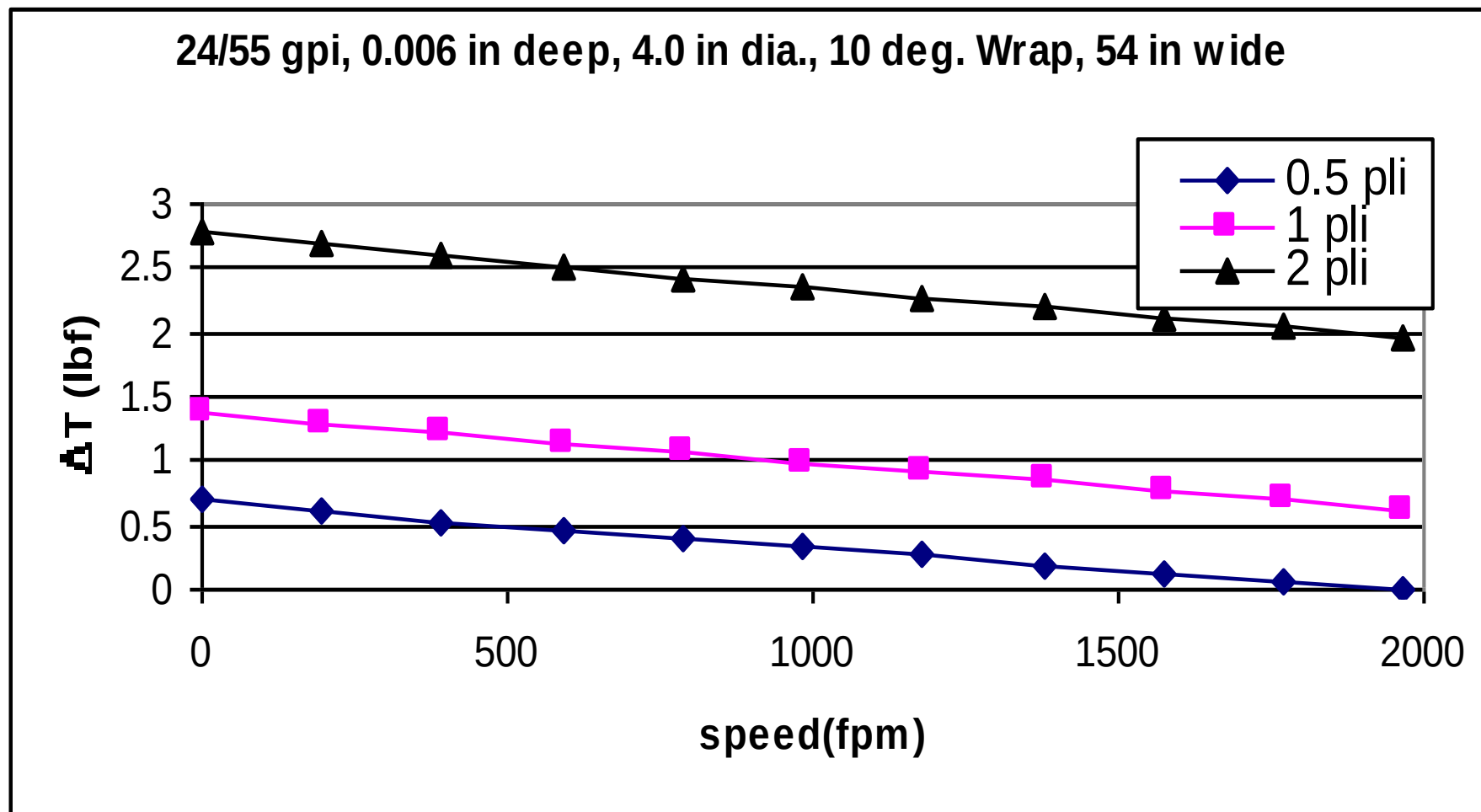
Also published as [EP0709328A3](#)

Inventors [Timothy J. c/o Minnesota Mining and Walker](#)

Applicant [Minnesota Mining And Manufacturing Company](#)

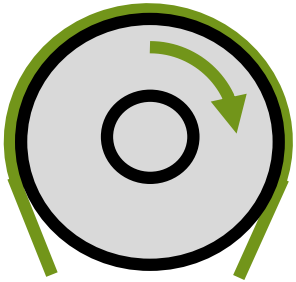


Effect of Tension on Traction



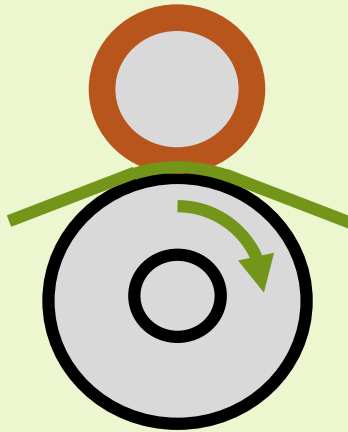
Information courtesy of Optimization Technologies, Inc.

Web Support / Transport / Tensioning Options



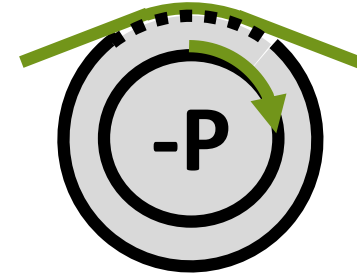
Wrapped
Roller

$$\frac{T_{HI}}{T_{LO}} = e^{\mu\theta}$$



Nipped
Roller

$$\Delta T = \mu N$$



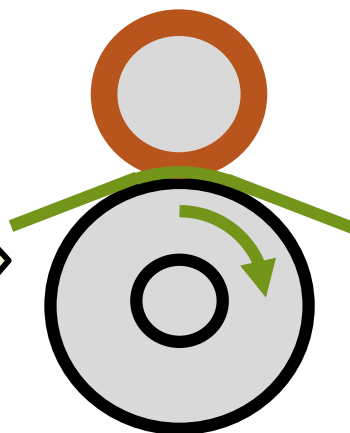
Vacuum
Roller

$$\Delta T = f(\mu, P, r, w, \theta, T)$$

Web Support / Transport / Tensioning Options

A nipped roller is the simplest options to create high traction:

- independent of tension,
- nearly independent of wrap angle,
- naturally reducing air lubrication, and
- Incorporated into many processes.



Nipped
Roller

$$\Delta T = \mu N$$

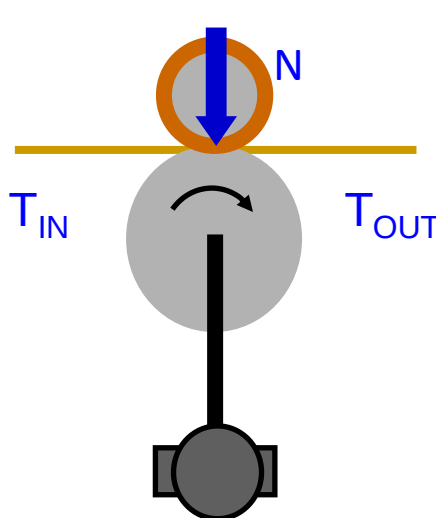
However, nipped rollers create problems, especially with:

- baggy web wrinkling,
- roller deflection
- variation and wear of rubber covered rollers.

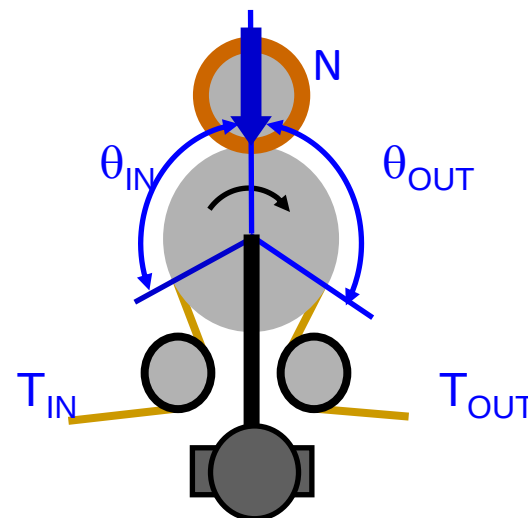
Nipped Pull Rollers: Available Traction

Nipped rollers are a simple way to increase friction,...

...but required contact with both sides of the web and may increase problems such as wrinkling, abrasion, and web breaks.



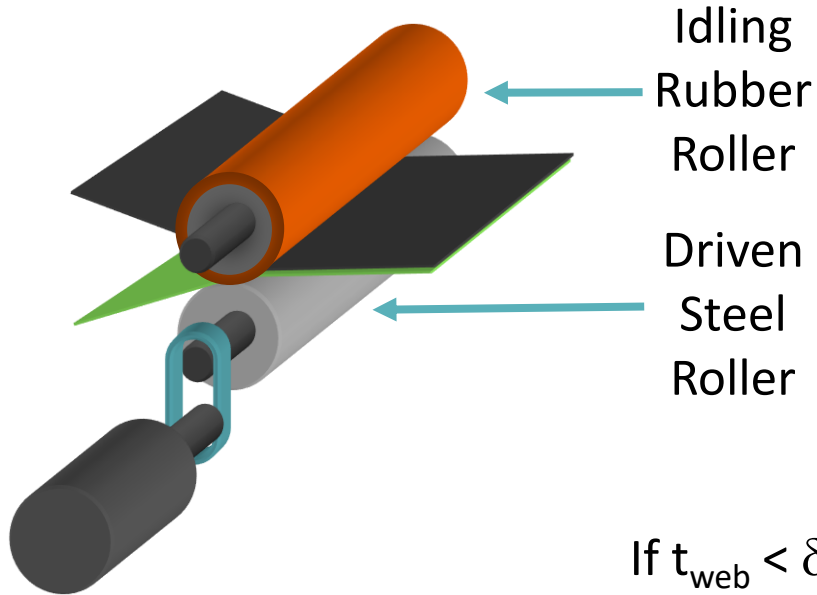
$$\Delta T = \mu N$$



$$\Delta T > \mu N$$

Friction is higher using both nip and wrap to create friction.

Nipped Roller Traction



What friction coefficient controls nipped roller traction?
Web-steel? Web-rubber?

t_{web} = web thickness

δ_{rubber} = rubber indentation

If $t_{web} < \delta_{rubber}$

Function of both web-steel
and web-rubber traction

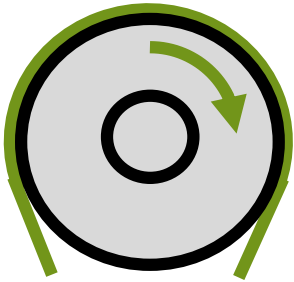


If $t_{web} > \delta_{rubber}$

Function of web-steel only.

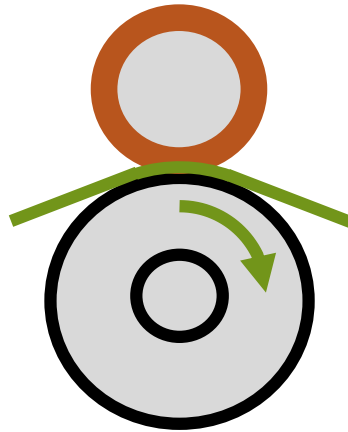


Web Support / Transport / Tensioning Options



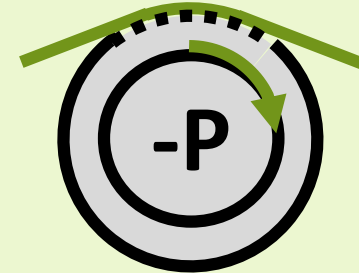
Wrapped
Roller

$$\frac{T_{HI}}{T_{LO}} = e^{\mu\theta}$$



Nipped
Roller

$$\Delta T = \mu N$$

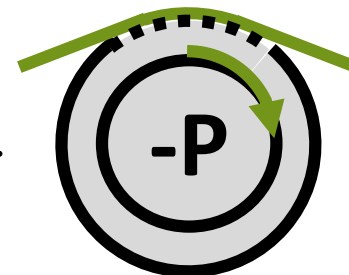


Vacuum
Roller

$$\Delta T = f(\mu, P, r, w, \theta, T)$$

Web Support / Transport / Tensioning Options

Vacuum rollers provide high traction without problems associated with nipped roller, but at increased cost and complexity.

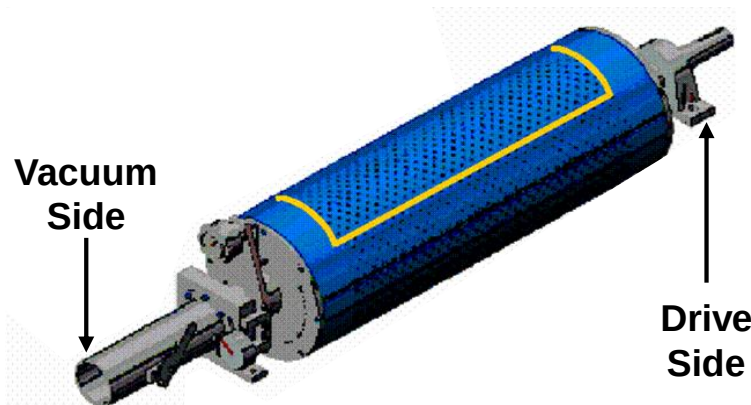


Vacuum
Roller

$$\Delta T = f(\mu, P, r, w, \theta, T)$$

Vacuum Rollers: Available Traction

Vacuum rollers are an effective one-side design to increase friction without the problems of nipped rollers.



$$\Delta P_{Eff} = \left(\frac{A_{EXP}}{A_{TOTAL}} \right) (P_{atm} - P_{VAC})$$

$$F = \mu N = \mu A(\Delta P)$$

$$F = \mu r w \theta \left(\Delta P_{Eff} + \frac{T}{r} \right)$$

- μ = Traction Coefficient
- r = Roller Radius, in.
- w = Web Width, in.
- θ = Wrap Angle, radians
- ΔP_{Eff} = Effective Pressure From Vacuum
- T = Web Tension, pli
- A_{EXP} = Area Exposed to Vacuum Pressure, in²
- A_{TOTAL} = Total Area of Vacuum Roller, in²
- P_{atm} = Atmospheric Pressure, psi.
- P_{VAC} = Internal Vacuum Roller Pressure, psi.

Vacuum Pull Roller Design

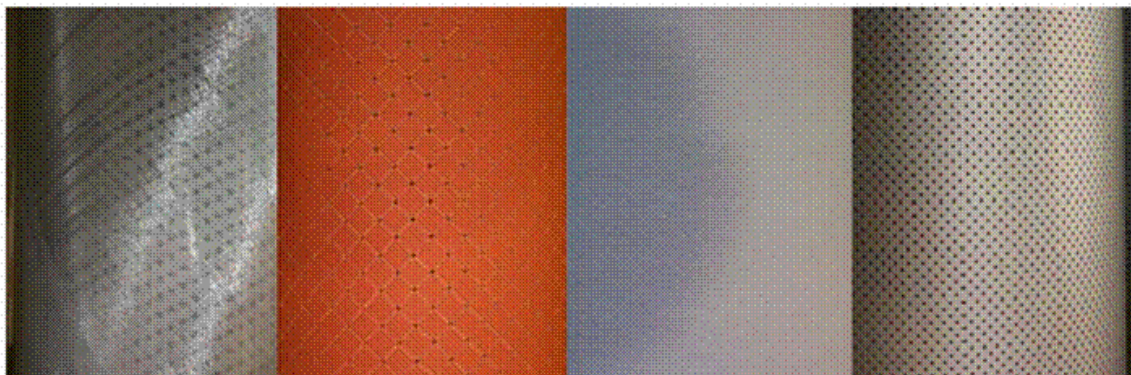
- Shell

- Coatings

- Tungsten carbide traction/release, rubber, nickel, custom

- Coverings

- Stainless steel screen, felt, etched seamless screen



Stainless Steel
Screen

Grooved
Rubber

Felt

Tungsten
Carbide

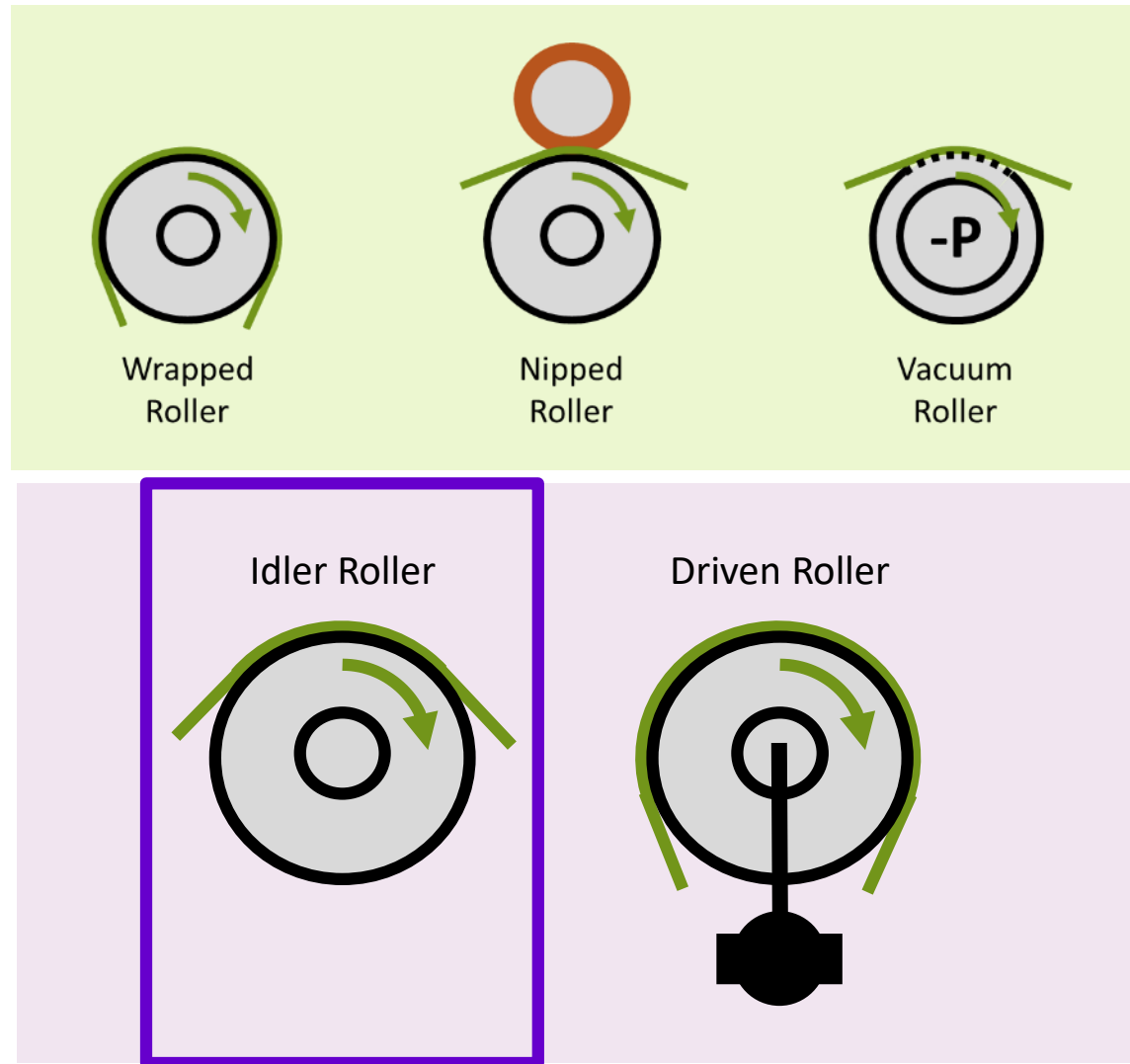


Image courtesy of New Era
Converting

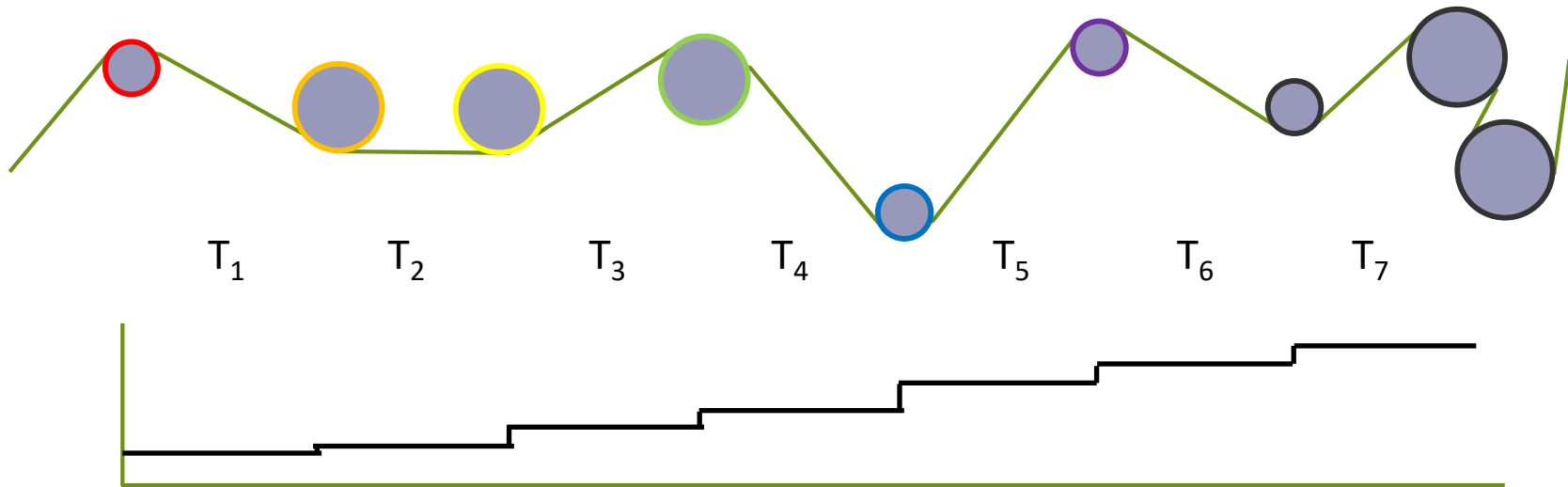
“Vacuum Pull Rollers”, Lee Fenema, Coating & Converting Resources, Inc.
AIMCAL Fall Technical Conference 2003

Traction Requirements

$$\text{TSF} = \frac{\text{Available Traction}}{\text{Required Traction}}$$

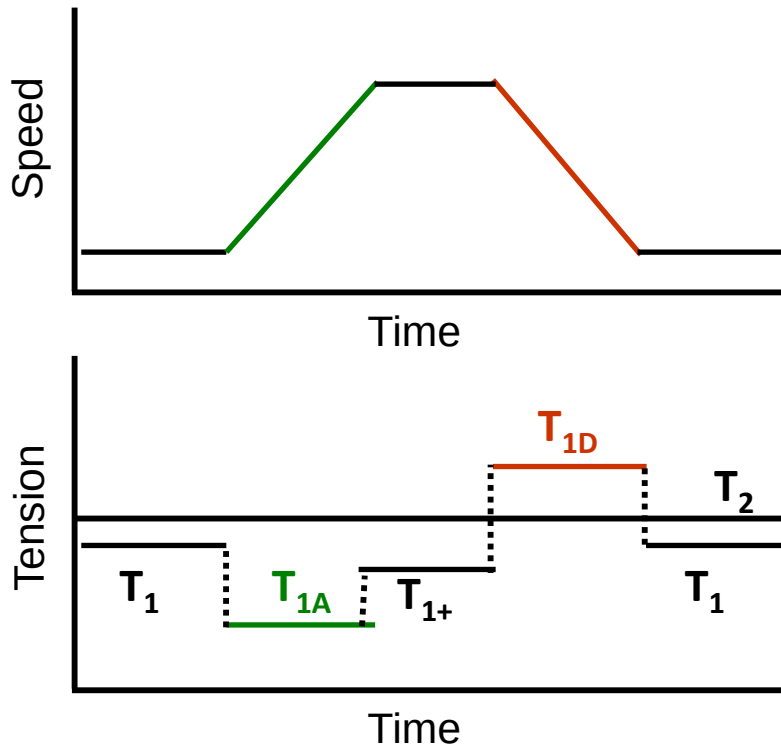


Rollers and Train Cars: Drag & Inertia



Idler Roller Tension Change

Tension change created by an idler roller is dependent on drag and inertia.

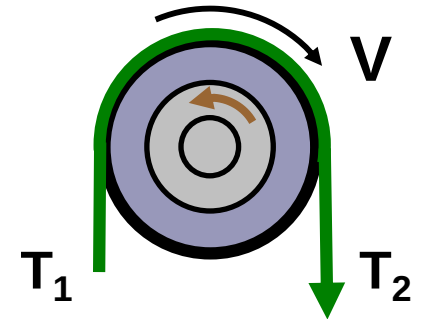


*Assume constant T_2

Steady Speed

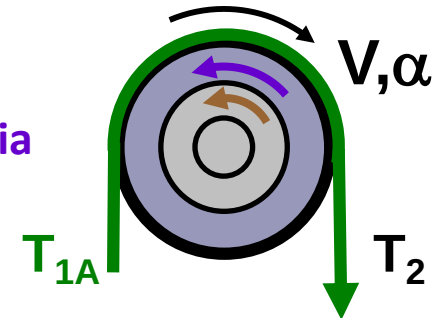
$$\Delta T = \text{Drag}$$

$$\text{Inertia} = 0$$



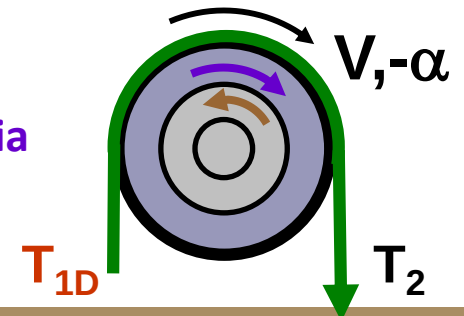
Accelerating

$$\Delta T = \text{Drag} + \text{Inertia}$$



Decelerating

$$\Delta T = \text{Drag} - \text{Inertia}$$



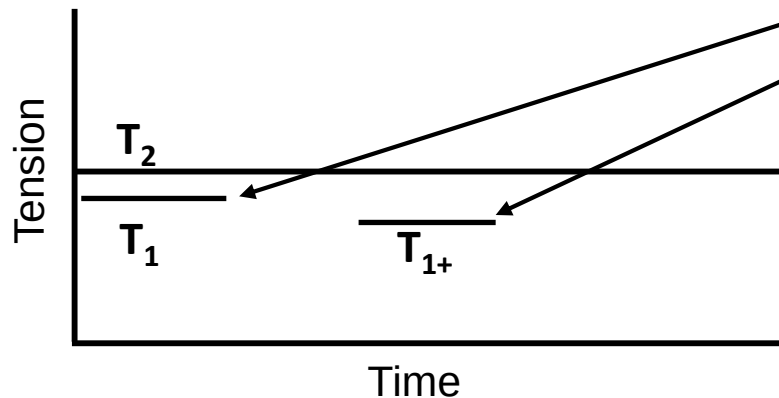
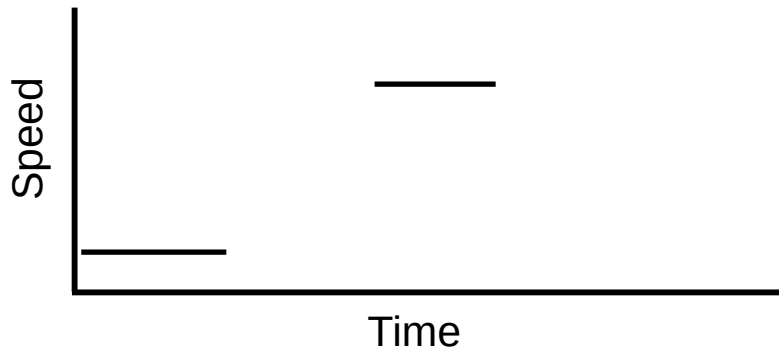
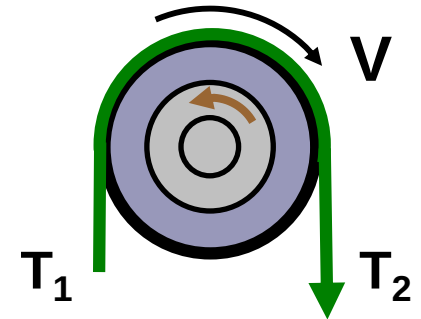
Idler Roller Tension Change: Steady-State

Tension change created by an idler roller is dependent on drag and inertia.

Steady Speed

$$\Delta T = \text{Drag}$$

$$\text{Inertia} = 0$$



*Assume constant T_2

$$\Delta T = T_2 - T_1 = \frac{M_B}{rw}$$

ΔT

Change in tension, pli

T_1

Upstream Tension, pli

T_2

Downstream Tension, pli

M_B

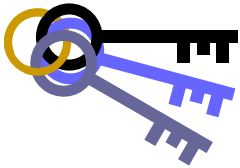
Drag torque of bearings

r

Roller radius, in.

w

Web width, in.



KEY CONCEPT



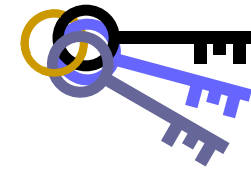
Roller Spin Down Testing

Timing idler roller deceleration times can be used as a simple qualitative test of roller performance or as part of a sophisticated calculation of roller drag and inertia.

Use: Compare performance of identical rollers

Example: Measure the time to stop from 100 rpm. Longer spin down times indicate lower bearing drag.

Use: Combined with roller inertia and traction models, spin down testing can predict when rollers will slip and scratch. Example: Prevent slip by ensuring available tension, wrap angle, and traction coefficient will overcome drag and inertial torques.

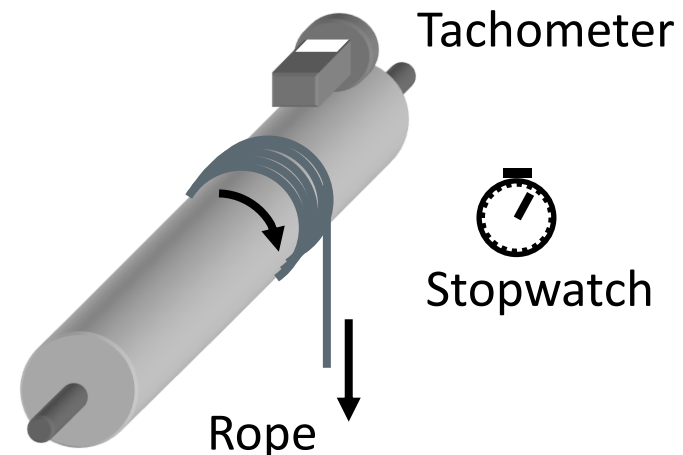
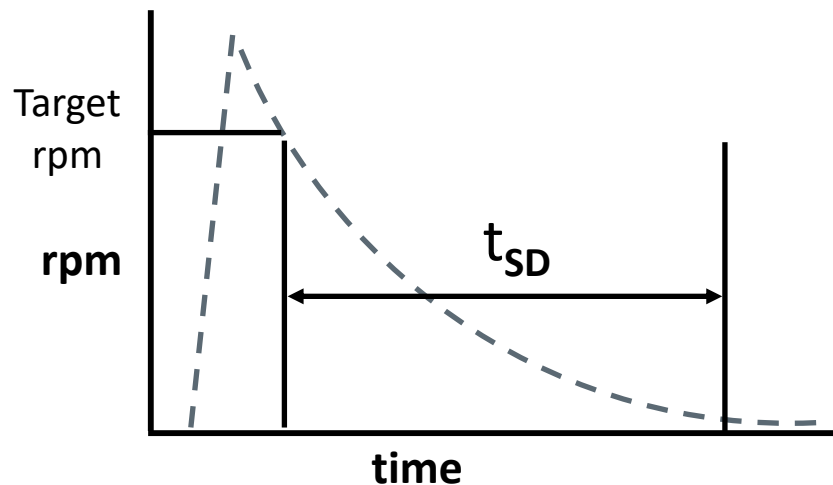


Bearing Drag Test

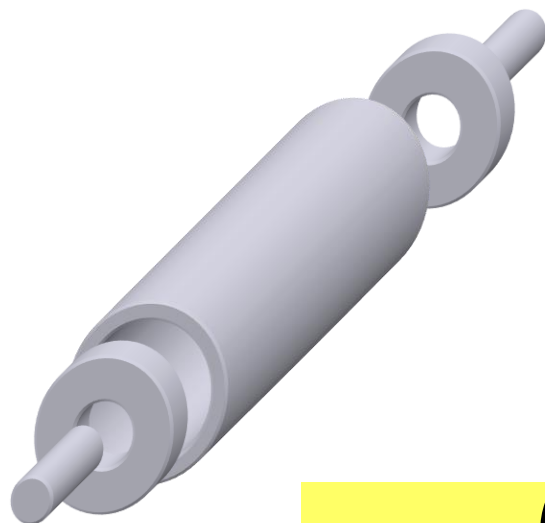
Use this test to compare identical idlers OR to calculate an idler roller's drag torque.

An idler roller's stopping time is a balance of roller inertia and bearing assembly drag torque.

1. Spin roller
2. Measure speed
3. Measure time to stop



Calculate Bearing Torque

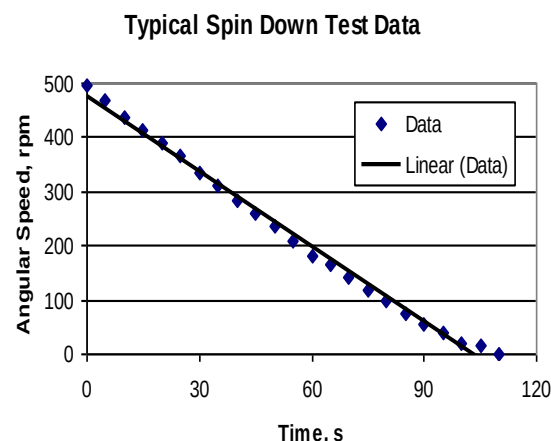


$$M_B = I \alpha_{SD}$$

$$I = \frac{\pi \rho w (r_O^4 - r_I^4)}{2}$$

$$\alpha_{SD} = \frac{2\pi(\Delta n)}{60 t_{SD}}$$

$$\alpha_{SD} = \frac{V}{r t_{SD}}$$



I Roller rotational moment of inertia (kg-m²)

r_I Roller shell inner radius (m)

r_O Roller shell outer radius (m)

w Roller width (m)

ρ Density (kg/m³)

α_{SD} Rotational deceleration (radians/s²)

Δn Change in rotation rate (rpm)

M_A Torque of inertia (N-m)

M_B Torque required to turn bearings

Idler Bearing Properties

Important Bearing Properties Include:

- Geometry
(inner diameter, outer diameter, width)
- Lubricant
(type, amount, maintenance)
- Capture Design
- Environment
- Spindown
(turning drag performance)
- Load Specification



Low Drag Bearing Example: SealMaster LO and XLO

Low Drag “LO”

- Felt seal cut down to reduce drag
- Provides excellent barrier that works to retain grease and acts as a filter to reduce the ingress of debris contaminants
- Features a channeling grease that reduces drag while lubricating the bearing
- Operating temperature from -20°F to 180°F
- A reduced housing fit to allow for the bearing to misalign more easily in the housing



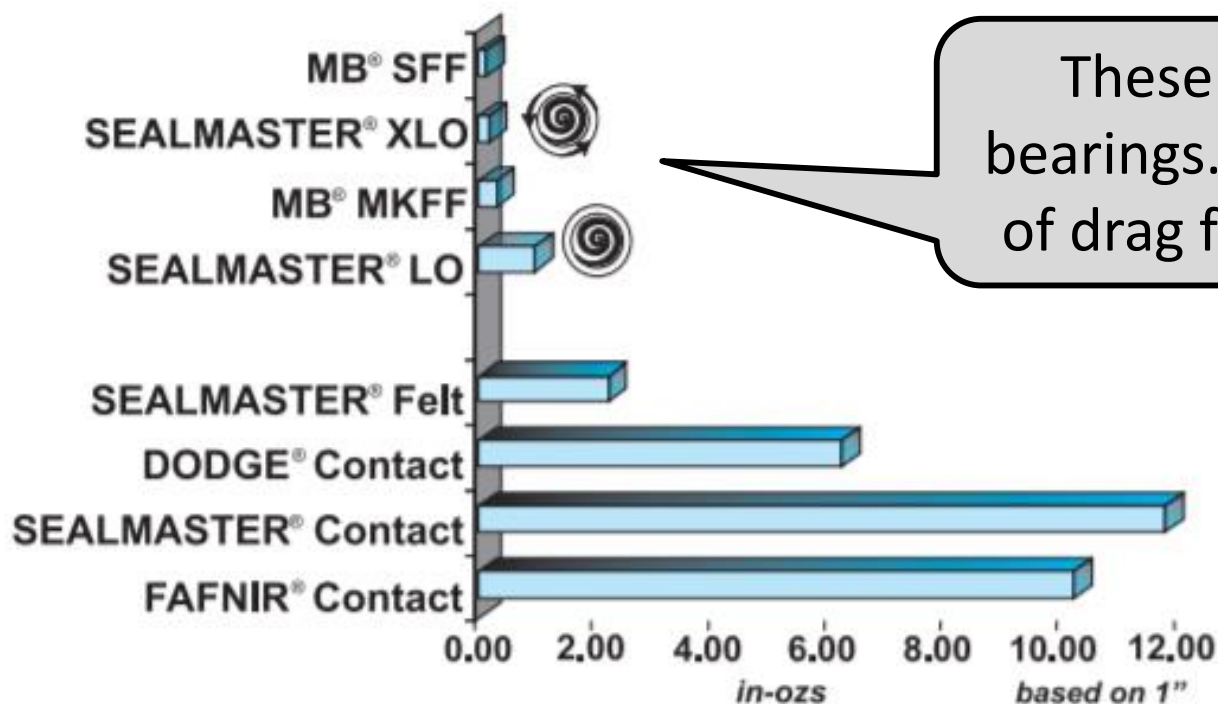
X-Tra Low Drag “XLO”

- Felt seal cut down and oil soaked for extra low drag
- Oil lubricated
- Operating temperature from -30°F to 100°F
- For dry and dusty environments
- A reduced housing fit to allow for the bearing to misalign more easily in the housing



Low Drag Bearing Example: SealMaster

BEARING ROTATIONAL TORQUE COMPARISON CHART



Courtesy of Sealmaster



Low Drag Bearing Example: Componex WINertia

WINERTIA IDLER APPLICATION ANALYSIS

Parameters (Customer to fill out gray sections)

Web Tension	1.00 lb/in
Web Width	60.00 in
Wrap Angle	90 deg
Nip Load	0 lb

Roller Data

Section Design	6.0 x 3 Std
Roll Face	66.00 in
Sectional Area	7.239 in ²
Weight	46.6 lbs
Area inertia _{xx}	24.225 in ⁴
Radius of Gyration, k _z	2.587 in
Rotational Inertia, Wk ²	311.8 lb-in ²
Resultant Tension Force	84.9 lbs
Defl by Gravity	0.0007 in
Defl by Tension	0.0014 in
Defl by nip	0.0000 in
Total Deflection	0.0021 in

Allowable Roller Deflection¹

Class A - precision or nipped	0.0053 in
Class B - most applications	0.0099 in
Class C - flexible materials	0.0198 in
Class D - conveyors & cores	0.0396 in

Shaft Data

Shaft Type	1.5" solid 1045 hard chrome pltd
Shaft Length	72.00 in
Shaft OD	1.500 in
Shaft ID	0.000 in
Sectional Area	1.767 in ²
Weight	36.13 lbs
Area Inertia _{xx}	0.249 in ⁴
Total idler load	131.4 lbs
Defl by Gravity	0.0236 in
Defl by Idler Load	0.0171 in
Total Deflection	0.0407 in
Brg Misalign by Weight	0.059 deg
Brg Misalign by IdlerLoad	0.050 deg
Total Brg Misalignment	0.109 deg

Note: Total deflection and misalignment results are worst case scenario, where all loads act in the same direction and sum together.

Deflection Calculator
(This calculator has been removed from Componex.net)

Componex calculates deflection angles at the bearing of both shell and shaft. To ensure long bearing life, rollers are designed to have less than a 1/4 degree of misalignment between shaft and shell.

¹ Recommendations published by David Roisum, PhD.

Idler Bearing Performance

Smaller bearings have less drag.

All bearing-roller assemblies have 7:1 performance variation.

	Bearing Inner Dia.	Bearing Drag (mode of pop.)	Bearing Drag Range
Bearing A	5 mm [0.197 in.]	0.56 N-mm [0.005 lbs-in] [0.08 oz-in]	0.23-1.69 N-mm [0.002-0.015 lbs-in]
Bearing B	12 mm [0.472 in.]	3.39 N-mm [0.030 lbs-in] [0.5 oz-in]	2.15-14.8 N-mm [0.019-0.131 lbs-in]
Bearing C	22 mm [0.875 in.]	7.91 N-mm [0.070 lbs-in] [1.1 oz-in]	4.52-32.6 N-mm [0.040-0.289 lbs-in]

Bearing Life Calculations

With small wrap angle and load, small bearings are predicted to have long life.

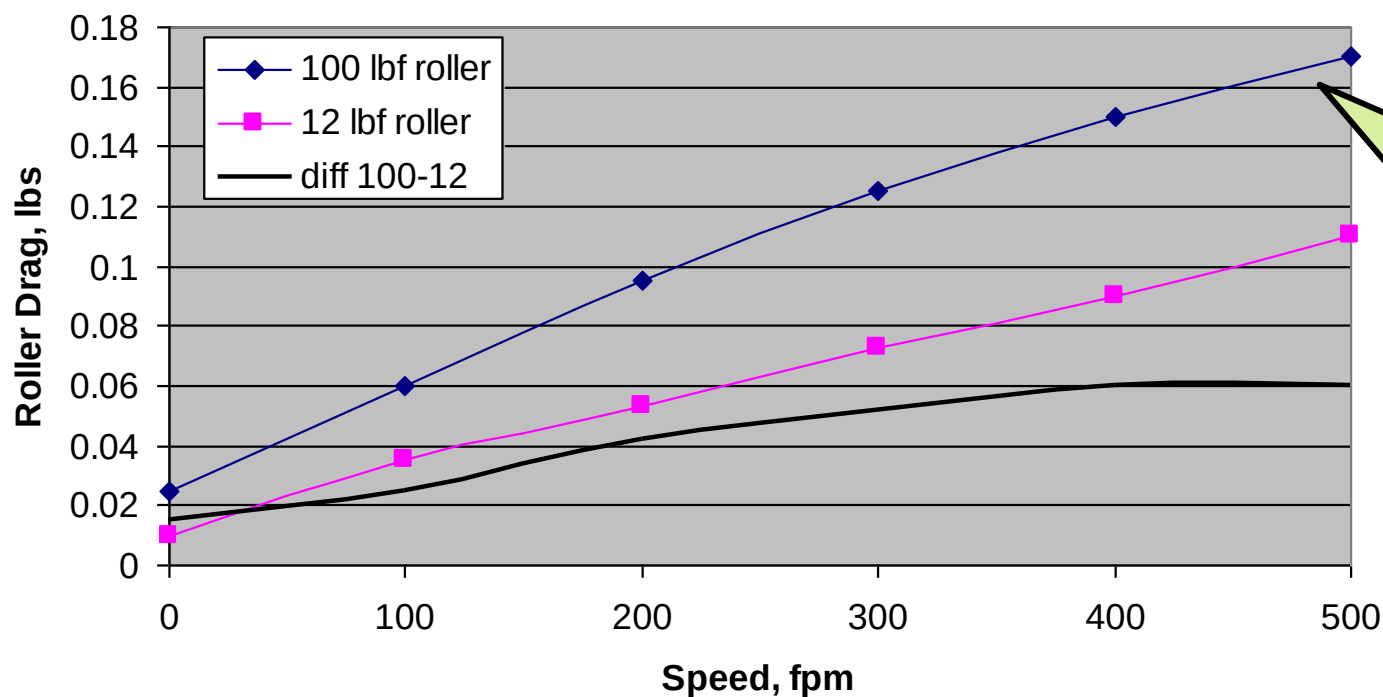
Beware, with higher load, small bearing life is too short.

Parameter	Bearing A (low wrap)	Bearing A (high wrap)	Bearing B	Bearing C
Wrap Angle	5 degr.	180	90	180
Static Load Rating	1600 N	1600	3780	9140
Equivalent Radial Load	32 N	253	187	258
Bearing Life, L_{10}	539 yrs	1.1 yrs	37	197

Bearing Drag vs. Radial Load



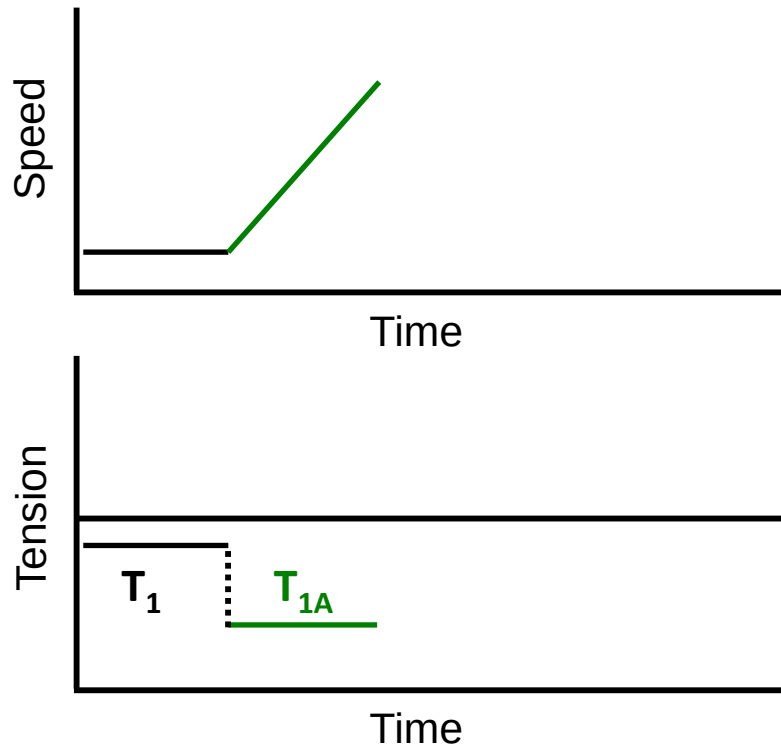
Bearing Drag vs. Radial Load (Roller Weight)



Roller drag may not increase proportional to load.

Idler Roller Tension Change

During acceleration, roller inertia adds to bearing drag to create the greatest losses to rollers.



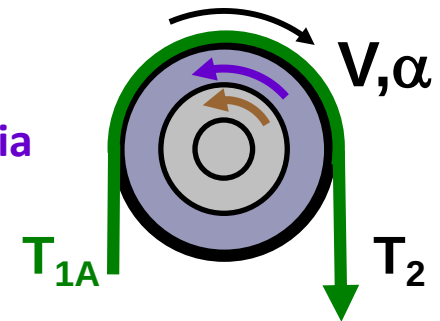
*Assume constant T_2

$$\Delta T = \frac{M_B + M_A}{rw}$$

$$M_A = I\alpha_A$$

$$\alpha_A = \frac{\Delta v}{rt_A}$$

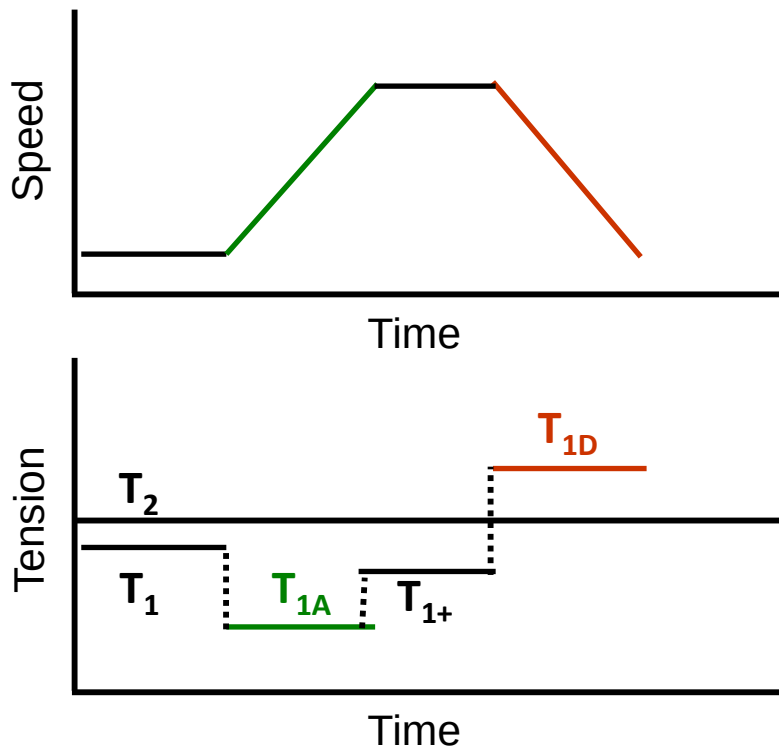
Accelerating
 $\Delta T = \text{Drag} + \text{Inertia}$



- α_A Process acceleration rate (radians/s²)
- Δv Process speed (m/s)
- r Roller radius (m)
- M_A Torque required to accelerate roller mass (N-m)

Idler Roller Tension Change

In deceleration, roller inertia drives the roller forward, opposite of bearing drags.



*Assume constant T_2

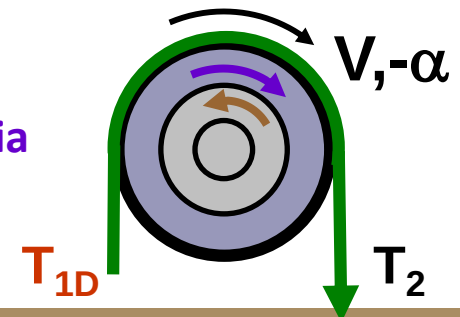
$$\Delta T = \frac{M_B - M_A}{rw}$$

$$M_A = I\alpha_A$$

$$\alpha_A = \frac{\Delta v}{rt_A}$$

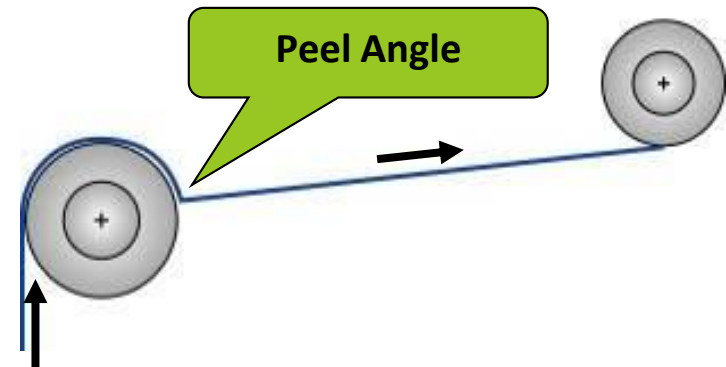
Under extreme conditions, deceleration inertia can create slack web.

Decelerating
 $\Delta T = \text{Drag} - \text{Inertia}$



Adhesion and Peeling

- ▶ Force is required to separate web from roller,
- ▶ Peel angle determined by adhesive force and web tension.
 - ▶ If peel force is zero, web releases tangentially (angle is zero).
 - ▶ If adhesive peel force is greater than tension, the web will fail to peel and wrap the roller.
- ▶ CD peel variations creates uneven tension, leading to shifting or wrinkling.
- ▶ Special roller release surfaces drop adhesive force.



$$\theta = \cos^{-1} \left(\frac{T_{ADH}}{T_{WH}} \right)$$

T_{ADH} = Peel tension (force per width)

T_{WH} = Web handling tension (force per width)

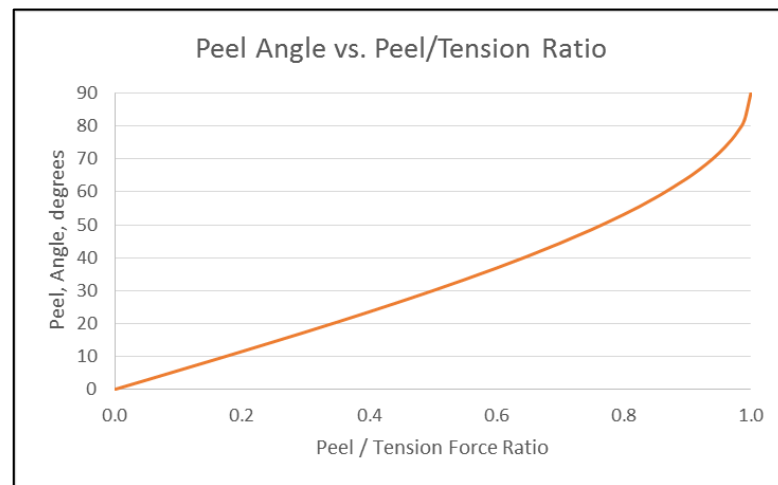
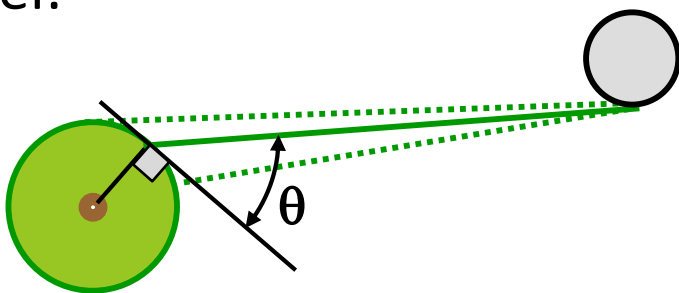
θ = Peel angle, radians

Adhesion and Peeling

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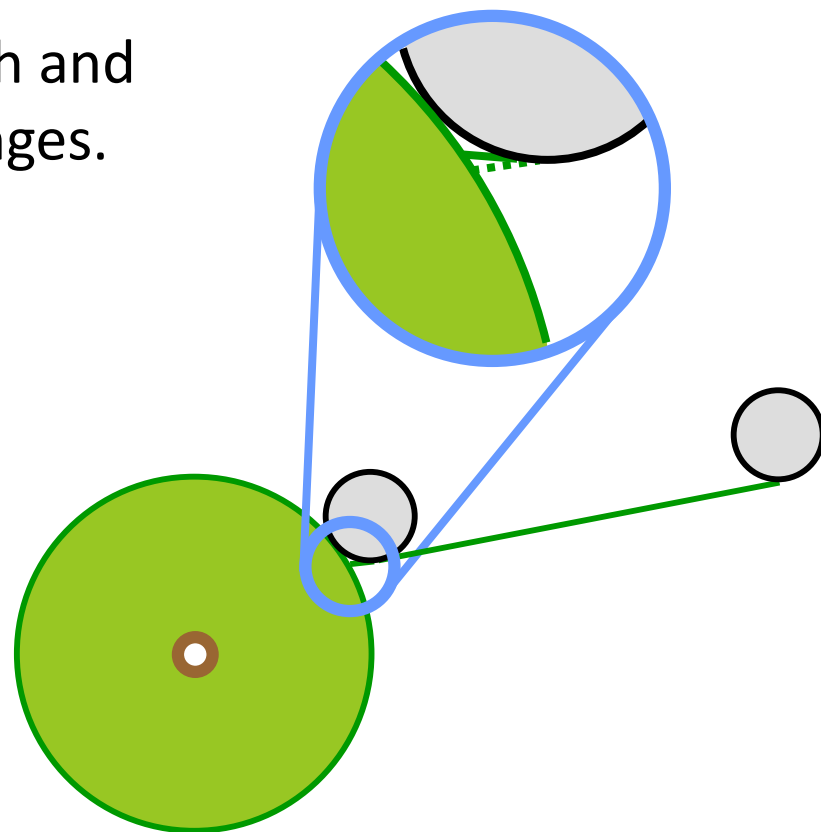
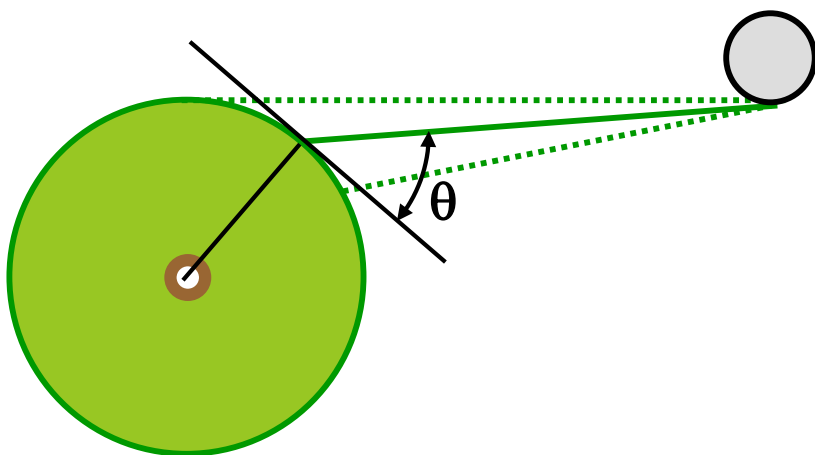
T_{WH} = Web handling tension (force per width)

θ = Peel angle, radians

Peel Angle and Peel Roller

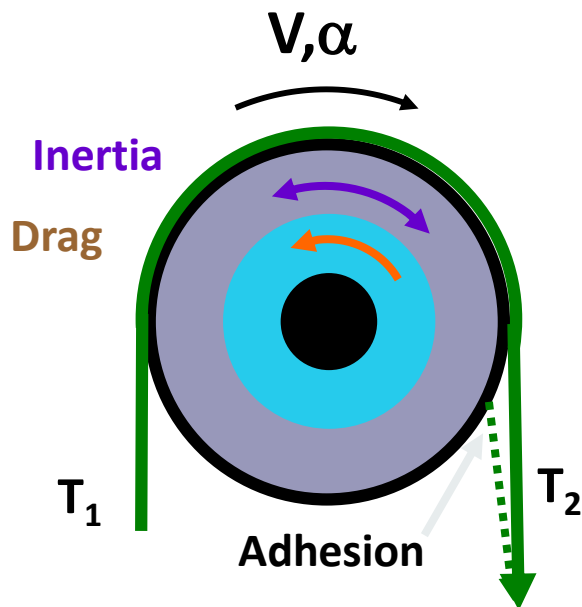
Peel angle will jump with variations in ratio of the forces of adhesion and tension. Without a peel controlling roller, this creates large web path variations.

A peel roller greatly reduces path length and tension variations from peel angle changes.
(Ignoring deflection, smaller is better.)



Adhesion and Idler Rollers

Any web-to-roller adhesion will add to the idler roller tension demand.



$$\text{Torque of Drag} = M_B$$

$$\text{Torque of Inertia} = I\alpha$$

$$\text{Torque of Adhesion} = f_{\text{peel}}r$$

$$\Sigma M = 0$$

$$T_2rw = T_1rw + M_B \pm I\alpha + f_{\text{peel}}r$$

$$\Delta T = \frac{M_B \pm I\alpha + f_{\text{peel}}r}{rw}$$

Roller Release Surfaces

Common “Release” Roller Surfaces:

- Teflon-Elastomer
 - Silicone Wrap or Paint
 - Plasma Sputtered
 - Sandblasted Surface
 - Release Liner Wrap
 - Chilled Roller
- } Reduced contact area of rough surfaces reduces adhesion.

Roller Release Surfaces



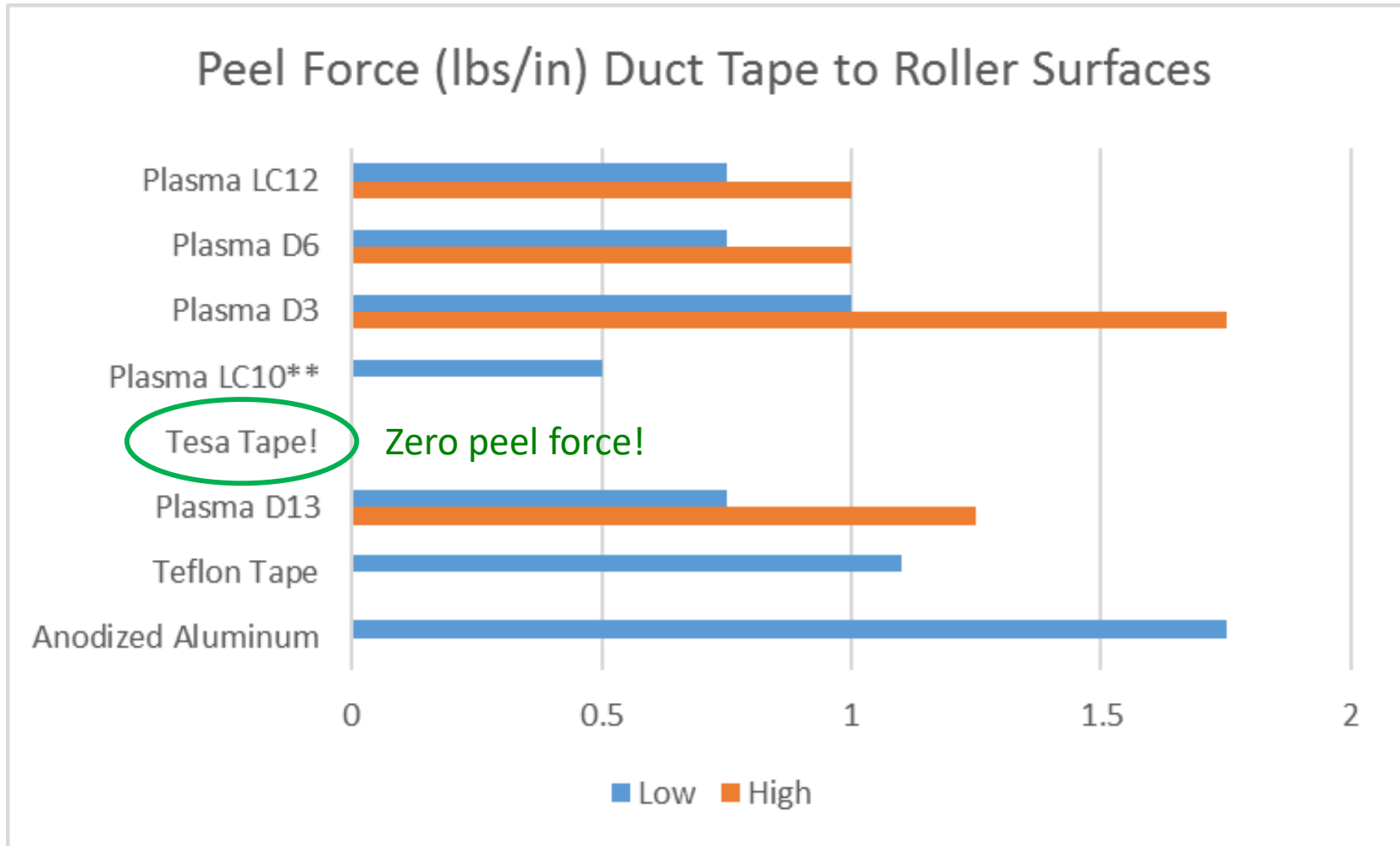
Warning:
Surfaces may lose
release properties
over time.



Glass cloth tape, silicone release,
service temp. 300F 9150C)



Measuring Peel Force of Adhesive Tape to Rollers

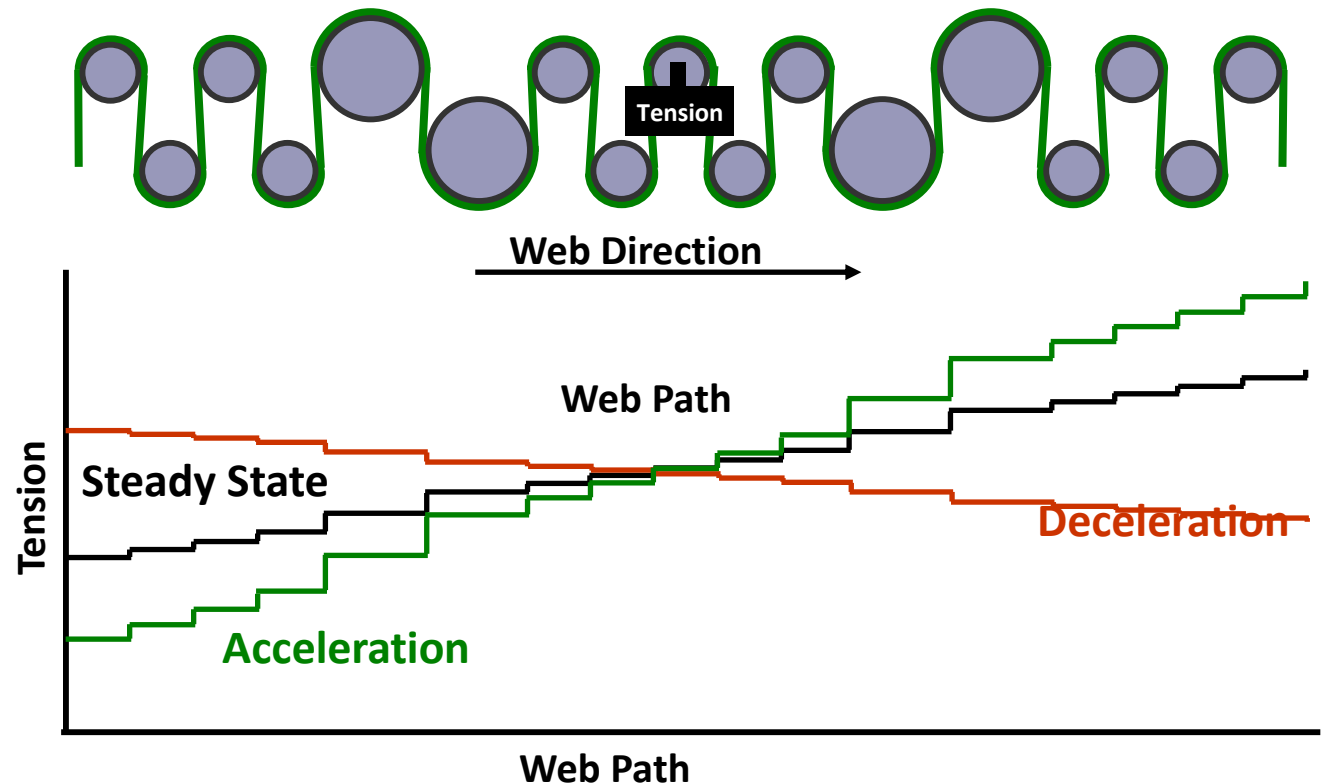
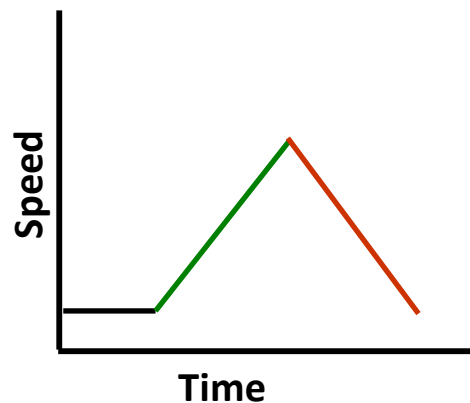


Tension in an Idler Roller Series

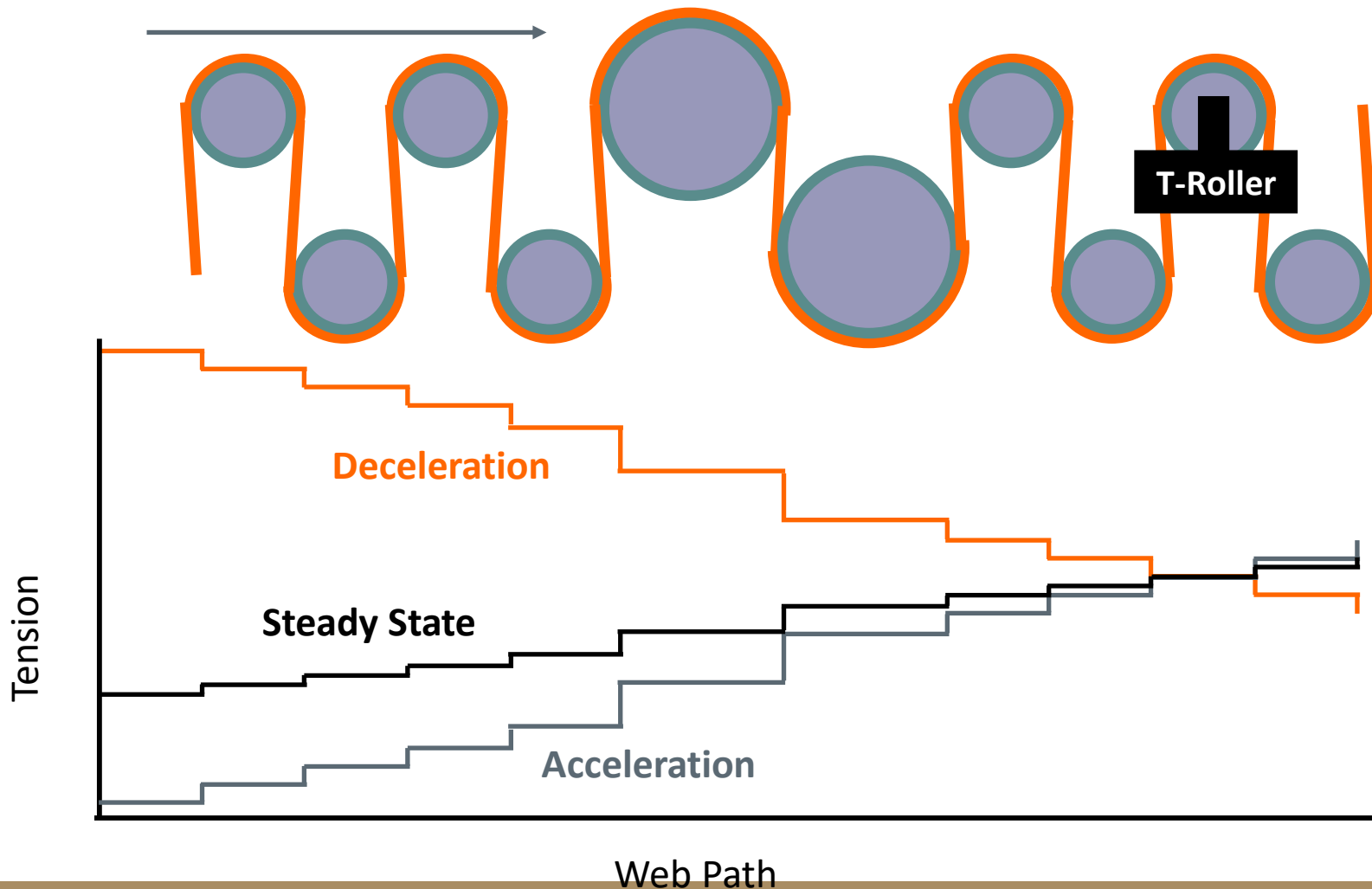
The tension through a series of rollers can change dramatically due to idler roller demands.

During acceleration,
upstream tension will
decrease.

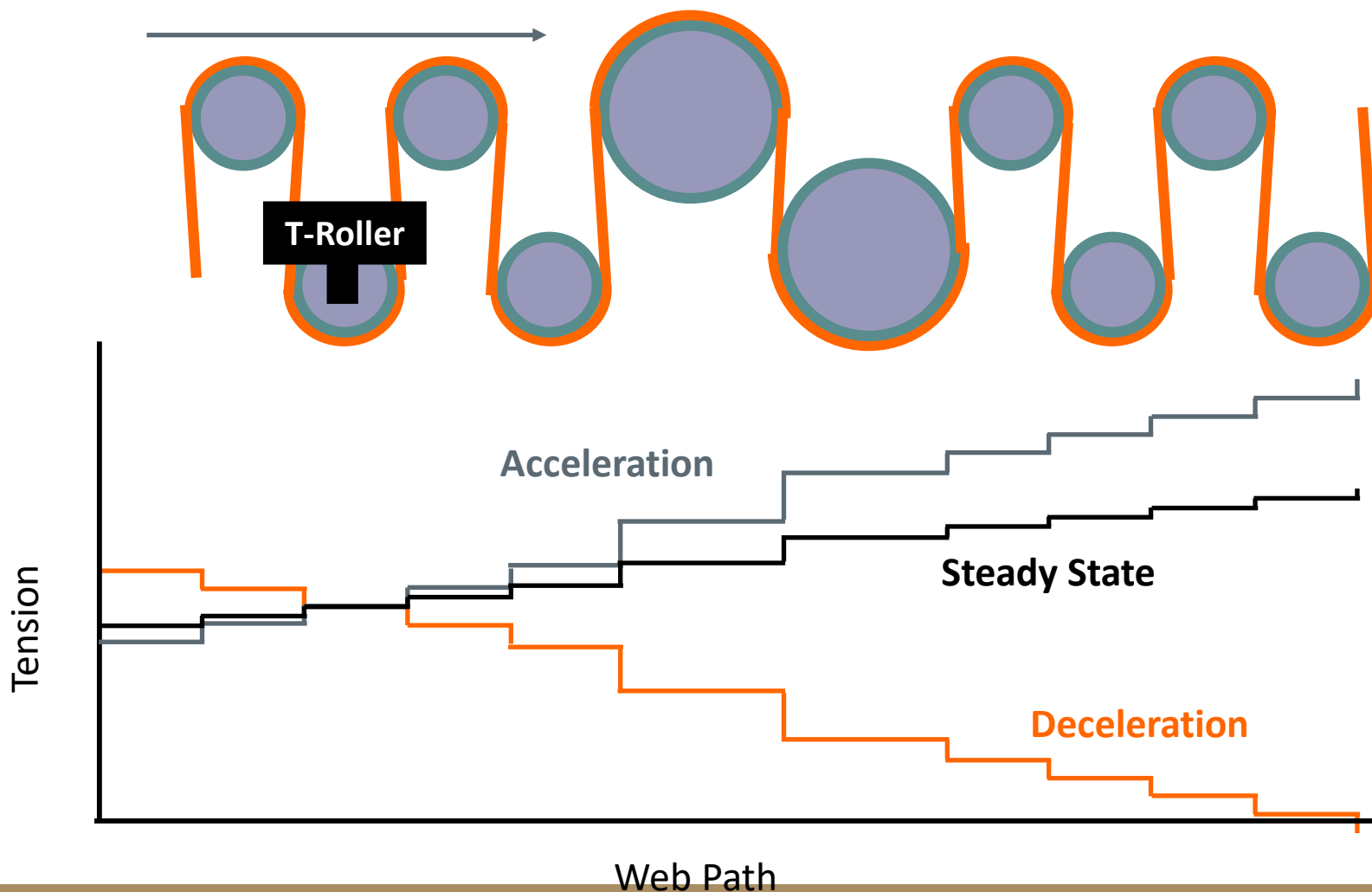
During deceleration,
upstream tension will
increase.



Tension Drop in Roller Series 1



Tension Drop in Roller Series 2



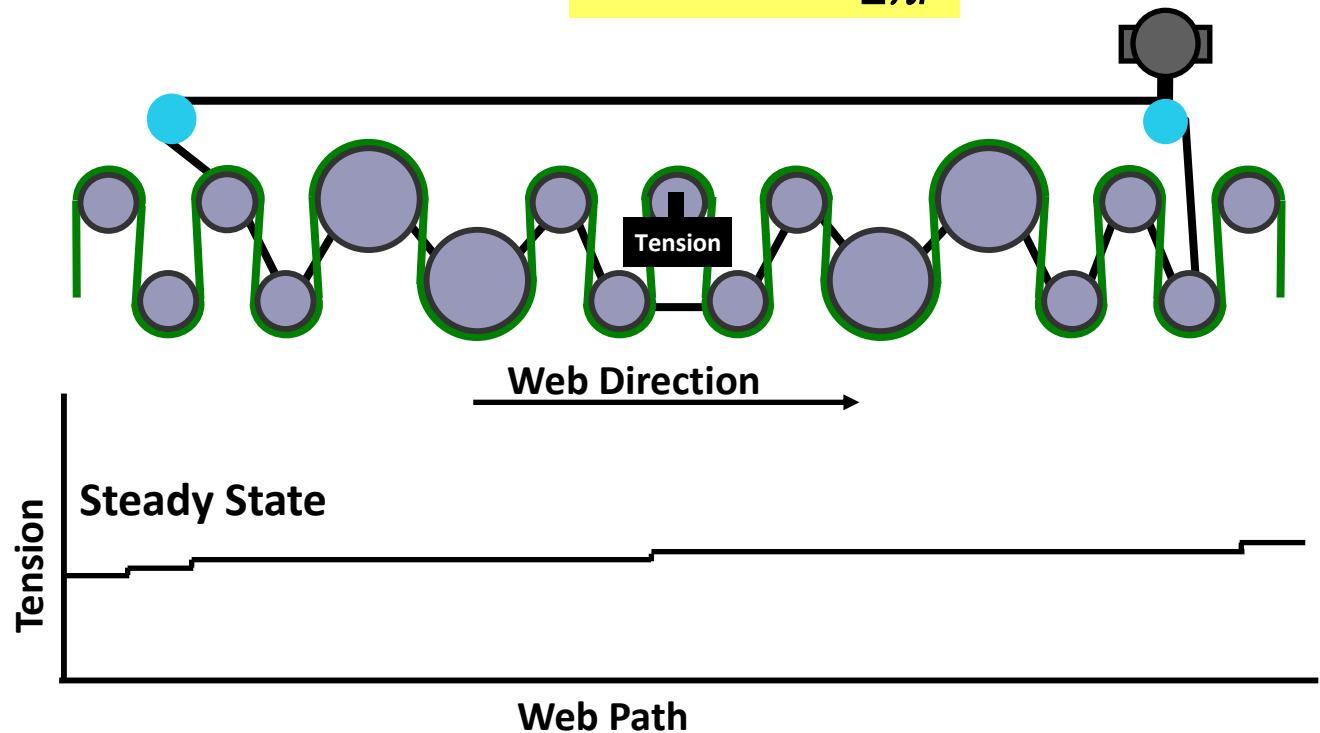
Tendency Driven Roller Series

The roller shaft is driven, using the internal bearings as a clutch to drive the roller shell to line speed.



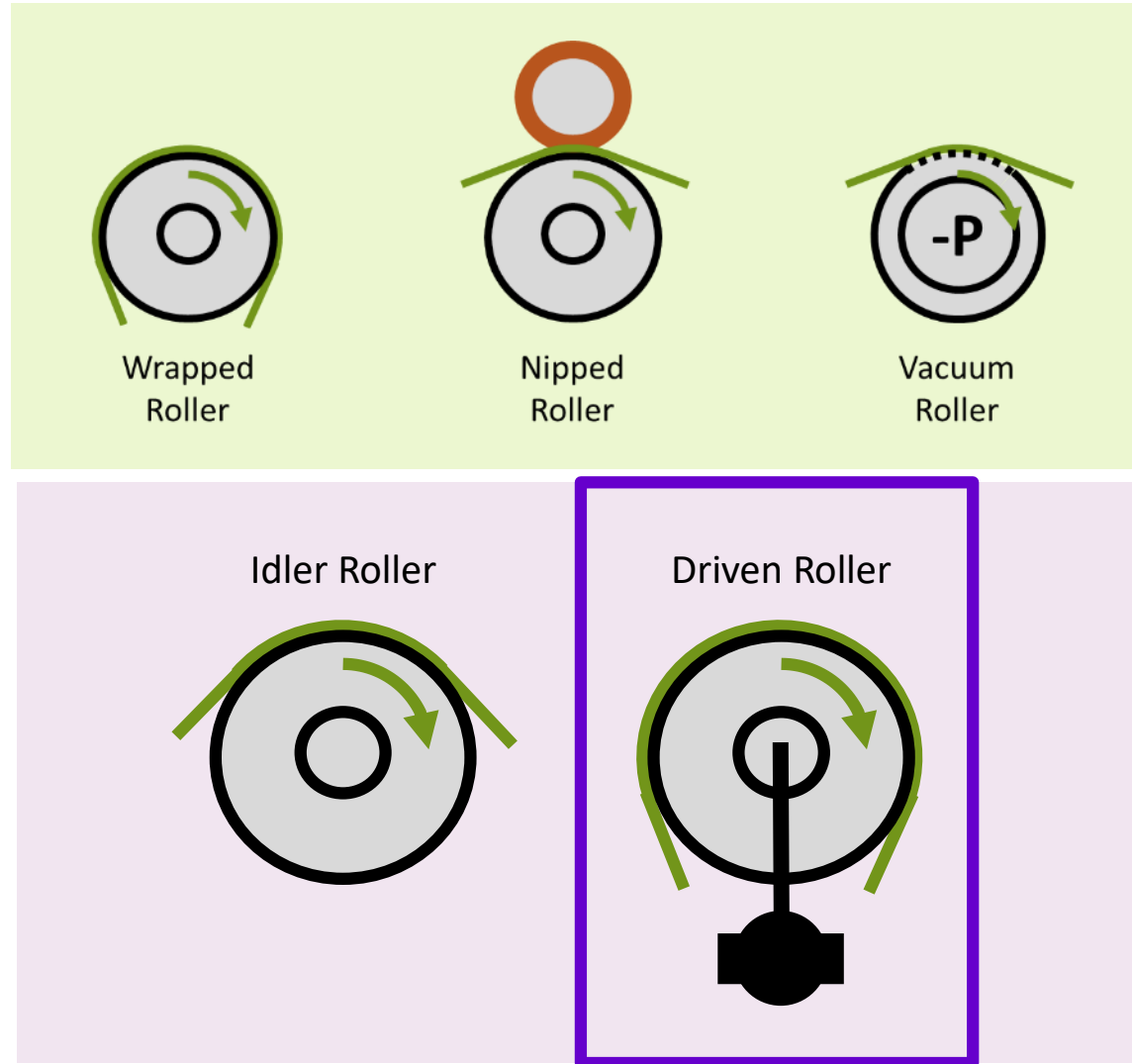
$$n(RPMs) = \frac{V}{2\pi r}$$

Tendency driven rollers prevent scratching and reduce MD tension variations.



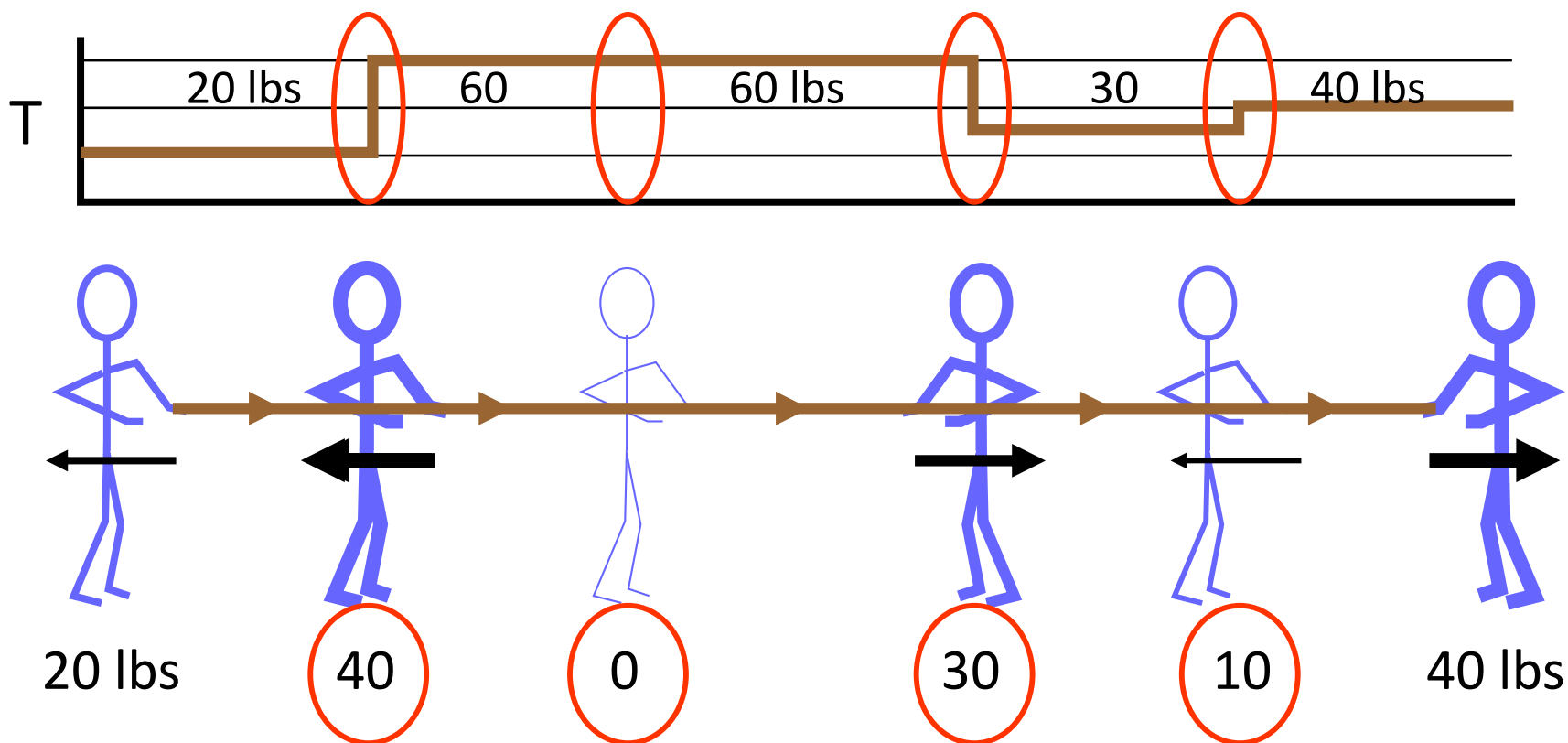
Traction Requirements

$$\text{TSF} = \frac{\text{Available Traction}}{\text{Required Traction}}$$



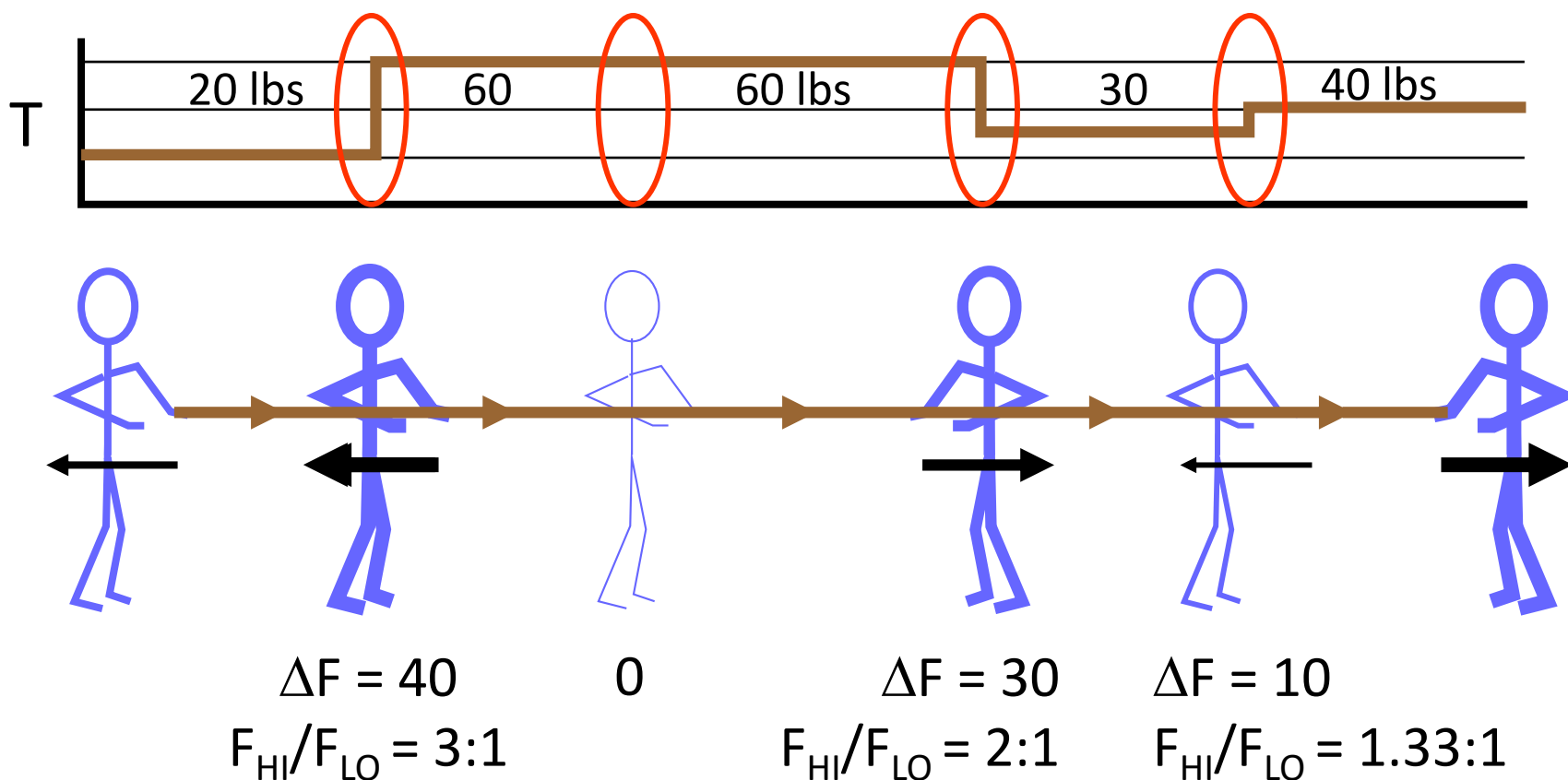
Traction Needed at Driven Rollers

The traction required by a driven roller is the difference in tensions between tension zones.

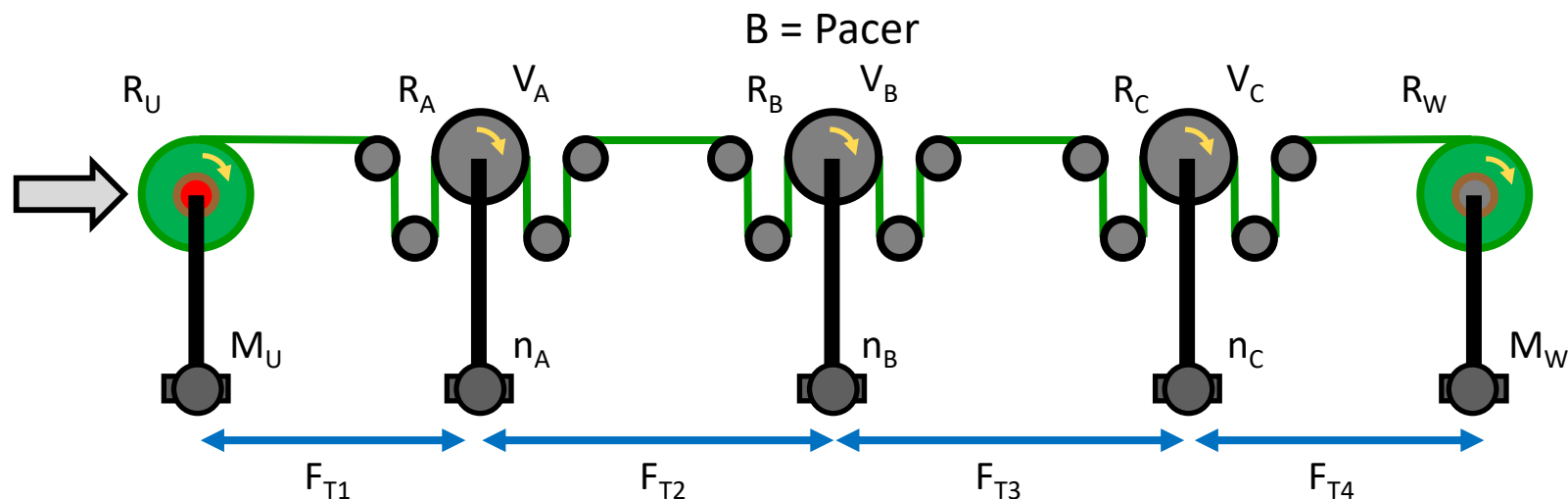


Traction Needed at Driven Rollers

Tension changes can be viewed as an absolute difference or a ratio



Tension and Slip



What determines tension if drive roller A slips?

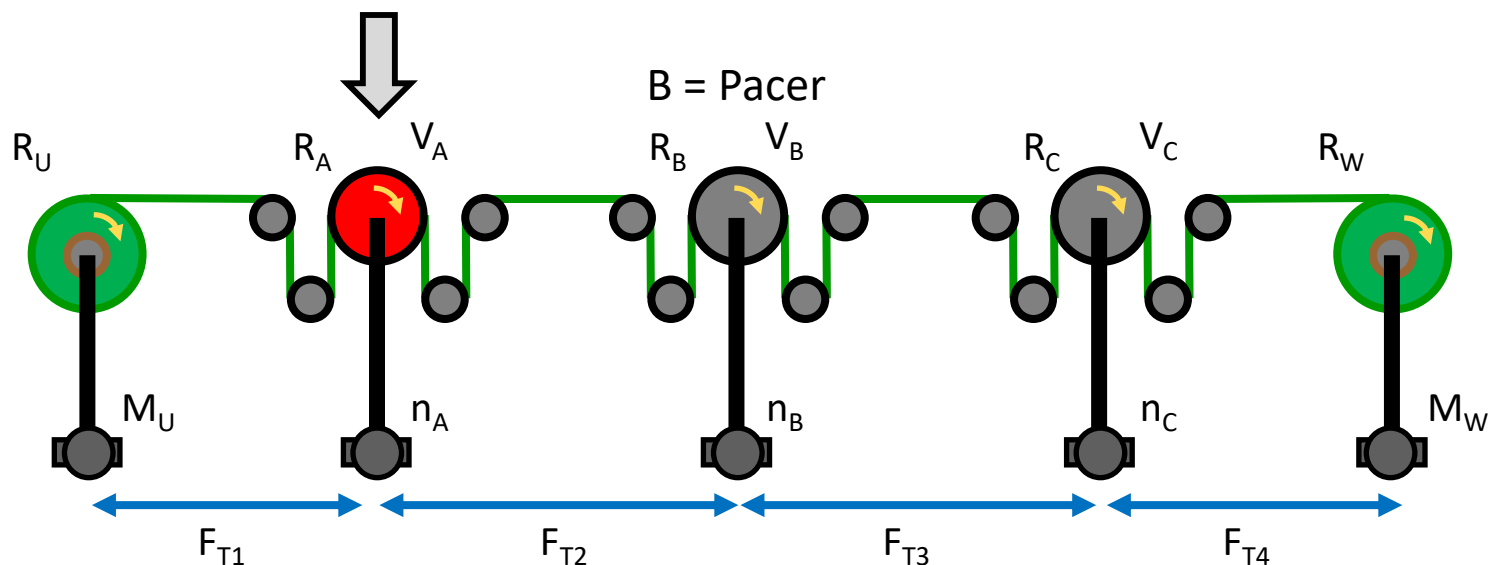
With nearly infinite wrap angle, the unwinding roll will be able to maintain tension as long as the core does not slip on the shaft or chucks.

Side load chucks torque transmission limit: $M_{CAP} = \mu_{CC} F_{SIDE} r$

Pneumatic shaft or chuck torque transmission limit: $M_{CAP} = \mu_{CC} P A r$

** Many chucks and shafts have raised features to bite into paper cores and increase traction beyond friction.

Tension and Slip



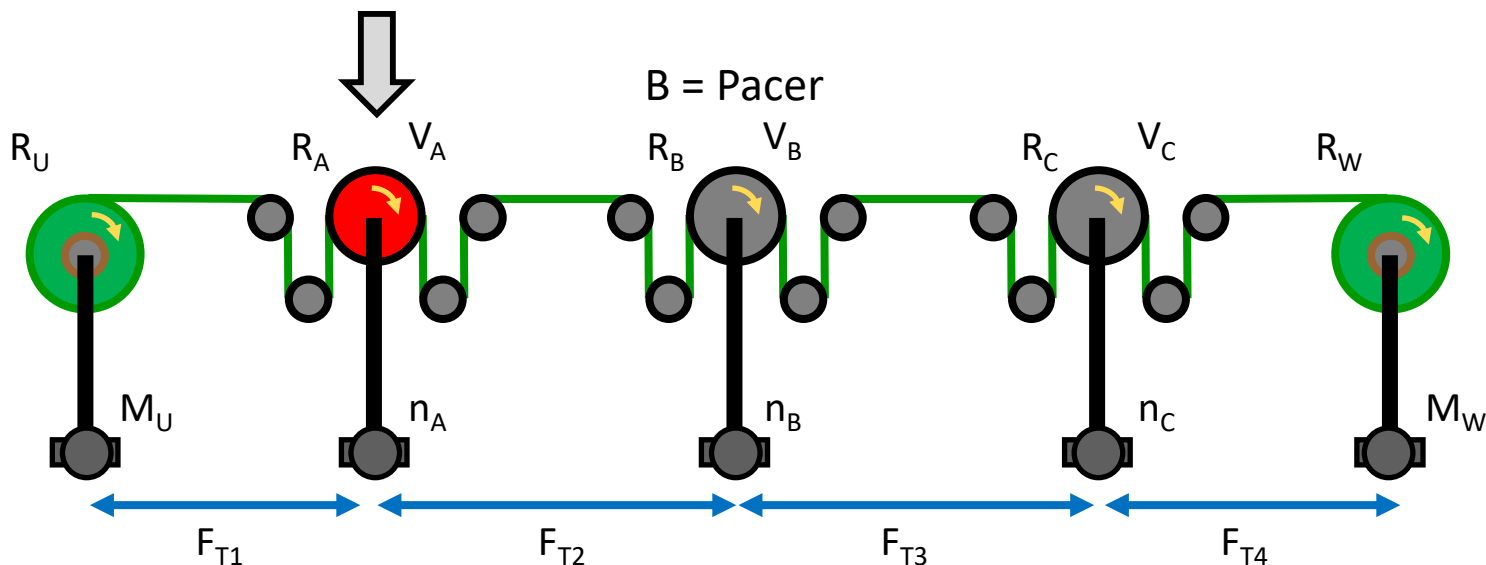
What determines tension if drive roller A slips?

The web will slip when the tension ratio exceeds the belt equation limit.

An wrapped roller will avoid slip if the tension ratio is between these two values determined by traction coefficient and wrap angle.

$$e^{-\mu\theta} < \frac{F_{T2}}{F_{T1}} < e^{\mu\theta}$$

Tension and Slip



What determines tension if drive roller A slips?

Slipping at drive roller A will limit the tension at F_{T2} .

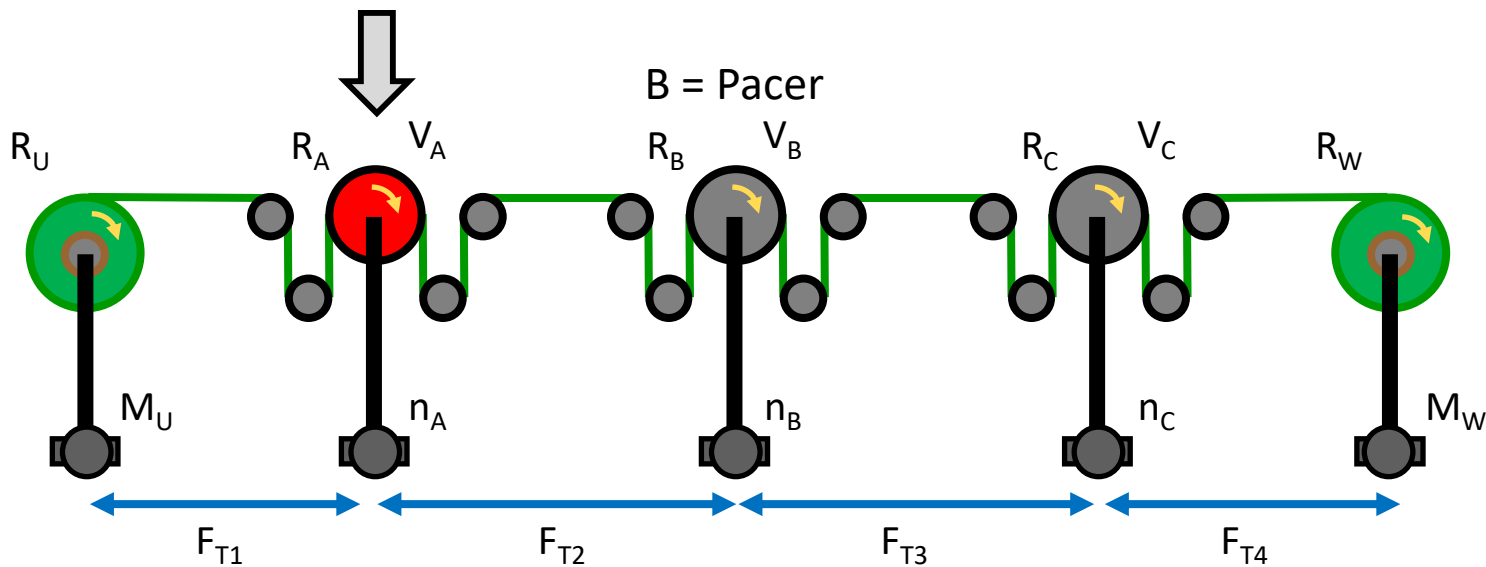
If A slips applying braking torque:

$$F_{T2} = F_{T1} e^{\mu\theta}$$

If A slips applying driving torque:

$$F_{T2} = \frac{F_{T1}}{e^{\mu\theta}}$$

Tension and Slip



What determines tension if drive roller A slips?

Example:

$\mu = 0.25$

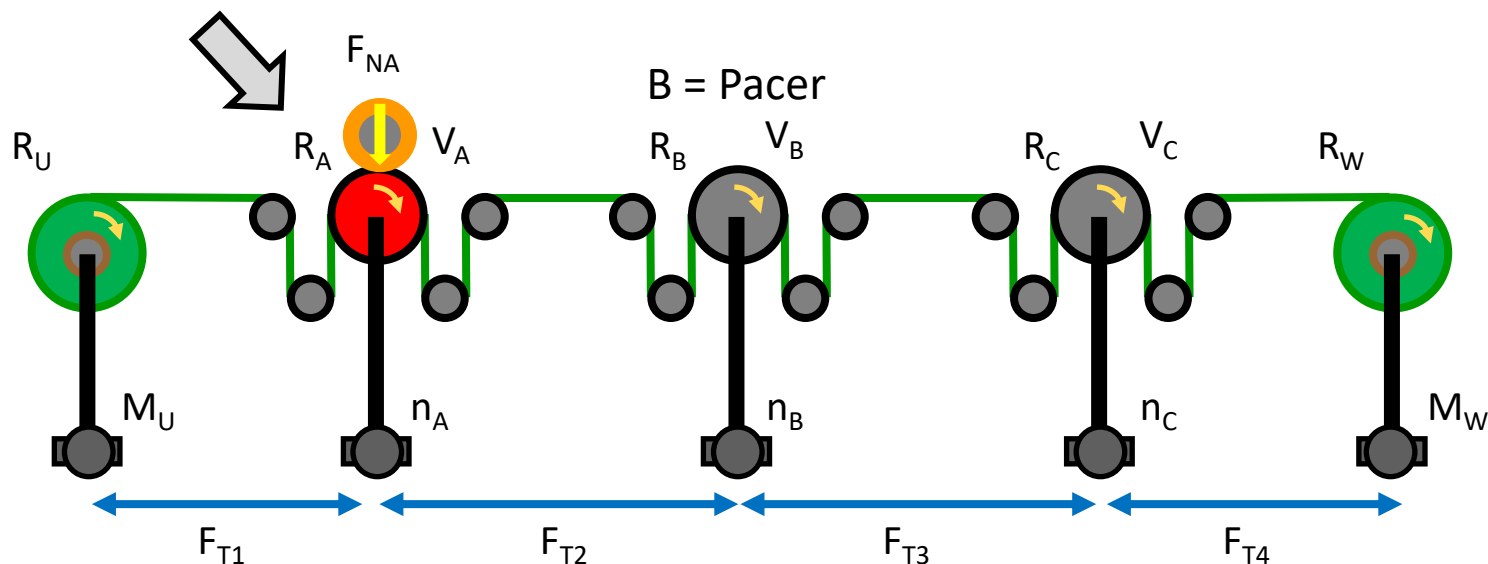
$\theta = 3.14$ radians (180 degr.)

Driven roller A will slip if it attempts to set F_{T2} to less than $0.46 F_{T1}$ or more than $2.19 F_{T1}$.

$$e^{-\mu\theta} < \frac{F_{T2}}{F_{T1}} < e^{\mu\theta}$$

$$0.46 < \frac{F_{T2}}{F_{T1}} < 2.19$$

Tension and Slip



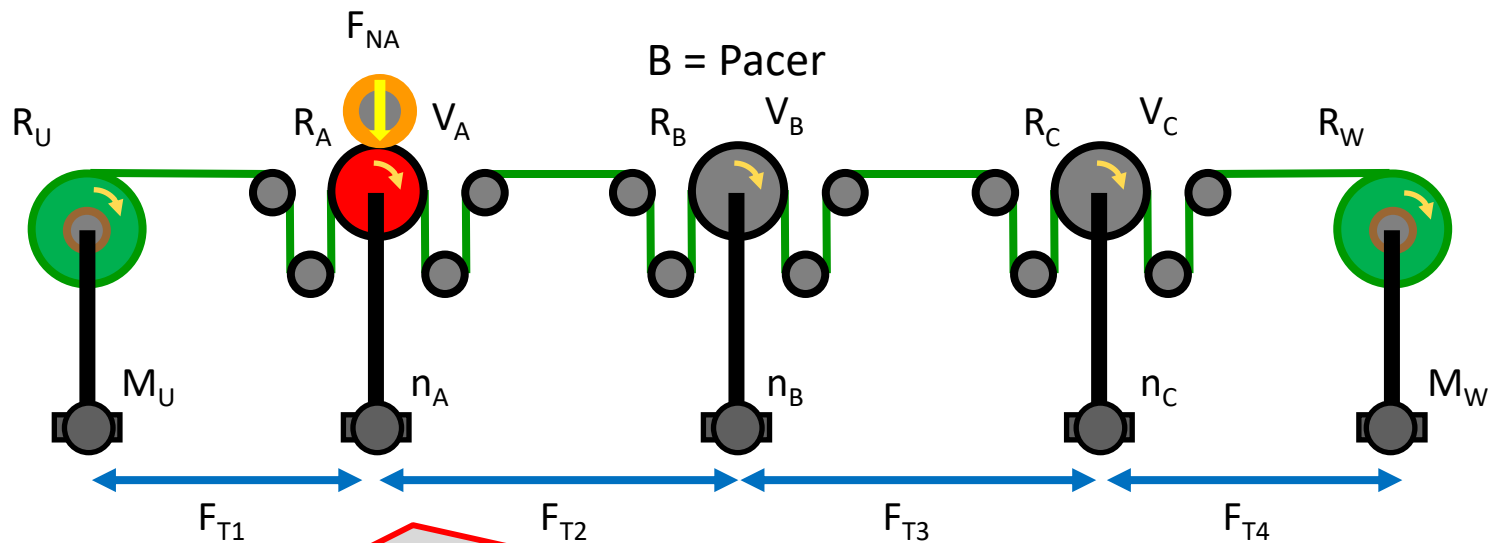
What determines tension if drive roller A slips?

Nipped rollers can create high traction and greatly reduce the risk of slipping. Nipped rollers with pre-nip or post-nip wrap angle will have even more slip-preventing traction.

Nip friction, $F_{NF} = \mu F_{NX}$

Example: $F_{NX} = 450 \text{ lbs}$, $\mu = 0.3$, $F_{NF} = 135 \text{ lbs}$
 $F_{NX} = 200 \text{ kgf}$, $\mu = 0.3$, $F_{NF} = 60 \text{ kgf}$

Tension from Speed Ratios (Draw)



! Nipped rollers can provide excellent no-slip control, but may cause other process problems, including: increase wrinkling with non-perfect (baggy) webs, lateral shifting, safety concerns, and performance changes and cost associated with wear of rubber coverings.

Tension Control and Slip

How will a tension control system behave during slip?

- ▶ Torque, draw, and tension control systems will all slip under the same conditions.
- ▶ **Braking torque:** The roller will slip and slow down to maximum under speed or stop.
- ▶ **Driving torque:** The roller will slip and speed up to maximum over speed.
- ▶ **Draw mode:** The roller will run at the programmed draw speed but the web will micro-slip over the entire wrap angle. This will be difficult to detect, except from micro-scratching, roller wear, or friction-limited tension (which may go undetected as most draw zones do not measure tension).

Tension Control and Slip

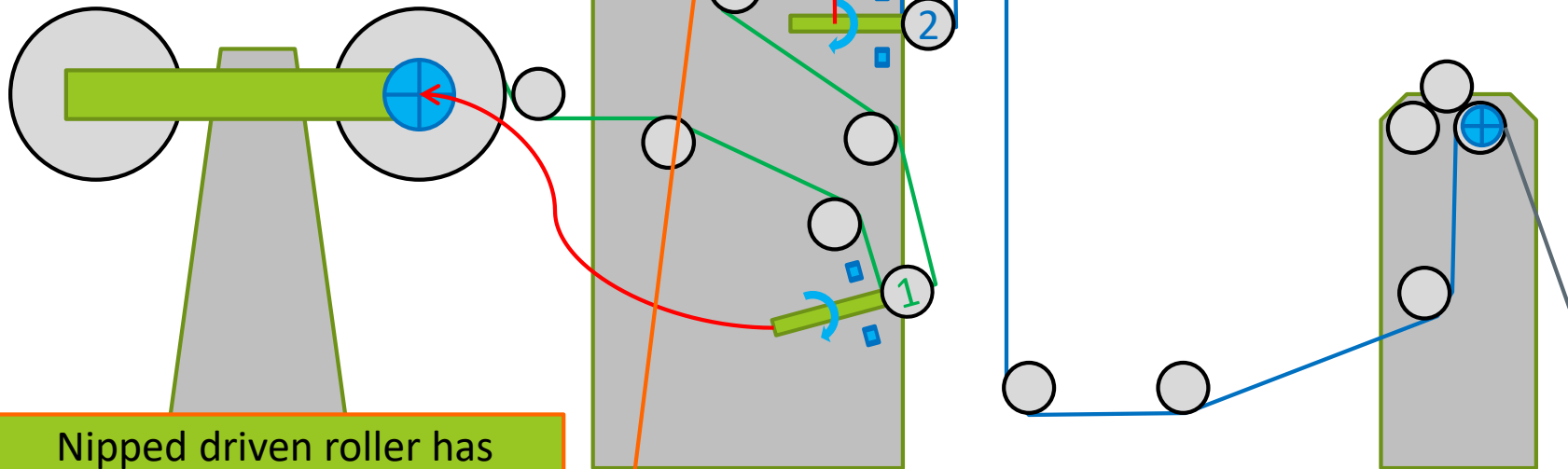
How will a tension control system behave during slip?

- ▶ **Closed-loop control mode:** Closed-loop control, whether controlling motor torque or speed, driven follower rollers should have a speed range trim limit of +/- 5-10% of master line speed.
- ▶ Slipping followers will slow down or speed up no more than 5-10% of line speed.
- ▶ Slipping pacer rollers can lead to confusing tension control behavior.
- ▶ A upstream or downstream follower will become the default master speed control, running the process at 5-10% over or under speed set point.
- ▶ Follower pairs running at maximum speed trim may lose their ability to control tension with potential to cause slack tension.

No Slip on Nipped Drive Roller

Unwind tension (T_1) is set by dancer #1 feedback. This is set low to avoid cinching/telescoping low friction unwinding rolls.

Infeed tension (T_2) is set by dancer roller #2 feedback. The preference is to have high tension here to pull out bagginess into print station 1.



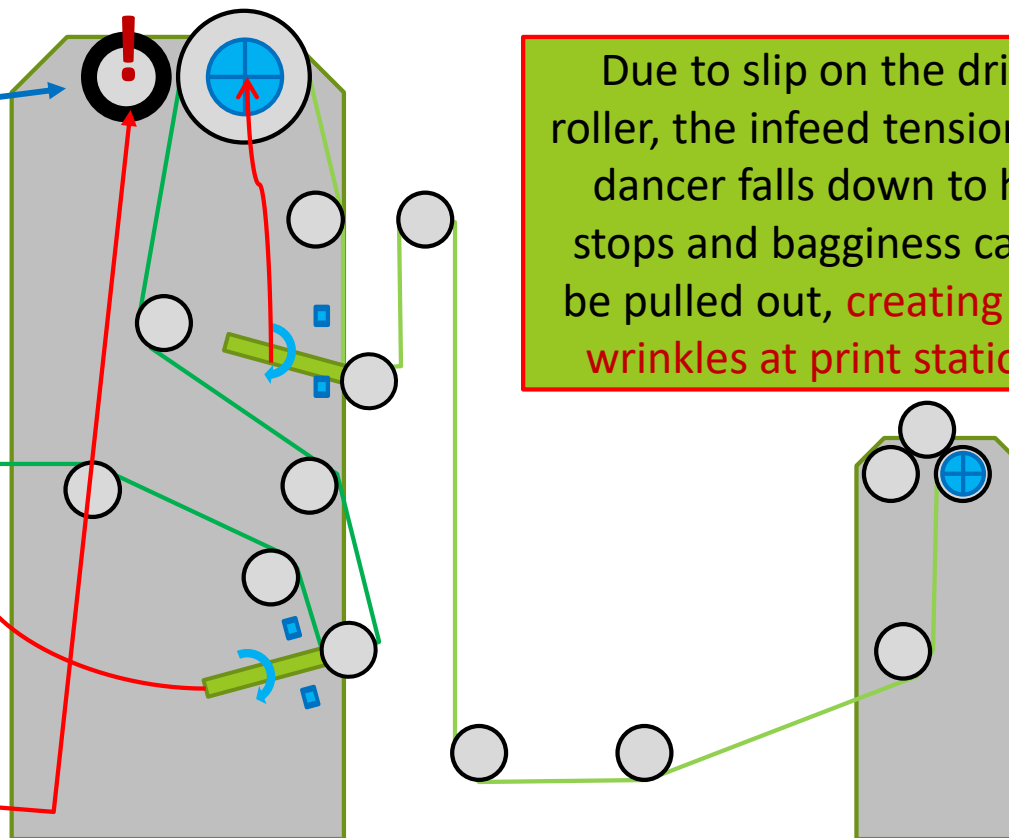
Nipped driven roller has sufficient traction to prevent slip between low unwind and high infeed tensions.

Slip on Unnipped Driven Roller

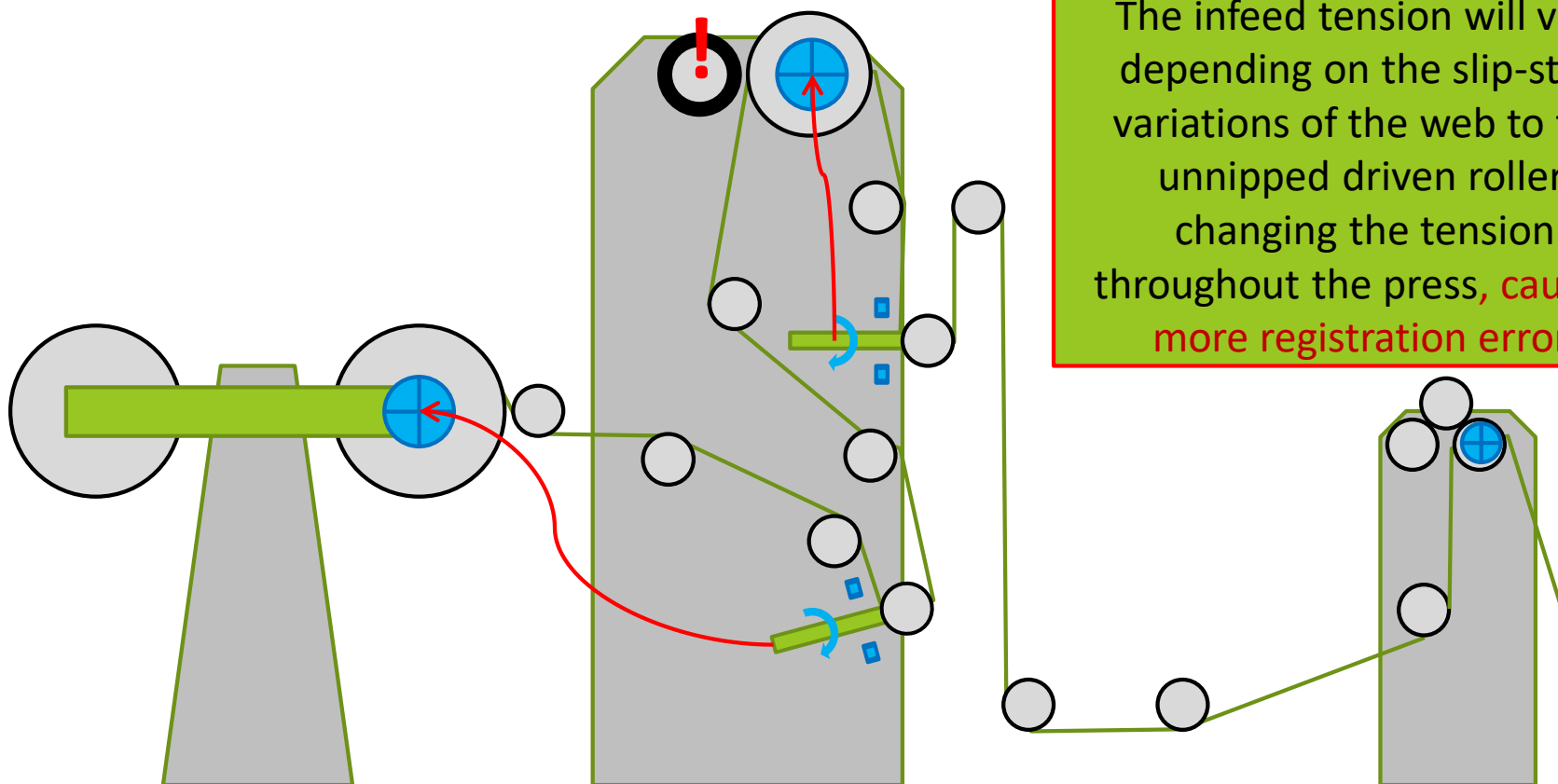
Nip is opened to avoid wrinkles with baggy high modulus webs.

Due to slip on the driven roller, the infeed tension zone dancer falls down to hard stops and bagginess cannot be pulled out, **creating more wrinkles at print station 1.**

The open nip driven roller lubricates with air at high speed/low tension conditions. The infeed tension control loop slows down the driven roller, but it slips and can't meet the high tension set point.



Slip on Unnipped Driven Roller



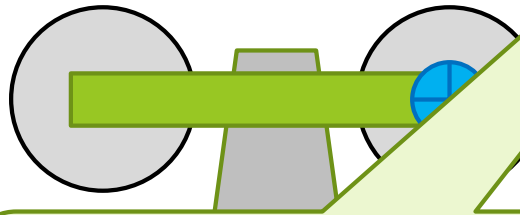
The infeed tension will vary depending on the slip-stick variations of the web to the unnipped driven roller, changing the tension throughout the press, **causing more registration error.**

Control with Textured Roller Surface

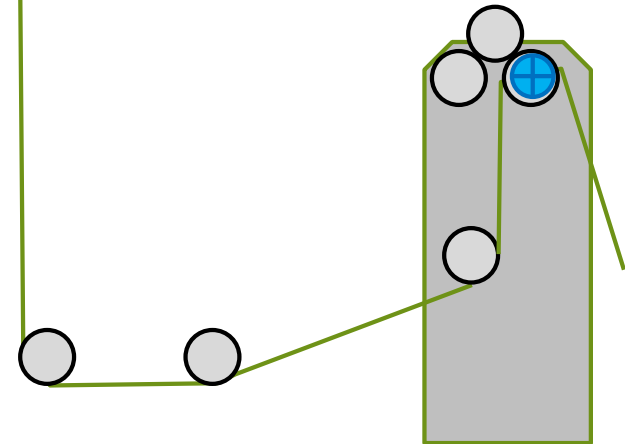
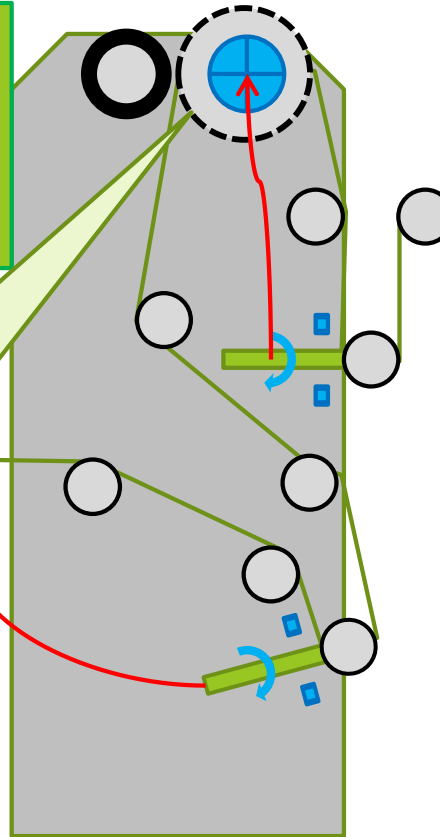
Remedy:

Grooving or increased roughness of the driven roller can prevent lubrication and maintain traction.

Using a rubber or other high friction surface (plus grooving / roughness) will provide the greatest traction, allowing for low unwind and high infeed tension



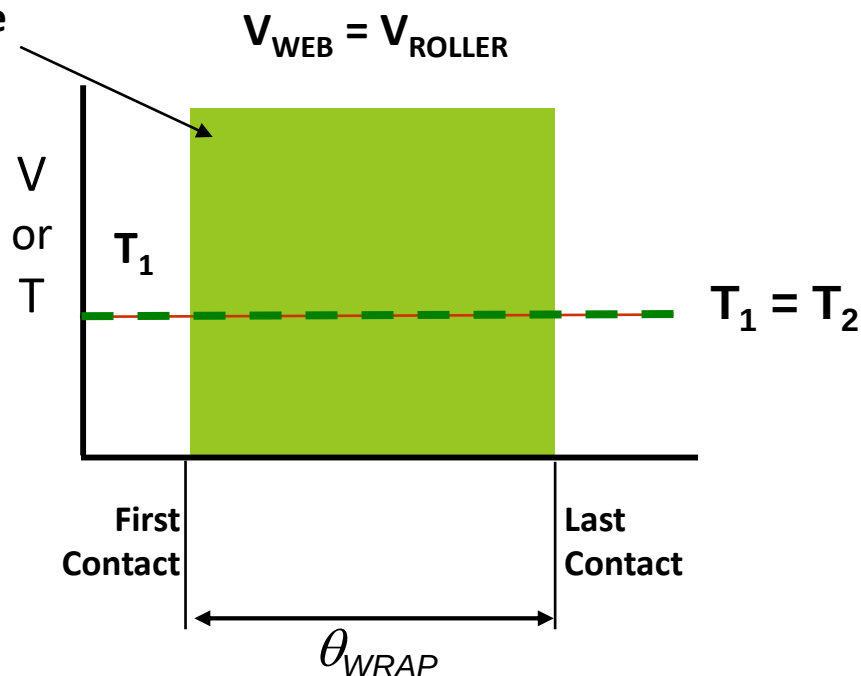
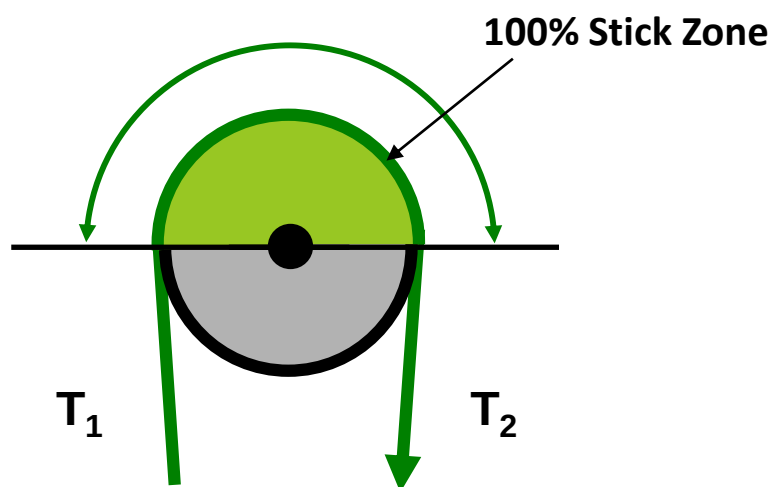
We proved the benefits of changing roller material and roughness in a short trial using masking tape on the driven roller.



Benefits: Control of cinching, less print station 1 wrinkles from baggy web, better registration control.

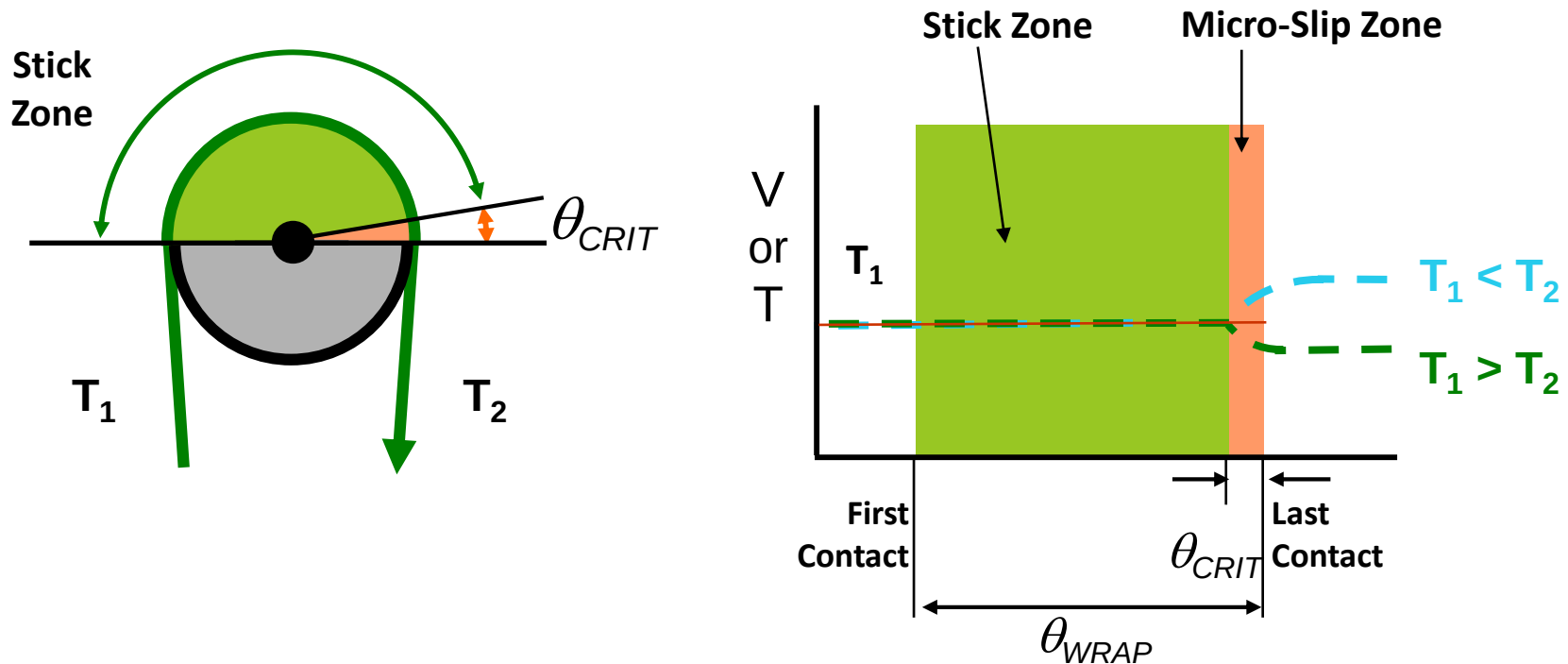
Stick

If $T_1 = T_2$, the web and roller have equal surface speed.



Micro-Slip

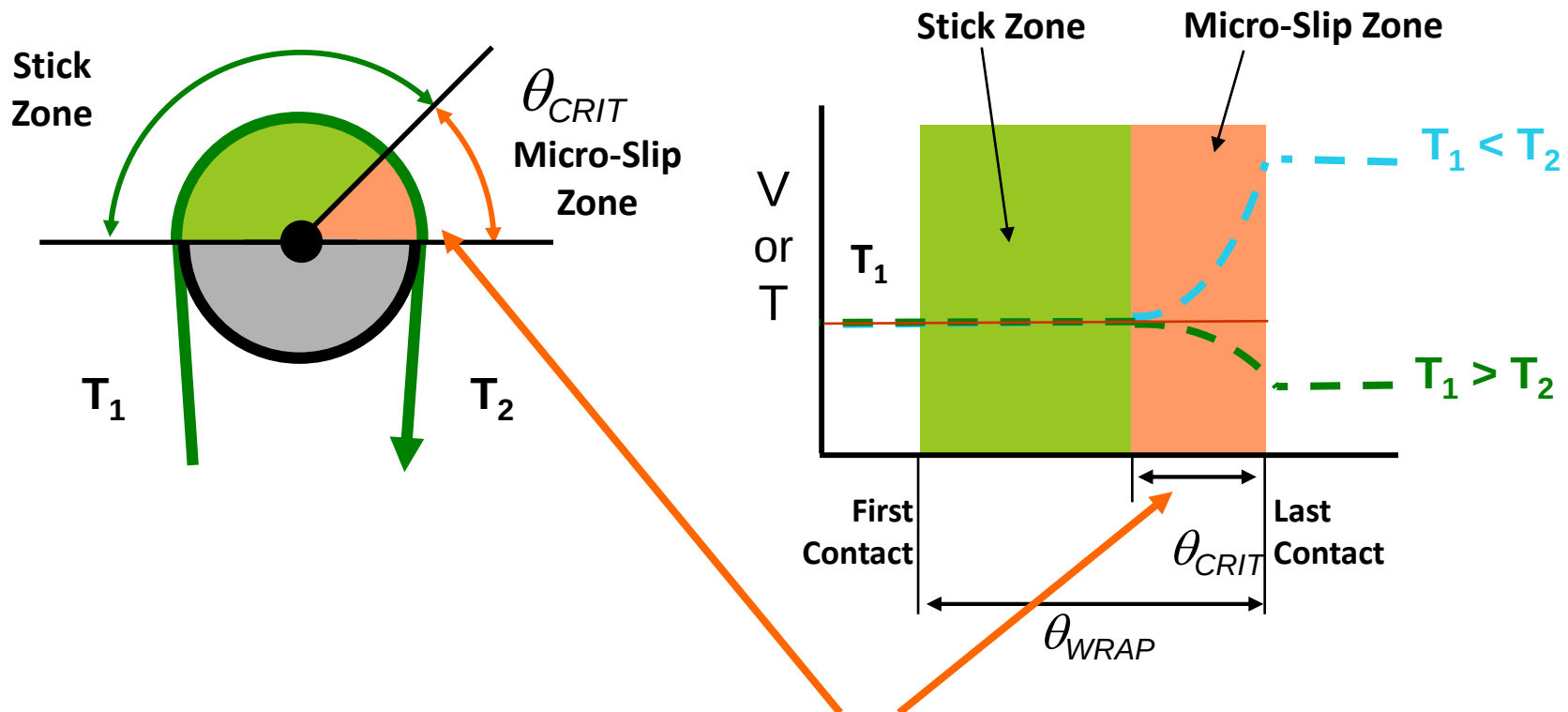
Even a small tension change creates a micro-slip zone.



$V_{WEB} = V_{ROLLER}$ until the micro-slip zone.

Micro-Slip

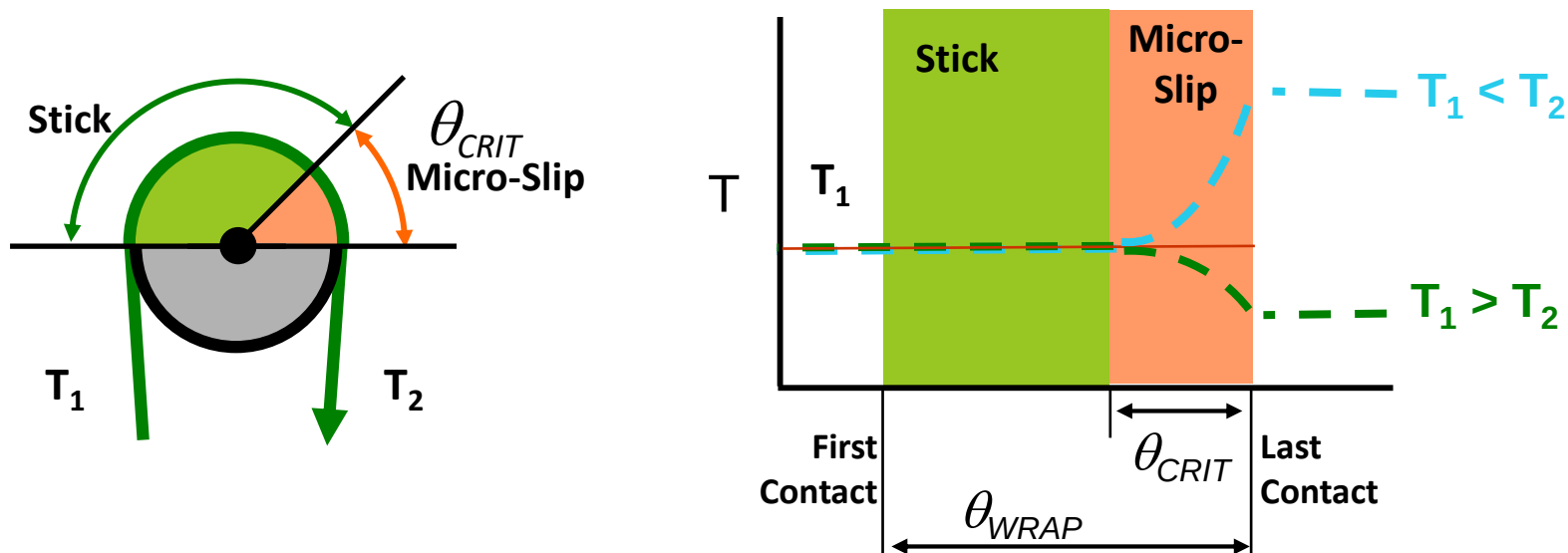
As the tension change increases, the micro-slip zone grows.



The micro-slip zone is on the exit side of the roller contact.

What is the Micro-Slip Dimension?

Slip between web and roller is proportional to strain change.



Micro-Slip Zone Length

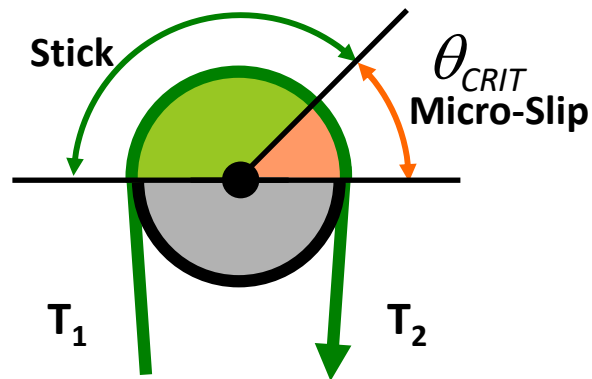
$$L_{M-S} = r \theta_{CRIT} = \frac{r \ln(T_{HIGH}/T_{LOW})}{\mu}$$

Distance of Slip

$$\Delta L_{SLIP} = (\Delta \varepsilon) L_{M-S} = \left(\frac{\Delta T}{tE} \right) L_{M-S}$$

Example: Micro-Slip Length

Slip between web and roller is proportional to strain change.



$$r = 100\text{mm}$$

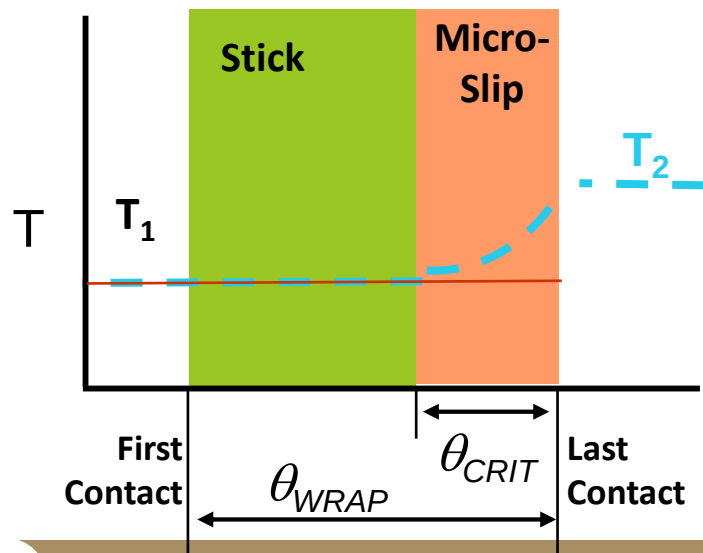
$$\mu = 0.3$$

$$t = 25\text{ mm}$$

$$T_1 = 200\text{ N/m}$$

$$T_2 = 250\text{ N/m}$$

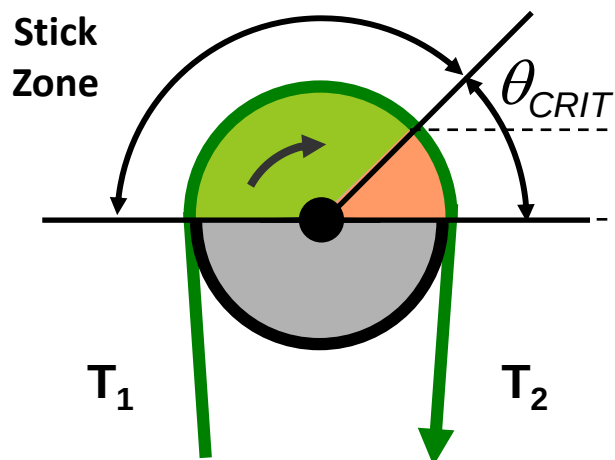
$$E = 3.4\text{ GPa}$$



$$L_{M-S} = \frac{(0.1)\ln(250/200)}{(0.3)} = 0.15\text{ m}$$

$$\Delta L_{SLIP} = \frac{(50\text{ N/m})(0.15\text{ m})}{(25\mu\text{m})(3.4\text{ GPa})} = 87\text{ }\mu\text{m}$$

Width Change from Micro-Slip



Web width will change in the micro-slip zone due to the tension change and the Poisson's effect.

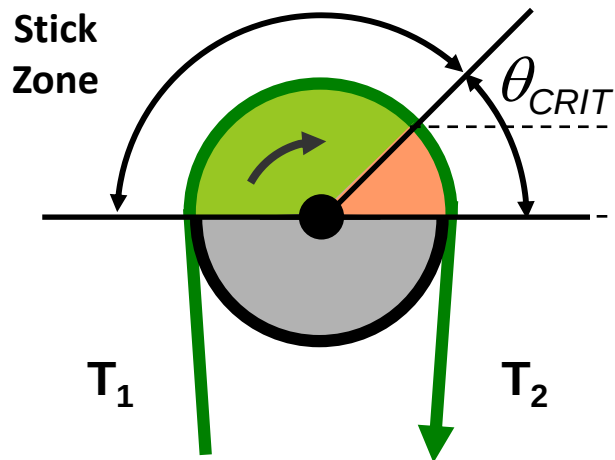
Low to High Tension Change

Micro-Slip Zone

High to Low Tension Change

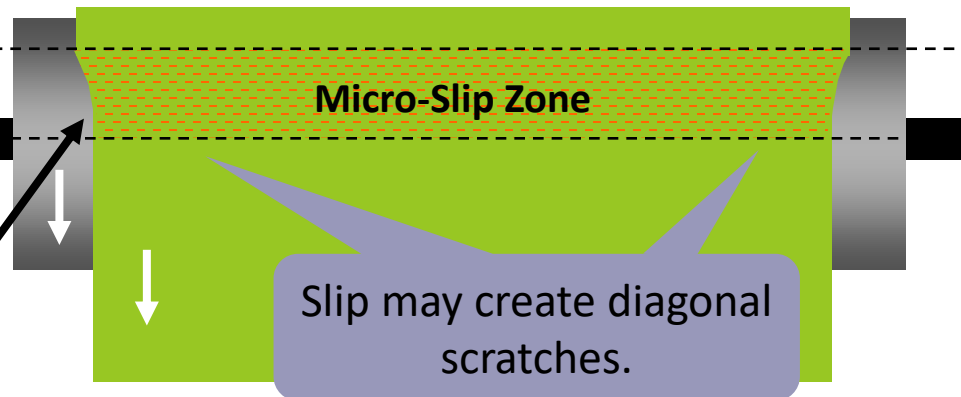
Micro-Slip Zone

Width Change from Micro-Slip

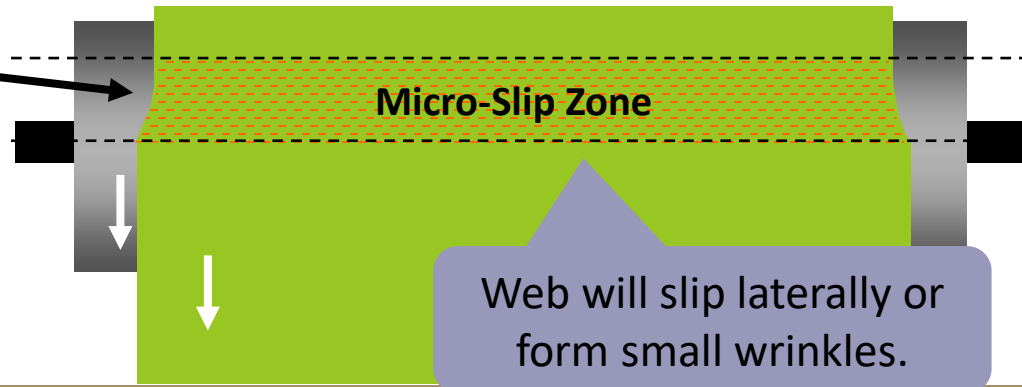


Web width will change in the micro-slip zone due to the tension change and the Poisson's effect.

Low to High Tension Change



High to Low Tension Change



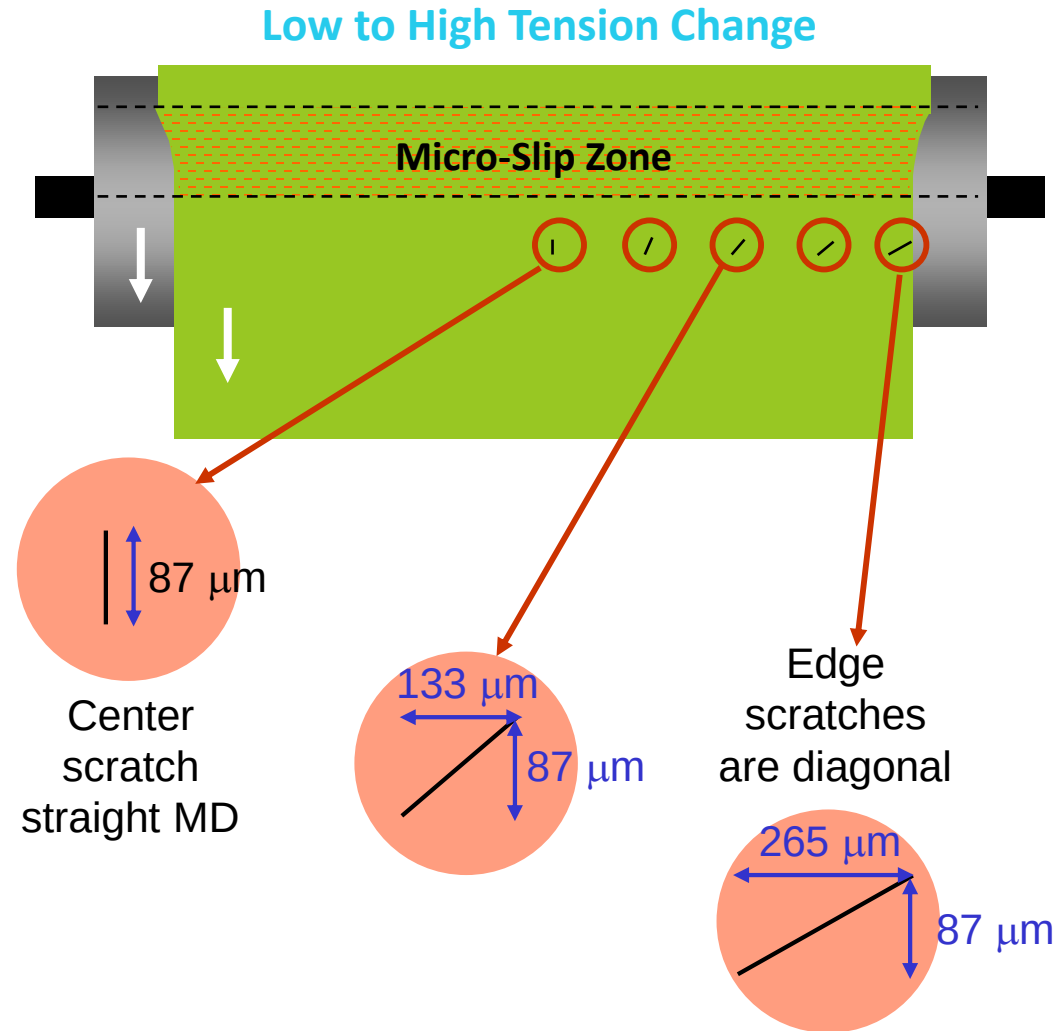
Calculate Lateral Shifting

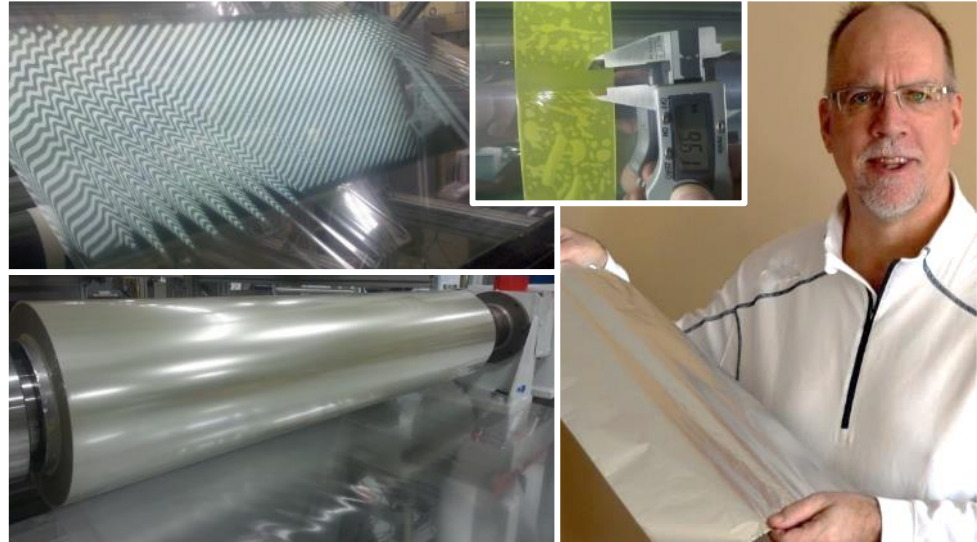
$$\begin{aligned}
 r &= 100\text{mm} & T_1 &= 200 \text{ N/m} \\
 \mu &= 0.3 & T_2 &= 250 \text{ N/m} \\
 t &= 25 \mu\text{m} & E &= 3.4 \text{ GPa} \\
 w &= 1.5 \text{ m} & \nu_{XY} &= 0.3
 \end{aligned}$$

$$\Delta w_{SLIP} = \nu_{XY} (\Delta \varepsilon) w = \frac{\nu_{XY} \Delta T w}{t E}$$

$$\Delta w_{SLIP} = \frac{(0.3)(50 \text{ N/m})(1.5 \text{ m})}{(25 \mu\text{m})(3.4 \text{ GPa})}$$

$$\Delta w_{SLIP} = 265 \text{ mm}$$





3.4 Turn Bars and Air Flotation

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3.4 Learning Objectives

- ▶ What pressure is needed to float a tensioned web?
- ▶ What is a web turn bar?
Why are turn bars used?
- ▶ What is an air reverser?
Why are air reversers used?
- ▶ What is air floatation?
Why is air floatation used?
- ▶ What methods are used for inline web folding?

Are Rollers Needed?

Fixed Bar or Plate

- ▶ Redirect the web over a bar, half-cylinder, or plate. There will be drag force, so tension in will be lower than tension out. The tension differential will be a function of web-roller coefficient of friction, wrap angle, and air lubrication (governed by speed, tension, radius, surface roughnesses, and porosity).

Fixed Forced-Air Lubricated Bar or Plate

- ▶ The drag force over any fixed bar or plate may be lowered with force-air lubrication. Forced-air lubrication devices have an internal cavity connected to a positive pressure supply and a hole or slot pattern for the air to flow out to the web to bar/plate contact zone. The forced air increases the web to surface lubrication, lowering the drag. These systems can simply assist lubrication, or with some engineering, can float the web at a designed height.

Folding

In-line folding is used to make envelopes, pouches, and other intentionally creased web products.

- Simple folding twists the web over long spans, minimizing side-to-side web path variations. Care should be taken not to increase edge stresses to the web's yield or break point.
- Controlled folding uses a 2:1:2 dual air bar geometry to maintain equal web paths and stresses on all lanes.

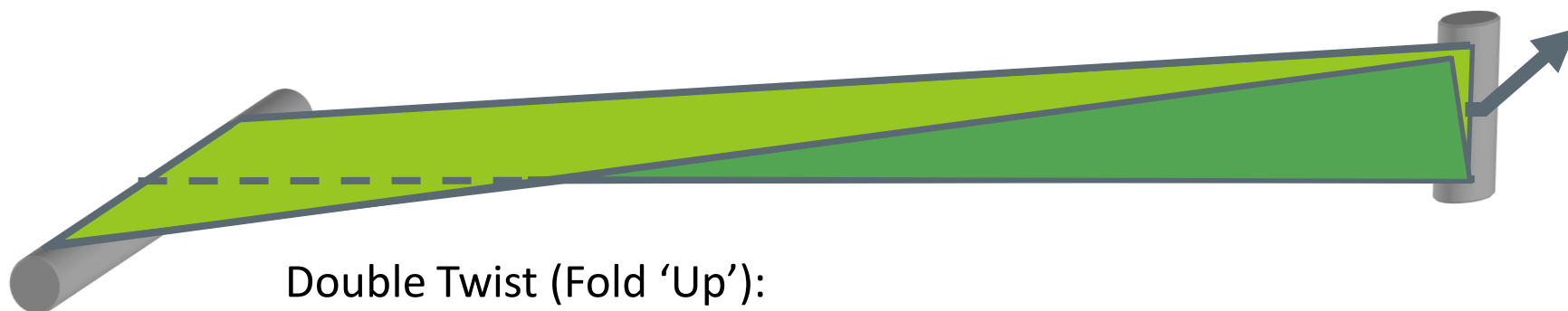
Folding

Simple folding can be completed over long spans, similar to web twisting.



Half Parallel (Fold 'Over'):

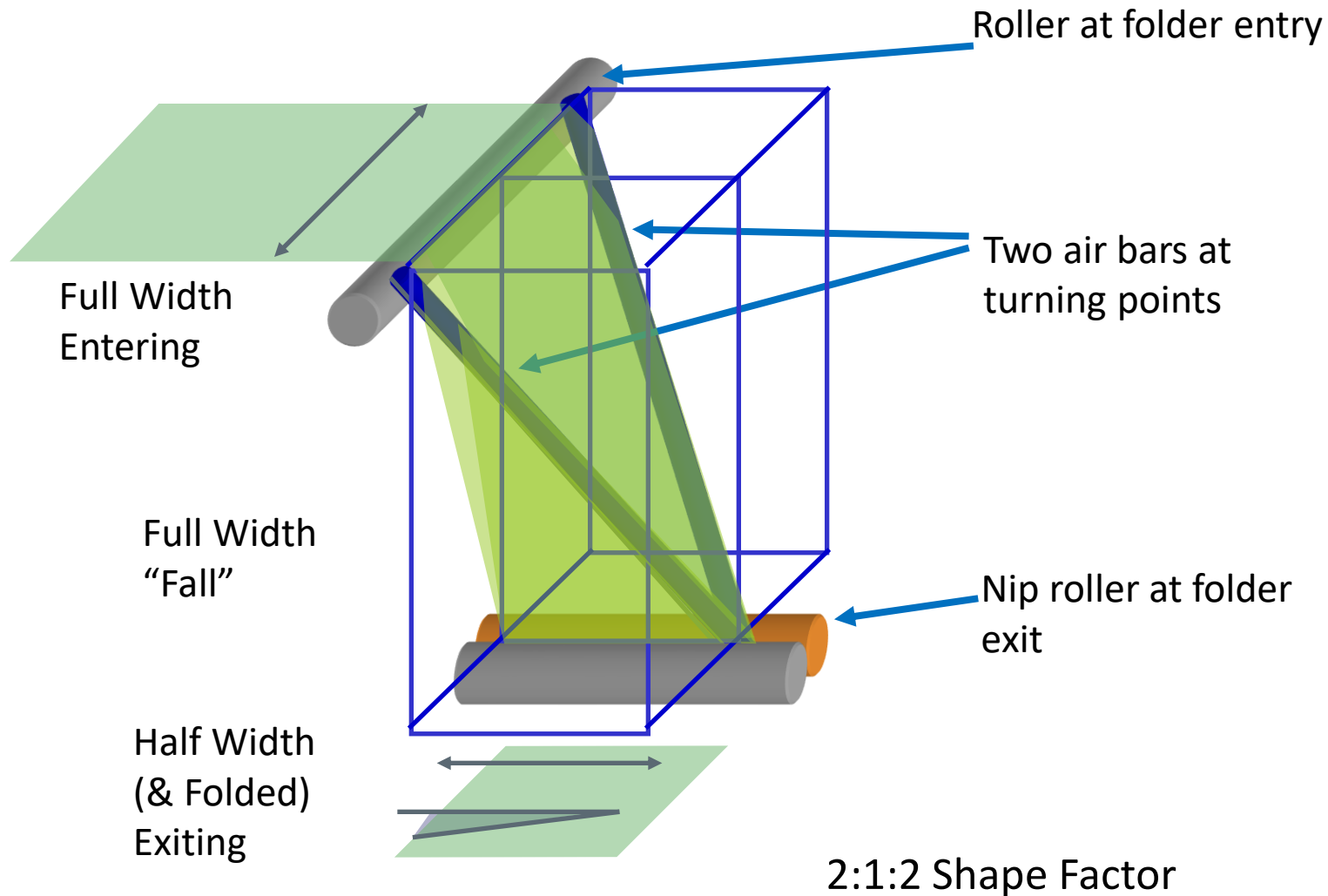
Half the web follows the parallel entry rule. The other half is forced over the top. A slide plate of shoe persuades the web to fold over.



Double Twist (Fold 'Up'):

Each edge twists 90 degrees out-of-plane. The center lane tracks straight.

Controlled Folding Geometry



Controlled Folding



Web Flip with Air Turns

1. Web enters 90-degree air turn.
2. Web wraps intermediate roller.
3. Web enters second 90 degree turn.
4. Web exits on centerline, but flipped from entry .

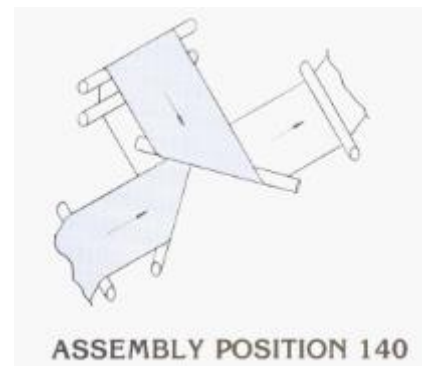
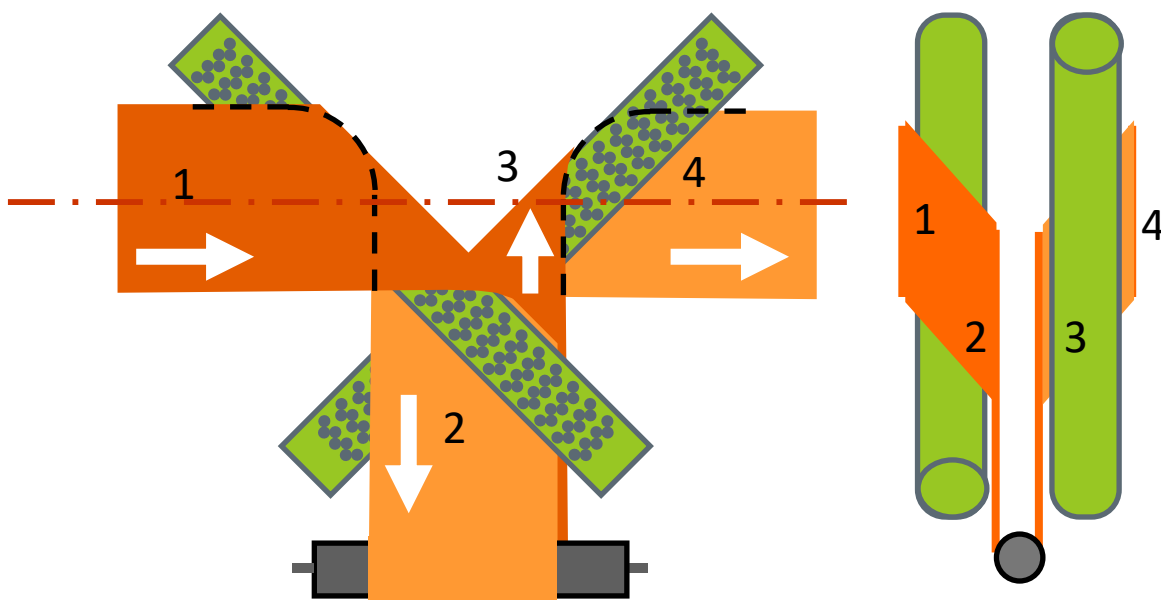


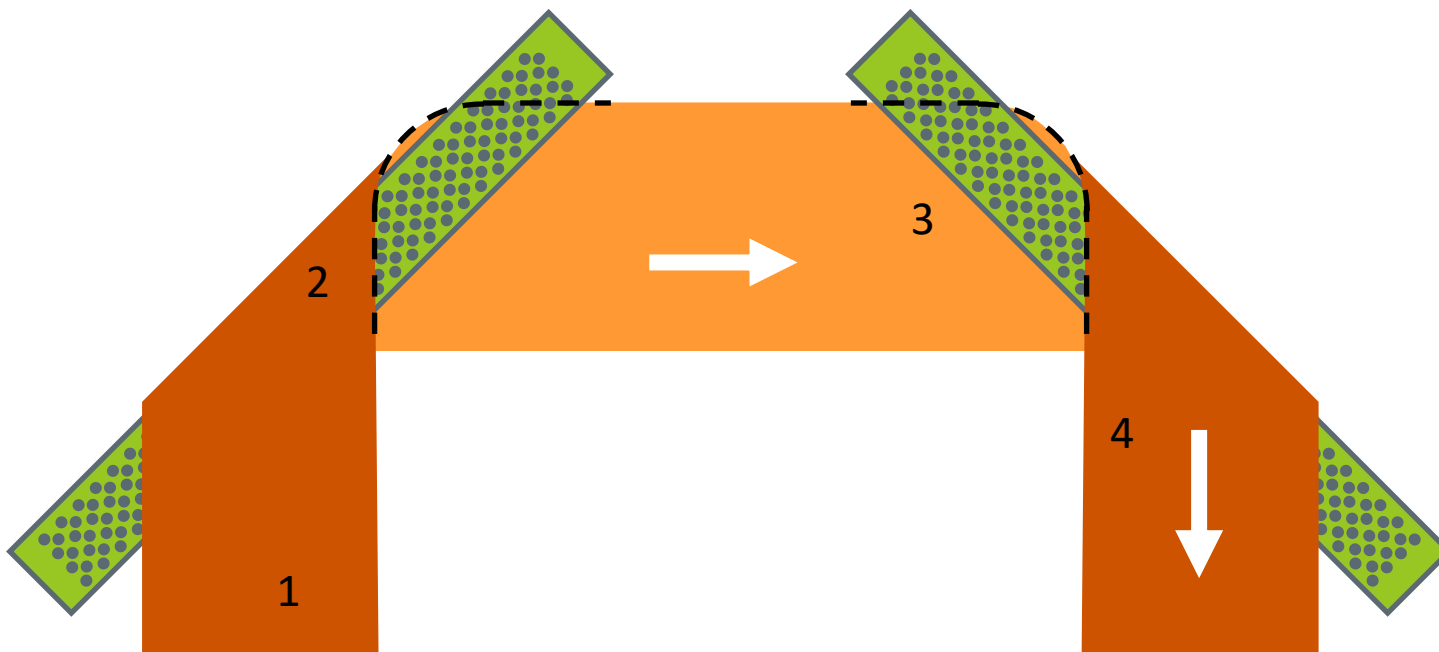
Image courtesy of Kohler Coating



Image courtesy of New Era Converting

Air Bar “U” Turn

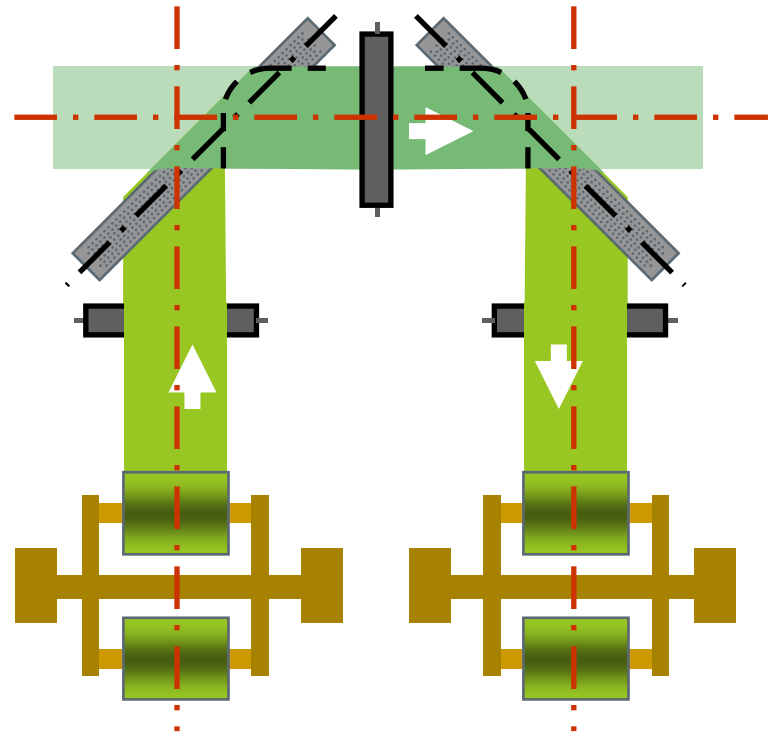
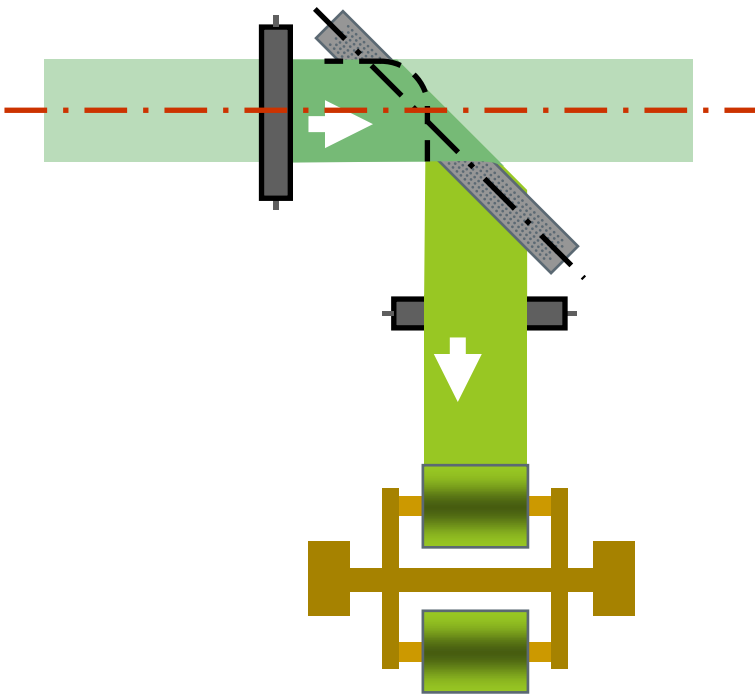
1. Two 90 degree turns can be used to make a “U” turn.
2. Web enters first 90 degree turn (flip one).
3. Web enters second 90 degree turn (flip two).
4. Web exits in reverse direction (same side up).



Line Layout Options Using Turn Bars

A single turn bar can move a winder 90° from the main process centerline.

Two turn bars can place the winder and unwind side-by-side.



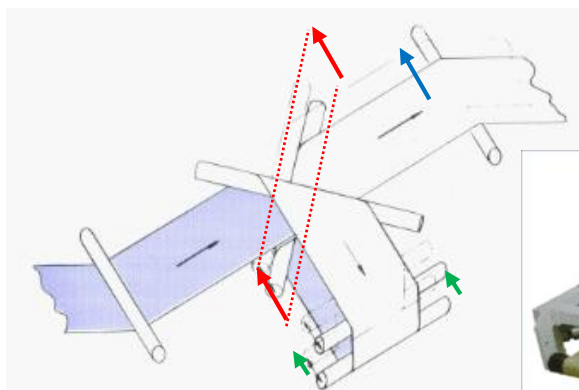
These options can shorten overall process length.

More Turn Bar Applications

Web Guiding

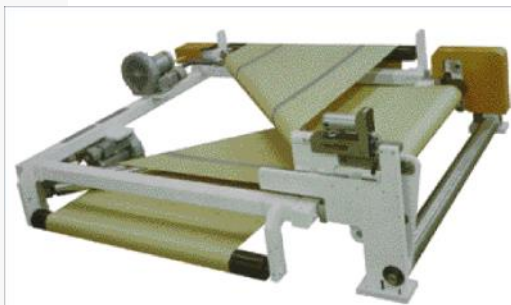
Shifting turn bar guides web to new centerline.

Side rollers move half as much as the turn bar to keep web path length constant.



GUIDEBAR UNIT

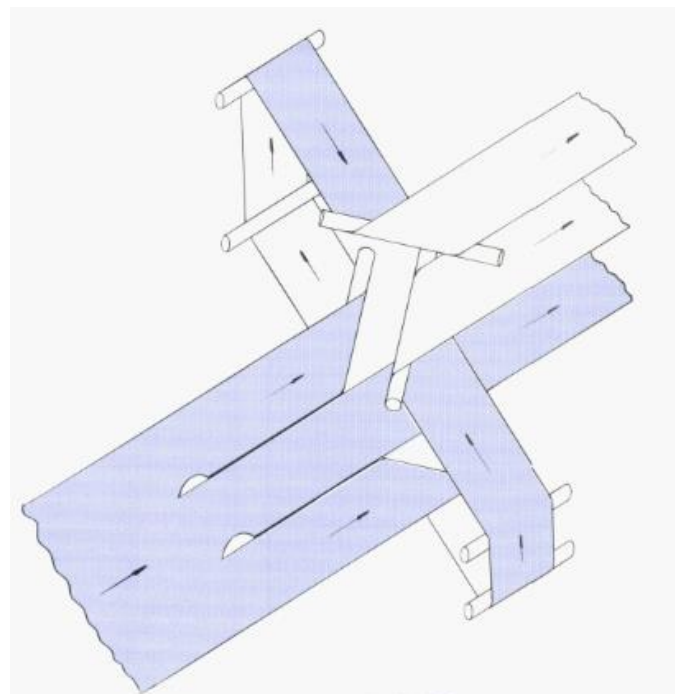
Utilizes a power pack and sensor from any guide manufacturer to provide single side contact guiding with zero change in web path length. Requires only 3 idler roll diameters of vertical height, regardless of web width. Can be designed to guide with or without inverting the web or to allow selection of either arrangement.



Images courtesy
of Kohler Coating

Web Slit and Stack

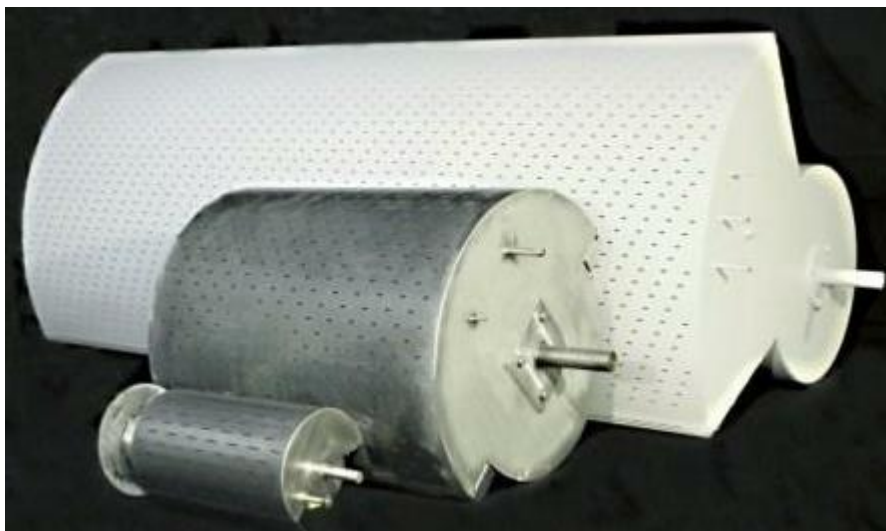
Slit lanes are guided to run atop each other.



WEB STACKER UNIT

Combines multiple slitwebs on a common centerline for slitting or laminating. Individual slit webs can be inverted or not as required.

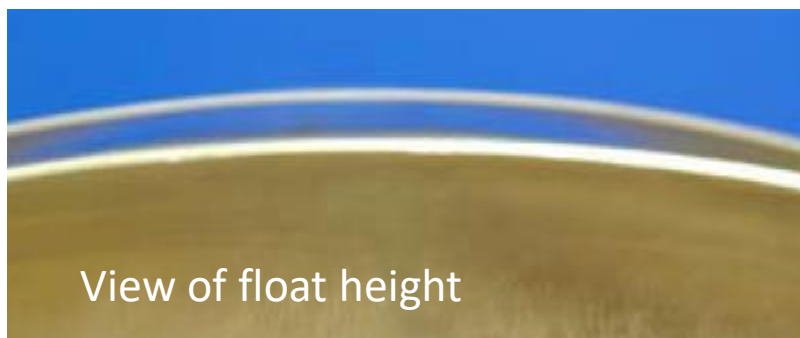
Air Turns / Reversers



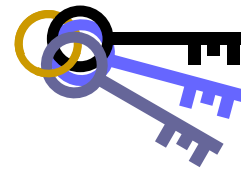
Images courtesy of Megtec Systems Inc.



Images courtesy of Advanced Systems Inc.



View of float height



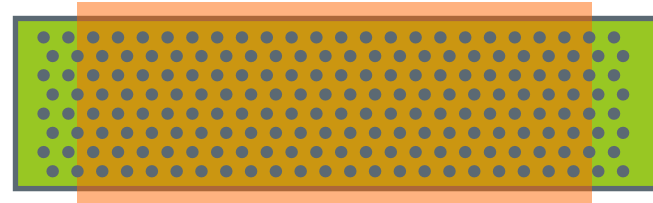
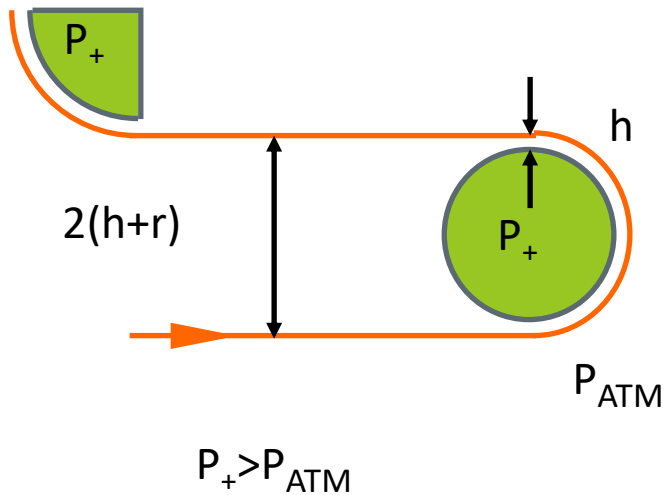
Lubricated Air Turns

$$P = \frac{T}{r}$$

$$T = 1PLI$$

$$r = 10''$$

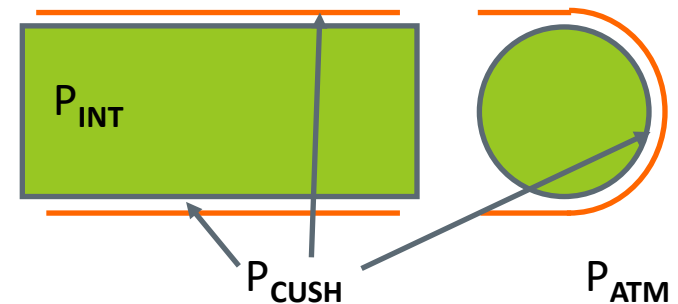
$$P = (1)/(10) = 0.1psi$$



Air Turn Calculations 1

1. Calculate air escape area from radius, width, wrap angle, and float height.

$$A_{ESC} = 2h(r\theta + w)$$



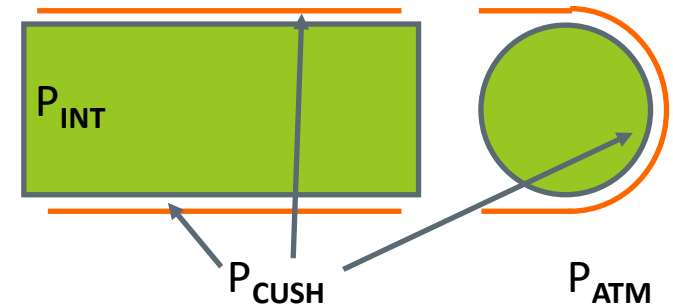
2. Calculate cushion pressure from web tension, radius, and width.

$$P_{CUSH} = \frac{T}{r}$$

Air Turn Calculations 2

- Calculate air escape velocity from cushion pressure, atmospheric pressure, and air density.

$$V_{ESC} = \sqrt{\frac{2(P_{CUSH} - P_{ATM})}{\rho}}$$



Based on:

$$\Delta P = \frac{\rho V^2}{2}$$

- Calculate air flow rate from escape velocity and escape area (this will determine fan capacity).

$$Q = V_{ESC} A_{ESC}$$

Air Turn Calculations 3

5. Calculate air velocity at holes from pressure drop and air density. (Internal pressure is typically 3 to 4 times cushion pressure for advanced air turns).

$$V_H = \sqrt{\frac{2(P_{INT} - P_{CUSH})}{\rho}}$$

6. Calculate area of all holes from escape flow rate and hole velocity.

$$A_{HOLES} = \frac{V_{ESC} A_{ESC}}{V_{HOLES}}$$

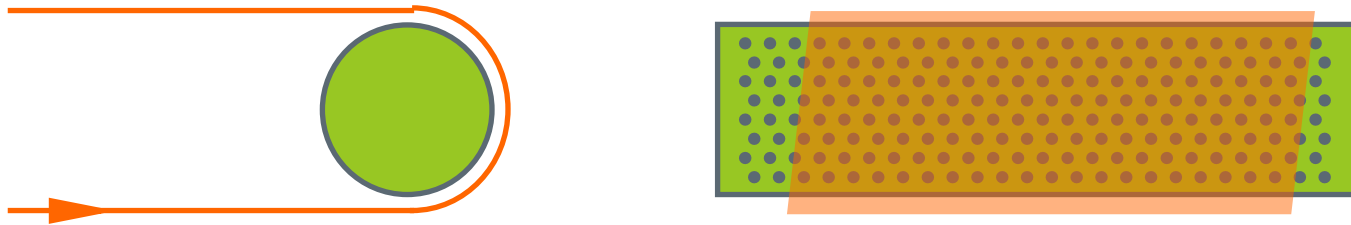
7. Calculate percent open area from hole area and total cushion area.

$$Percent_Holes = \frac{A_H}{rw\theta}$$

Air Turn Lateral Motion

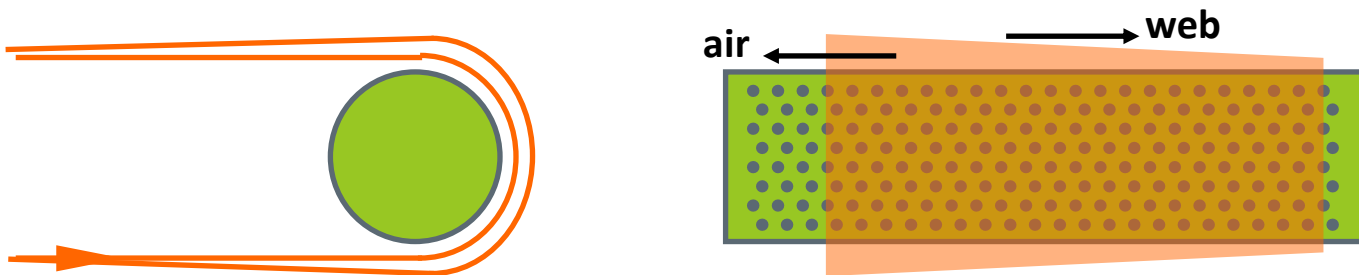
The parallel entry rule does not apply to air turns.

The web may take a helical path, changing lateral position.



Baggy web or uneven air flow may float higher on one side.

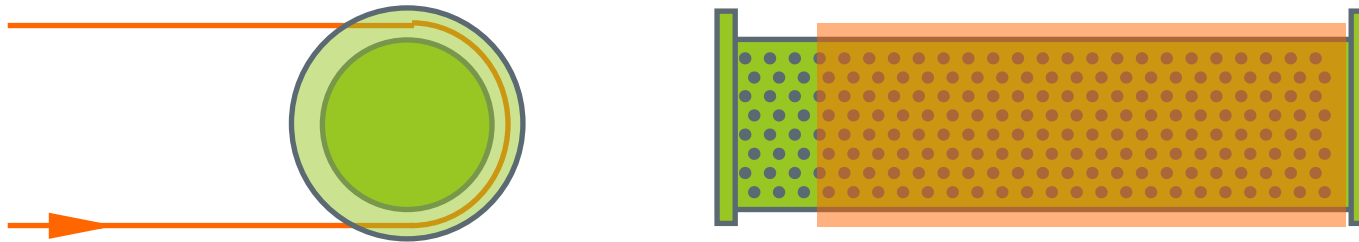
This creates a lateral force capable of shifting the web.



Air Turn Added Features

Some air turns have edge dams to limit lateral motion.

This works better with stiffer webs, but may wrinkle thin webs.



Thinner webs are more prone to flutter at the air turn exit.

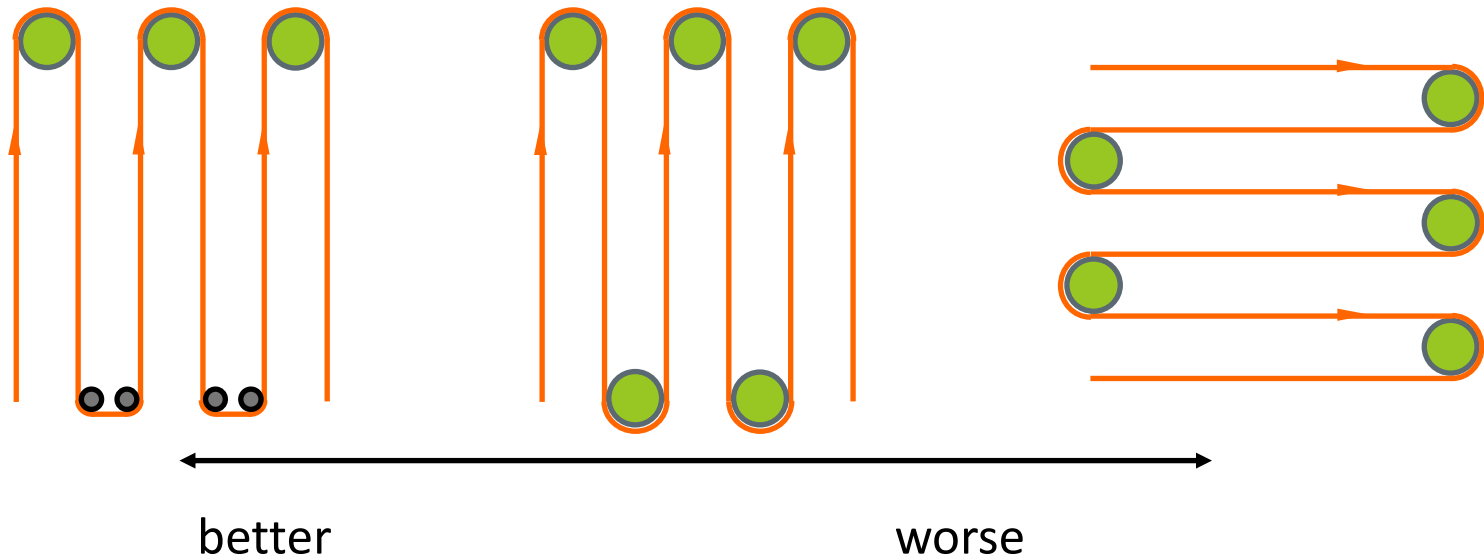
Some air turns have a extension on the exit side to reduce flutter.



Air Turn Series

Air turns have no centering tendency.

- Avoid excessive sequential air turns.
- For extensive one-sided drying, alternate between rollers and air turns.
- Vertical spans are preferred to horizontal (less catenary and alignment problems).



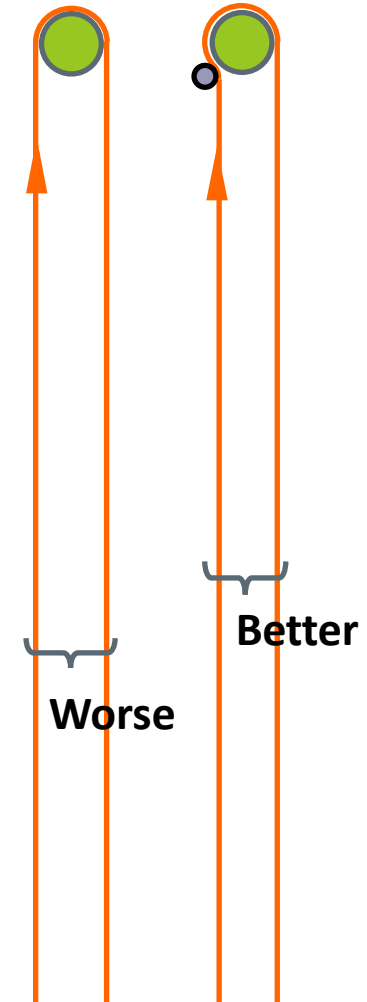
Air Turn Spans

Avoid long spans into air turns.

A pre-air turn idler with 10-20° wrap can greatly improve air turn lateral stability.



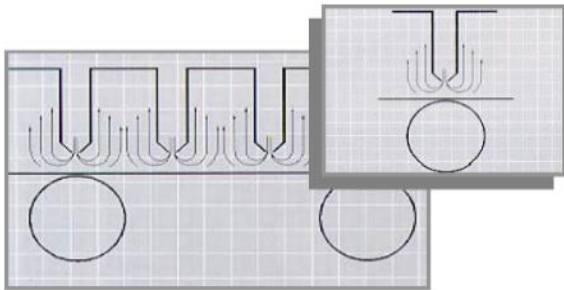
In long horizontal spans, catenary sag must be considered.



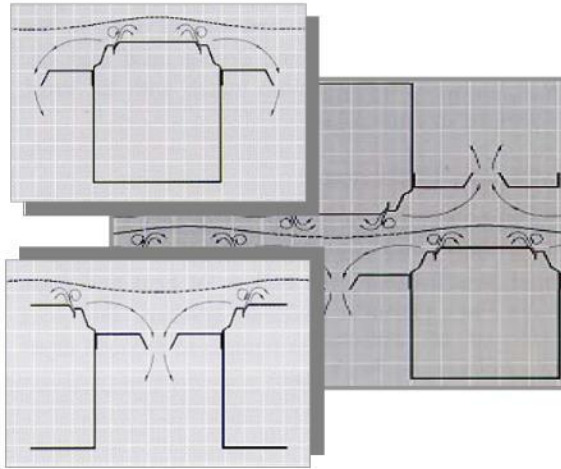


Nozzle Options

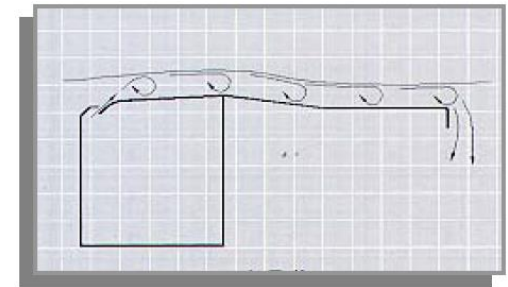
Slot Nozzles



Impingement Nozzles

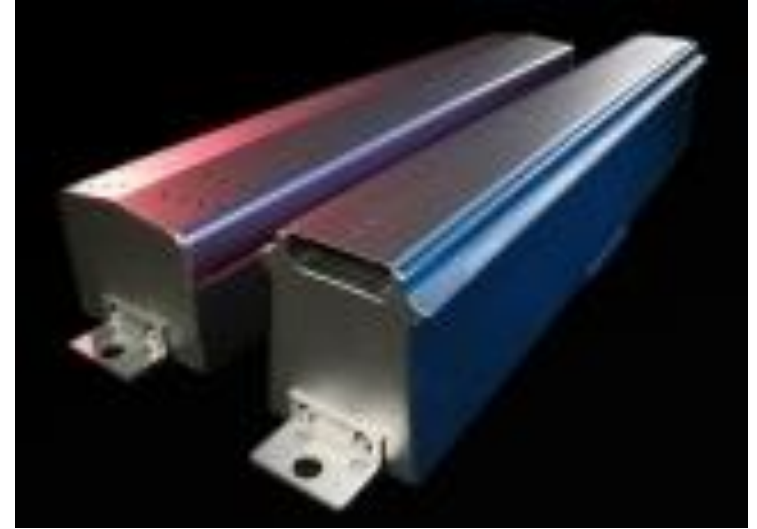


Coanda Nozzles



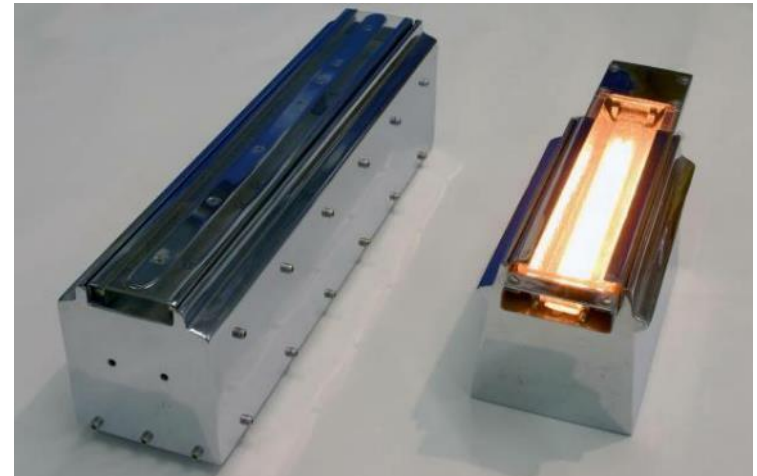
Images courtesy of Advanced Systems Inc.

Air Nozzle Options



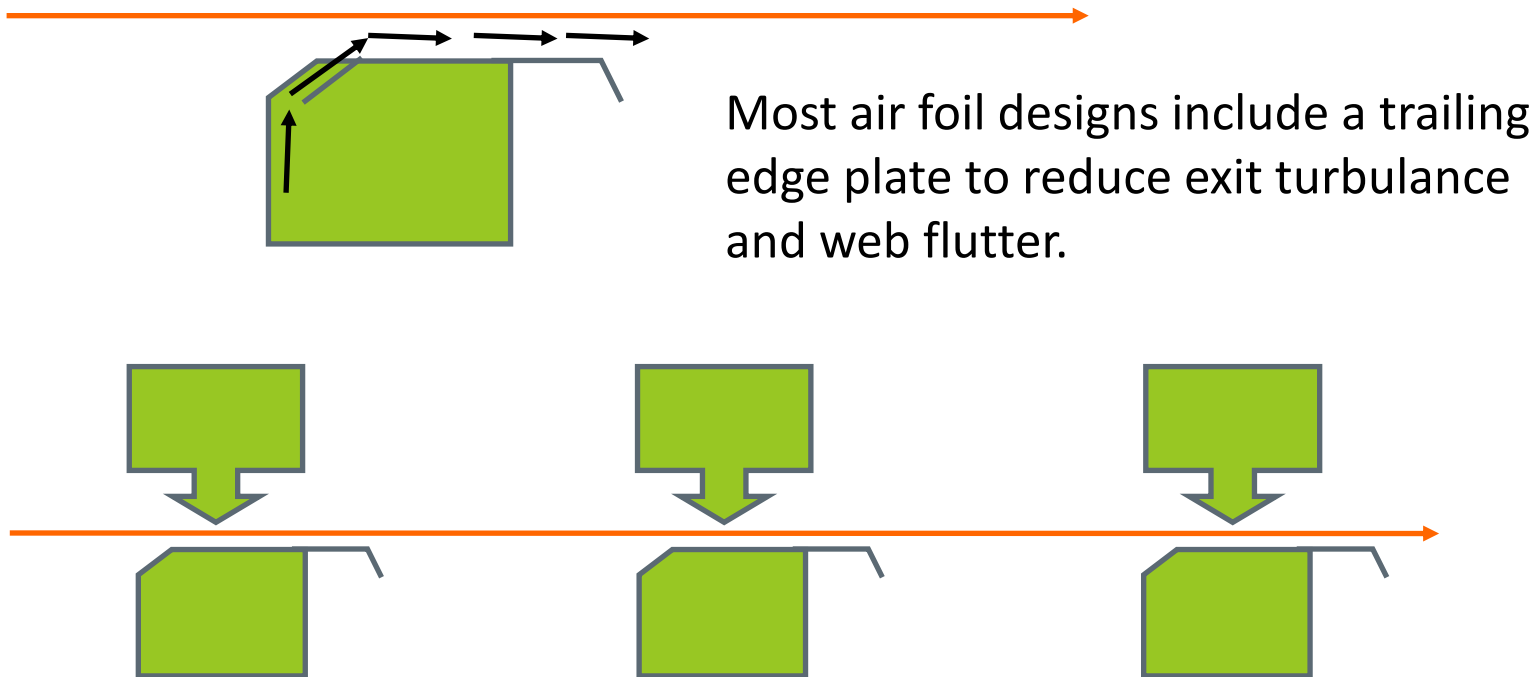
Images courtesy of Megtec Systems Inc.

Nozzles with
integrated RF or IR

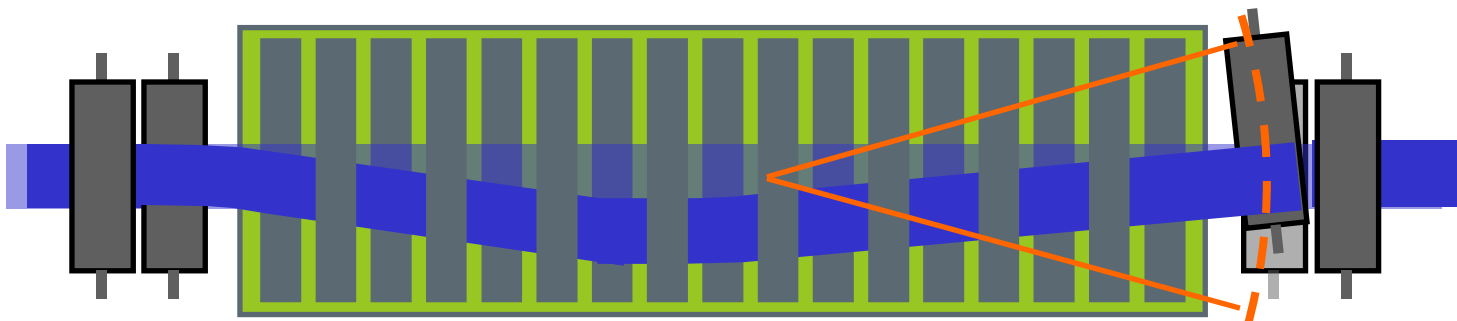


Air Foils

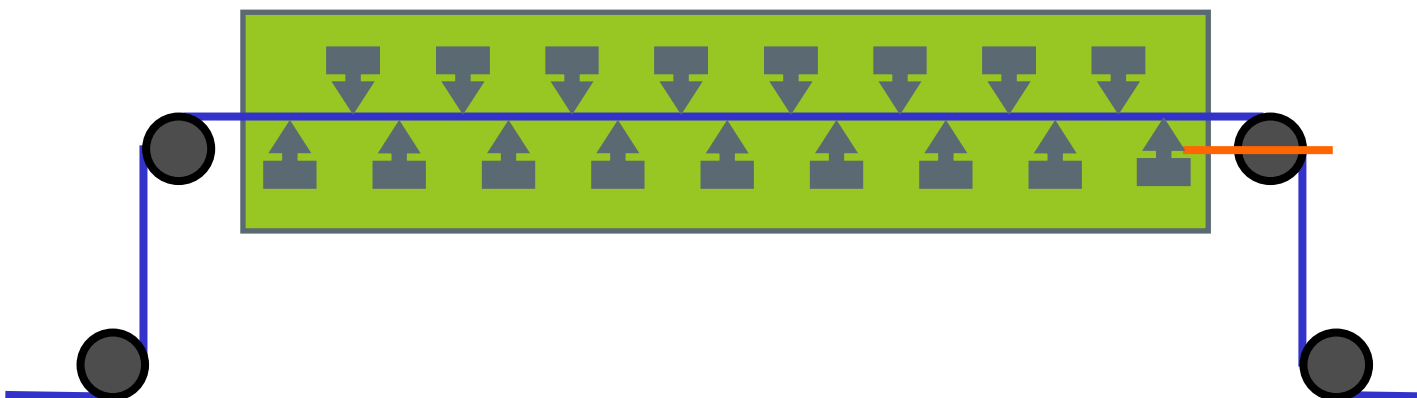
Air flow is mostly parallel to web, resulting in small lateral forces



Floatation Ovens Tracking

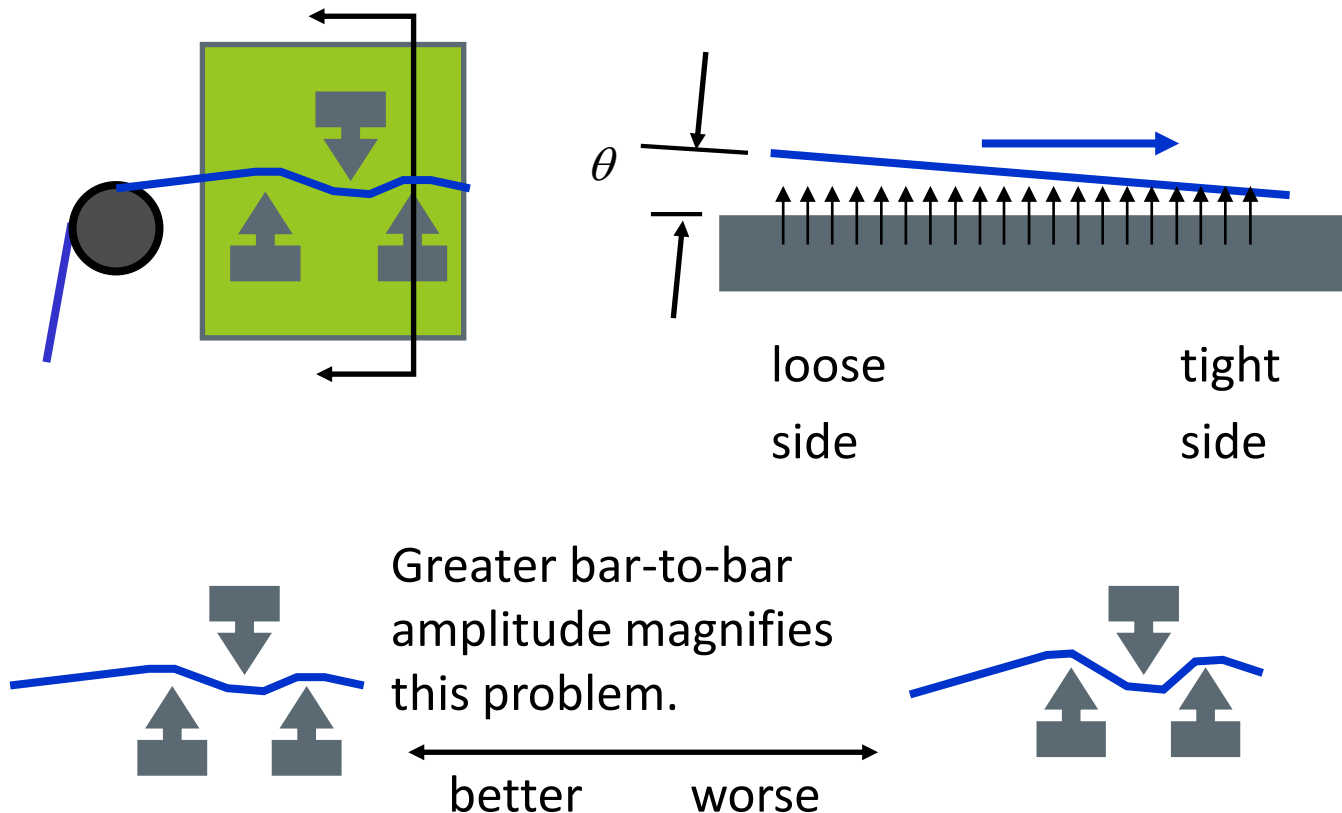


As mentioned earlier, air floatation ovens commonly create lateral tracking problems.

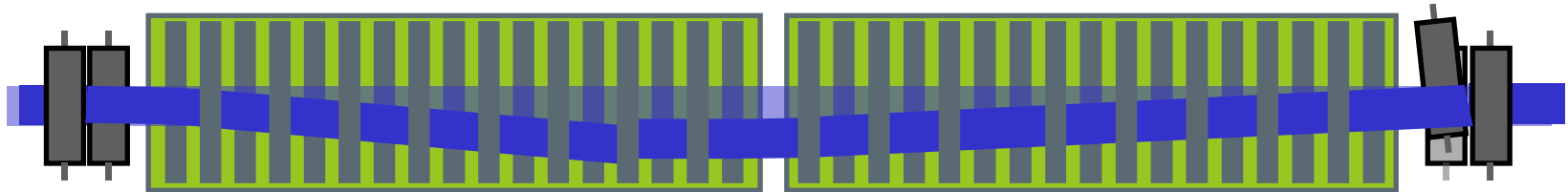


Floatation Web Amplitude

Baggy web in air floatation is prone to large lateral motion toward the tight side.



Mapping Lateral Motion



Mapping the web position vs. web length for various process conditions (tension, air flow by zone) can identify problem areas.

